



Non-commutative deformations of quotient singularities.

Lien Cartaya Bermúdez

A thesis submitted in partial fulfillment of the requirements
for the Master's degree in Mathematics.

Advisor: Stephen Griffeth, PhD.

Institute of Mathematics
University of Talca

February 2024

Contents

Abstract	4
Chapter 1. Introduction	5
Chapter 2. Preliminaries.	10
1. Filtration and grading.	10
1.1. Graded algebras.	10
1.2. Filtered modules.	11
1.3. Associated graded algebra.	12
1.4. Induced filtration.	13
2. Tensor products, tensor algebras and semidirected product.	13
2.1. Tensor product.	13
2.2. Tensor algebra.	14
2.3. Semidirect product.	15
3. Dimensions.	16
3.1. Gelfand-Kirillov dimension.	16
3.2. Classical Krull dimension.	19
4. Categories, functors and natural transformations.	19
4.1. Categories.	19
4.2. Functors.	21
4.3. Natural transformations.	22
4.4. Adjunctions.	23
5. Tor.	24
6. Some results of Algebraic Geometry.	25
6.1. Fiber product of affine schemes.	25
6.2. Flatness.	27
6.3. Linearly reductive groups.	28
7. Representations	29
7.1. Representation of groups.	29
7.2. Representations of quivers.	36

Chapter 3. Non-commutative deformations of Kleinian singularities.	44
1. The algebras $\mathcal{S}^\lambda$ and $\mathcal{O}^\lambda$.	44
1.1. Example using the cyclic group of order 2. Part 1.	45
2. Deformed preprojective algebras.	46
3. Morita equivalence.	48
3.1. Example using the cyclic group of order 2. Part 2.	57
4. Representations of deformed preprojective algebras.	58
4.1. Example using the cyclic group of order 2. Part 3.	60
5. Reflection functors.	61
5.1. An example using the cyclic group of order 2. Part 4.	67
6. Roots for a quiver.	68
6.1. An example using the cyclic group of order 2. Part 5.	71
7. The case when $\lambda \cdot \delta \neq 0$.	72
8. The case when $\lambda \cdot \delta = 0$	76
8.1. The map $\phi_\lambda : \mathcal{O}^\lambda \rightarrow \mathbf{A}^\lambda$.	78
8.2. Situation over the K -algebra $\mathbf{R}$.	85
Final comments.	96
Bibliography	97

Acknowledgements

First and foremost, I want to express my sincere thanks to my advisor, Professor Stephen Griffeth. His choice of reading Crawley-Boevey and Holland's article not only revealed a useful tool in his current research line but also allowed me to grow as a mathematician by studying the necessary topics for understanding the article. I owe much of what I've learned in mathematics to him, and I am grateful for the beautiful ideas that he has introduced me to.

I appreciate Professors William Crawley-Boevey and Daniel Juteau for their valuable comments and explanations regarding the results studied in the thesis.

Special thanks to Carlos Ajila, who has been willing to teach me and help improve my mathematical skills since I started studying representations. Studying with him has been greatly beneficial and, at times, crucial to my understanding of Crawley-Boevey and Holland's article.

Overall, I acknowledge and thank the excellent faculty of academics at the Institute of Mathematics at the University of Talca, who significantly contributed to my education during the program.

I express my gratitude to Adrian Vidal, who has listened to me over all these years and helped me make the right decisions. To my mother, Lien Bermúdez, and Ignacio Vidal, without whom I would not have been able to pursue this master's degree.

Finally, I thank to the University of Talca and to the Agencia Nacional de Investigación y Desarrollo, ANID, Chile, for supporting this work through a fee scholarship and the FONDECYT Proyecto Regular 1231355.

Abstract

This thesis constitutes a review of the article "Non-commutative deformations of Kleinian singularities" by Crawley-Boevey and Holland [10]. We will study a family of non-commutative deformations, defined by Crawley-Boevey and Holland, of the coordinate ring of the Kleinian singularity K^2/Γ , with $K = \overline{K}$, $\text{char} K = 0$ and $\Gamma \leq SL_2(K)$ non-trivial finite subgroup. This family is the following:

$$\mathcal{O}^\lambda := e \frac{K\langle x, y \rangle \otimes K\Gamma}{(xy - yx - \lambda)} e \text{ for } \lambda \in Z(K\Gamma), \text{ where } e = \frac{1}{|\Gamma|} \sum_{\gamma \in \Gamma} \gamma,$$

and $K\langle x, y \rangle$, is the ring of non-commuting polynomials with coefficients in K . If λ has trace 0 on $K\Gamma$, then $\mathcal{O}^\lambda$ is commutative. On the other hand, let Q be the extended Dynkin quiver obtained by the choice of an orientation in the McKay graph associated to Γ ; we relate $\mathcal{O}^\lambda$ to the deformed preprojective algebra of Q , called $\Pi^\lambda(Q)$. We classify the simple $\Pi^\lambda(Q)$ -modules according to the position of λ relative to the root system associated with the McKay graph of Γ . For λ of trace 0 on $K\Gamma$ we construct an isomorphism between $\mathcal{O}^\lambda$ and the ring of functions on the moduli space of $\Pi^\lambda(Q)$ -modules of dimension δ .

CHAPTER 1

Introduction

This thesis constitutes a review of the article "Non-commutative deformations of Kleinian singularities" written by Crawley-Boevey and Holland in 1998 [10]. While not covering its entirety, the focus is on the key results that are of interest in the study of the ring of polynomial functions on the zero-fiber of the quotient map under the action of a quaternionic reflection group W , continuing the work initiated in [21], [22], [16], [17] and [19]. We specifically studied sections 1 through 8 of the article, which consists of 10 sections.

This thesis consists of three chapters: the current introductory chapter, a second chapter with preliminary results, and a third chapter elaborating on the findings studied in [10].

Let K be an algebraically closed field of characteristic zero (we will mainly consider $K = \mathbb{C}$) and let Γ be a non-trivial finite subgroup of $SL_2(K)$. There is a projection from K^2 to the Kleinian singularity $K^2 \twoheadrightarrow K^2/\Gamma = \text{Spec}(K[x, y]^\Gamma)$. We study a family of deformations of the Kleinian singularity. We are interested in subsequently generalizing to deformations of V/W , where V is a symplectic K -vector space, and W is a symplectic reflection group.

The deformations of the Kleinian singularity: In section 1 of chapter 3 we introduce our deformations, denoted by $\mathcal{O}^\lambda$ for $\lambda \in Z(K\Gamma)$, which are in general non-commutative. The group Γ acts naturally on the ring $K\langle x, y \rangle$ of non-commuting polynomials, so one can form the *skew group algebra* $K\langle x, y \rangle \rtimes \Gamma = K\langle x, y \rangle \otimes K\Gamma$, whose multiplication is linearly generated by

$$(f \otimes \gamma)(f' \otimes \gamma') = f(\gamma \cdot f') \otimes \gamma\gamma'$$

for all $f, f' \in K\langle x, y \rangle, \gamma, \gamma' \in \Gamma$, where $\cdot$ denotes the action of Γ on $K\langle x, y \rangle$. When confusion will not result, we may omit the tensor symbol and write multiplication as juxtaposition. For $\lambda \in Z(K\Gamma)$ let

$$S^\lambda := (K\langle x, y \rangle \rtimes \Gamma) / (xy - yx - \lambda).$$

We then define $\mathcal{O}^\lambda := eS^\lambda e$, where $e = \frac{1}{|\Gamma|} \sum_{\gamma \in \Gamma} \gamma$.

The ring $\mathcal{O}^\lambda$ is naturally filtered (see section 1 in the chapter 2), with a filtration derived from the degree filtration of $K\langle x, y \rangle$. The graded algebra associated with this filtration is isomorphic to $K[x, y]^\Gamma$ by the map

$$K[x, y]^\Gamma \longrightarrow \text{gr}\mathcal{O}^\lambda = e(K[x, y] \rtimes \Gamma)e$$

$$f \mapsto fe = ef = efe.$$

We need the McKay graph and a family of deformations of the preprojective algebra to study how $\mathcal{O}^\lambda$ depends on λ .

The McKay graph: Let $N_0, N_1, \dots, N_n$ be the irreducible representations of Γ (see sections 7.1 in the chapter 2), with N_0 trivial, and let V be the natural 2-dimensional representation of Γ . The *McKay graph* of Γ is the graph with vertex set $I = \{0, 1, \dots, n\}$, and with as many edges between i and j as the multiplicity of N_i in $V \otimes N_j$ (which is the same as the multiplicity of N_j in $V \otimes N_i$, because $V^* = V$). This graph is an extended Dynkin diagram of type ADE (see subsection 7.2.5 in section 7.1 of the preliminary chapter 2 or for more details see [15]).

Let Q be the quiver obtained from the McKay graph by choosing any orientation of the edges (see section 7.2 in chapter 2), and let $\delta \in \mathbb{N}^I$ be the vector with $\delta_i = \dim N_i$.

We always identify

$$Z(K\Gamma) = K^I$$

$$\lambda \leftrightarrow (\lambda_i),$$

where λ_i is the trace of λ on N_i . For example, $\lambda \cdot \delta = \sum_{i \in I} \lambda_i \delta_i$ is the trace of $\lambda \in Z(K\Gamma)$ on the regular representation $K\Gamma$.

The deformed preprojective algebra: Now let Q be an arbitrary quiver (the most important case is when Q is an orientation of the McKay graph of Γ). In section 2 of chapter 3 we define the preprojective algebra and its deformations. We write $\overline{Q}$ for the double of Q , i.e. the quiver obtained by adding a reverse arrow $a^* : j \rightarrow i \in Q^{op}$ for each arrow $a : i \rightarrow j \in Q$, and we write $K\overline{Q}$ for its path algebra. $K\overline{Q}$ has basis the paths in $\overline{Q}$, including a *trivial path* e_i for each vertex i . If $\lambda = (\lambda_i) \in K^I$, the *deformed preprojective algebra of Q of weight λ* is

$$\Pi^\lambda(Q) = \Pi^\lambda := K\overline{Q} / \left(\sum_{a \in Q} [a, a^*] - \lambda \right),$$

where λ is identified with the element $\sum_{i \in I} \lambda_i e_i \in K\overline{Q}$, and $[a, a^*] = aa^* - a^*a$ is a commutator.

In section 3 of chapter 3, we prove the first significant result that relates deformations $\mathcal{O}^\lambda$ and Π^λ :

THEOREM 0.1. $\mathcal{O}^\lambda \cong e_0 \Pi^\lambda e_0$, $\forall \lambda \in Z(K\Gamma) = K^I$ and $S^\lambda \xrightarrow{M} \Pi^\lambda$, which means that the categories $S^\lambda\text{-mod}$ and $\Pi^\lambda\text{-mod}$ are equivalent. (See Theorem 3.8 and Corollary 3.9).

Thus, we can see $\mathcal{O}^\lambda$ as a subring of the deformed preprojective algebra $\Pi^\lambda(Q)$.

Representations of the deformed preprojective algebra: In section 4 of chapter 3 we study the representations of the deformed preprojective algebra Π^λ . Each element of the K -vector space

$$\text{Rep}(\overline{Q}, \delta) = \bigoplus_{\substack{a:i \rightarrow j \\ a \in Q}} \text{Hom}(K^{\delta_i}, K^{\delta_j}) \oplus \text{Hom}(K^{\delta_j}, K^{\delta_i})$$

determines a representation of $\overline{Q}$ with dimension vector δ and the isomorphism classes of representations of dimension vector δ are in bijection with the orbits in $\text{Rep}(\overline{Q}, \delta)$ under the action of the group $\text{GL}(\delta) = \prod_{i \in I} \text{GL}(K^{\delta_i})$. Letting

$$\text{End}(\delta)_0 = \left\{ (\theta_i) \in \prod_{i \in I} \text{End}(K^{\delta_i}) \mid \sum_{i \in I} \text{tr}(\theta_i) = 0 \right\},$$

there is a map (the moment map $\text{Rep}(\overline{Q}, \delta) = T^*(\text{Rep}(Q, \delta)) \longrightarrow \text{Lie}(\text{GL}(\delta)) \supseteq \text{End}(\delta)_0$)

$$\mu : \text{Rep}(\overline{Q}, \delta) \rightarrow \text{End}(\delta)_0$$

$$x = (x_a, x_{a^*}) \mapsto \sum_{a \in Q} [x_a, x_{a^*}].$$

If $\lambda \in K^I$ satisfies $\lambda \cdot \delta = 0$ then λ can be identified with an element in the center of $\text{End}(\delta)_0$, $\text{Rep}(\Pi^\lambda(Q), \delta) = \mu^{-1}(\lambda)$ is the space (and naturally a scheme) of $\Pi^\lambda(Q)$ -modules of dimension vector δ , and we write $K[\text{Rep}(\Pi^\lambda(Q), \delta)]$ for its coordinate ring.

Another important result is proven in the section 8:

THEOREM 0.2. *For Q an orientation of the McKay graph of Γ , we have: $\lambda \cdot \delta = 0 \implies \mathcal{O}^\lambda \cong K[\text{Rep}(\Pi^\lambda(Q), \delta)]^{\text{GL}(\delta)}$. (See section 8).*

In order to understand this theorem, the previous sections of the chapter 3 are necessary. Now, we return to the case of an arbitrary quiver Q .

In section 5 of chapter 3, we study certain reflection functors which are the key tool for the study of the representation theory of the deformed preprojective algebra.

In section 6 of chapter 3 we introduce the root system associated to the quiver and prove fundamental results on the classification of $\Pi^\lambda(Q)$ -modules, such as:

THEOREM 0.3. *Let $\lambda \in K^I$ and let Q be an arbitrary quiver.*

- $\lambda_i = 0$ if and only if $\Pi^\lambda(Q)$ has a representation, denoted S_i , with dimension vector ϵ_i , which is the vector with 1 in position i and zeros in the other positions (see Lemma 6.7).
- $\lambda \cdot \alpha = 0$ for some real root $\alpha \implies \Pi^\lambda(Q)$ has a simple representation with dimension vector a real root (see Theorem 6.8).
- $\lambda \cdot \alpha \neq 0 \forall \alpha$ real root $\implies$ any finite-dimensional indecomposable representation of $\Pi^\lambda(Q)$ has dimension vector an imaginary root (see Theorem 6.8).

THEOREM 0.4. *Let Q be a Dynkin quiver or disjoint union of Dynkin quivers. Then $\Pi^\lambda(Q) = 0$ if and only if $\lambda \cdot \alpha \neq 0 \forall \alpha$ real root (see Corollary 6.10).*

In the section 7 using a notion of dominance we study $\mathcal{O}^\lambda$ when $\lambda \cdot \delta \neq 0$ and only the case $K = \mathbb{C}$ is of interest to us. Its properties depend on the affine root system associated to the McKay graph (see [4]). The main result in this section is the following classification of the finite-dimensional Π^λ -modules if $\lambda \cdot \delta \neq 0$.

THEOREM 0.5. *Let Q be a quiver, let $I_\lambda = \{i \in I \mid \lambda_i = 0\}$ for λ dominant and let Q_λ be the full-subquiver of Q with vertex set I_λ . If Q_λ is a Dynkin quiver (or disjoint union of Dynkin quivers), then the S_i , with $i \in I$, are the finite-dimensional simple Π^λ -modules (see part 2 of the Theorem 7.3).*

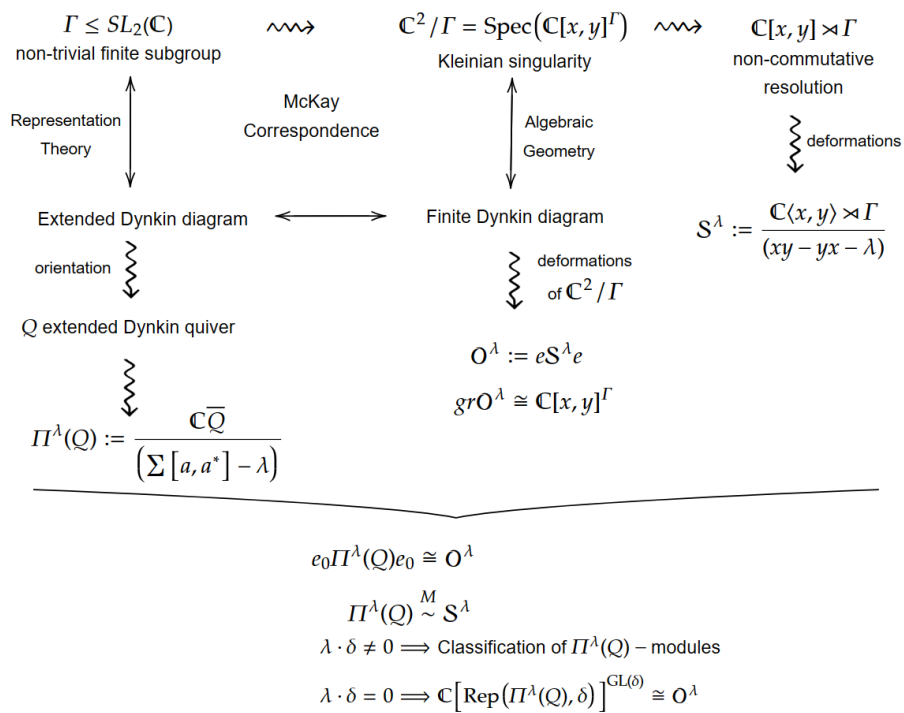
And also there is the following theorem for extended Dynkin quivers:

THEOREM 0.6. *Let Q be an extended Dynkin quiver, Π^λ has no non-zero finite-dimensional modules if and only if $\lambda \cdot \alpha \neq 0$ for all roots α .*

The results are most complete for extended Dynkin quivers in the case when $\lambda \cdot \delta \neq 0$. We write R_λ for the set of roots α (see section 6) with $\lambda \cdot \alpha = 0$, a finite set, and let Σ_λ be the set of minimal elements of R_λ .

THEOREM 0.7. *If Q is extended Dynkin and $\lambda \cdot \delta \neq 0$ then the isomorphism classes of finite-dimensional simple Π^λ -modules are in 1 – 1 correspondence with the set Σ_λ , under the assignment sending a simple module to its dimension vector (see Theorem 7.7).*

The following diagram illustrates the described relationship between the Kleinian singularity, its deformations, and the deformed preprojective algebra, particularly for our case of interest, $K = \mathbb{C}$.



CHAPTER 2

Preliminaries.

1. Filtration and grading.

Let K be any field. Let R be a K -algebra.

1.1. Graded algebras.

DEFINITION 1.1. We say that R is *graded* if there are K -vector subspaces R_i , $i \in \mathbb{N}$, such that

- (1) $R = \bigoplus_{i \in \mathbb{N}} R_i$,
- (2) $R_i \cdot R_j \subseteq R_{i+j}$

They R_i are called the *homogeneous components* of R . The elements of R_i are the *homogeneous elements of degree i* .

If $R_i = 0$ when $i < 0$ then we say that the grading is *positive*.

DEFINITION 1.2. Let $S = \bigoplus_{i \geq 0} S_i$ another graded algebra over K . Let be $\phi : R \rightarrow S$ a homomorphism of K -algebras. We say that ϕ is a *graded homomorphism of K -algebras* if $\phi(R_i) \subseteq S_i$.

Thus, a graded homomorphism is one that preserves the degree.

DEFINITION 1.3. A two-side ideal I of R is a *graded ideal* if $I = \bigoplus_{i \geq 0} (I \cap R_i)$.

Thus, homogeneous elements generate a graded ideal. And the converse is also true: an ideal generated by homogeneous elements must be graded.

The next proposition shows the concepts of graded homomorphism and graded ideal are related:

PROPOSITION 1.4. Let $R = \bigoplus_{i \geq 0} R_i$ and $S = \bigoplus_{i \geq 0} S_i$ be graded algebras over K .

- (1) The kernel of a graded homomorphism of K -algebras $\phi : R \rightarrow S$ is a graded two-sided ideal of R .
- (2) If I is a graded two-side ideal of R then R/I is a graded K -algebra

PROOF. (1) Let $a = a_0 \oplus \dots \oplus a_s$ be an element of the kernel of ϕ , with a_i of degree i . Thus,

$$\phi(a) = \phi(a_0) + \dots + \phi(a_s)$$

Since ϕ is graded, the degree of $\phi(a_i)$ is i . Thus $a_i \in \ker(\phi)$ for $i = 0, \dots, s$. Hence, $a \in \ker(\phi) \cap E_i$.

- (2) Let $I = \ker(\phi)$ be a graded two-sided ideal. Then we may decompose the quotient ring R/I into a direct sum of K -vector spaces,

$$R/I \cong \bigoplus_{i \geq 0} (R_i / (I \cap R_i)).$$

If $a_i \in R_i$ and $a_j \in R_j$ then $(a_i + I)(a_j + I) = a_i a_j + I$ corresponds to an element of $R_{i+j} / (I \cap R_{i+j})$ under this isomorphism. Hence R/I is a graded ring, which proves that R/I is a graded K -algebra. □

1.2. Filtered modules. A graded algebra admits a special kind of module:

DEFINITION 1.5. A left R -module M is a *graded module* if there are K -vector spaces M_i , for $i \geq 0$, such that

- (1) $M = \bigoplus_{i \geq 0} M_i$,
- (2) $R_i \cdot M_j \subseteq M_{i+j}$

Then the M_i are the *homogeneous components of degree i of M* .

Note that the definition of graded module depends on the graded structure chosen for the K -algebra R .

DEFINITION 1.6. Let M be a graded left R -module. A submodule N of M is a *graded submodule* if $N = \bigoplus_{i \geq 0} (N \cap M_i)$

DEFINITION 1.7. Let M, M' be graded left R -modules. R -module homomorphism $\theta : M \rightarrow M'$ is *graded* if $\theta(M_i) \subseteq M'_i$.

It follows from the previous definitions that $\ker(\theta)$ is a graded submodule and the quotient module M/N is also a graded R -module.

DEFINITION 1.8. Let R be a K -algebra. A family $\mathcal{F} = \{F_i\}_{i \geq 0}$ of K -vector spaces is a *filtration* of R if

- (1) $F_0 \subseteq F_1 \subseteq F_2 \subseteq \dots \subseteq R$.
- (2) $R = \bigcup_{i \geq 0} F_i$ and
- (3) $F_i \cdot F_j \subseteq F_{i+j}$.

In this case R is called a *filtered algebra*. We use the convention that $F_j = \{0\}$ if $j < 0$.

FACT 1.9. Every graded algebra is filtered.

In effect, suppose that $G = \bigoplus_{i \geq 0} G_i$ is a graded algebra. Consider the vector spaces

$F_k = \bigoplus_{i=0}^k G_i$. Note that $F_k \subseteq F_{k+1}$ and their union is the whole of G . Since,

$$F_k \cdot F_m = \bigoplus_{i+j \leq k+m} G_i G_j,$$

and $G_i G_j \subseteq G_{i+j}$, we have that $F_k F_m \subseteq F_{k+m}$. Thus, $\{F_k\}_{k \geq 0}$ is a filtration of G .

On the other hand, there are filtered algebras which do not have a natural grading.

1.3. Associated graded algebra. We use a filtration of a K -algebra to construct a graded algebra. This construction will be very useful because many properties of this graded algebra pass on to its parent filtered algebra.

Suppose that $\mathcal{F} = \{F_i\}_{i \in \mathbb{N}}$ is a filtration of R .

DEFINITION 1.10. The *symbol map of order k* , which is the canonical projection of vector spaces

$$\sigma_k : F_k \longrightarrow F_k / F_{k-1}$$

DEFINITION 1.11. The *graded algebra of R associated with the filtration $\mathcal{F}$* is the K -vector space

$$gr^{\mathcal{F}} R = gr R := \bigoplus_{i \geq 0} (F_i / F_{i-1})$$

that is a K -algebra with the following multiplication:

Let be $\sigma_k(a)$ and $\sigma_m(b)$ two homogeneous elements, for some $a \in F_k$ and some $b \in F_m$, their product is defined by

$$\sigma_k(a) \sigma_m(b) = \sigma_{k+m}(ab)$$

We extend this multiplication by linearity for arbitrary elements in $gr^{\mathcal{F}} R$.

PROPOSITION 1.12. *Let A and B be two filtered algebras, with filtrations F and G , respectively, such that $\bigcup_d F^d = A$, $\bigcup_d G^d = B$ and $F^{-1} = 0 = G^{-1}$. Let $\phi : A \rightarrow B$ be a map such that $\phi|_{F^d} : F^d \rightarrow G^d$ for all d . Then, if the graded map $gr\phi : gr A \rightarrow gr B$ is an isomorphism, then the map $\phi : A \rightarrow B$ is also an isomorphism.*

PROOF. • Assume that $gr\phi$ is surjective. It is sufficient to verify that

$\phi|_{F^d} : F^d \rightarrow G^d$ is surjective for all d . We proceed by induction on d .

The base case is when $d = 0$. In this case $gr\phi = \phi|_{F^0} : F^0 = gr A \rightarrow G^0 = gr B$, so, $\phi|_{F^0}$ is surjective.

Now, let $g \in G^{d+1} \Rightarrow \bar{g} \in G^{d+1}/G^d \Rightarrow \bar{g} = gr\phi(\bar{f})$, where $\bar{f} \in F^{d+1}/G^d \Rightarrow f \in F^{d+1}$. Then $\phi(f) = g \bmod G^d \Rightarrow \phi(f) - g = \phi(f') \in G^d$, for some $f' \in F^d$. Then, by the induction hypothesis $g = \phi(f - f')$. Thus, ϕ is surjective.

- Assume that $\text{gr}\phi$ is injective. Let $f \in A$ a nonzero element and take the minimal d such that $f \in F^d$, then $\bar{f} \in F^d/F^{d-1}$ is also nonzero. So, by injectivity of $\text{gr}\phi$, $\text{gr}\phi(\bar{f}) \neq 0$. But $\text{gr}\phi(\bar{f}) = \overline{\phi(f)}$, then $\phi(f) \neq 0$.

□

1.4. Induced filtration. Let R be a K -algebra with a filtration $\mathcal{F}\{F_i\}_{i \geq 0}$, and I an ideal of R . Then the quotient R/I has a filtration given by $\{\pi(F_i)\}_{i \geq 0}$, where $\pi : R \rightarrow R/I$ is the canonical surjection. We can also define a filtration on I by $\{F_i \cap I\}_{i \geq 0}$

LEMMA 1.13. *In the previous context, the following sequence is exact*

$$0 \longrightarrow \text{gr}I \longrightarrow \text{gr}R \longrightarrow \text{gr}(R/I) \longrightarrow 0$$

LEMMA 1.14. *An ideal is homogeneous if and only if it can be generated by a homogeneous generator.*

(see Chapter 7 in [6] or section 6 of Chapter 7 in [31])

2. Tensor products, tensor algebras and semidirected product.

2.1. Tensor product.

DEFINITION 2.1. Let R be a ring, let N be a right R -module and let M be a left R -module. The *tensor product of M and N over R* , $N \otimes_R M$, is the abelian group which is the quotient of the group $N \times M$ freely generated by formal symbols $n \otimes m$, $n \in N$, $m \in M$, modulo the relations

$$(n_1 + n_2) \otimes m - n_1 \otimes m - n_2 \otimes m,$$

$$n \otimes (m_1 + m_2) - n \otimes m_1 - n \otimes m_2,$$

$$nr \otimes m - n \otimes rm,$$

where $n, n_1, n_2 \in N$, $m, m_1, m_2 \in M$ and $r \in R$.

The elements $n \otimes m \in N \otimes M$, for $n \in N$, $m \in M$ are called *pure tensors* or *simple tensors*. Note that in general, there are elements of $N \otimes M$ which are not simple tensors. Every tensor can be written (non-uniquely in general) as a finite sum of simple tensors.

DEFINITION 2.2. Let N be a right R -module, let M be a left R -module and let L be an abelian group (written additively). A map $\varphi : N \times M \rightarrow L$ is called *R -balanced* if

$$\varphi(n_1 + n_2, m) = \varphi(n_1, m) + \varphi(n_2, m)$$

$$\varphi(n, m_1 + m_2) = \varphi(n, m_1) + \varphi(n, m_2)$$

$$\varphi(n, rm) = \varphi(nr, m)$$

for all $n, n_1, n_2 \in N$, $m, m_1, m_2 \in M$ and $r \in R$.

With this terminology, it follows immediately from the relations in 2.1 that the map $\iota : N \times M \longrightarrow N \otimes_R M$ is R -balanced. The next theorem proves the extremely useful *universal property of the tensor product* with respect to balanced maps.

THEOREM 2.3. *Suppose R is a ring with 1, N is a right R -module, and M is a left R -module. Let $N \otimes_R M$ be the tensor product of N and M over R and let $\iota : N \times M \longrightarrow N \otimes_R M$ be the R -balanced map defined above.*

- (1) *If $\phi : N \otimes_R M \longrightarrow L$ is any group homomorphism from $N \otimes_R M$ to an abelian group L then the composite map $\varphi = \phi \circ \iota$ is an R -balanced map from $N \times M$ to L .*
- (2) *Conversely, suppose L is an abelian group and $\phi : N \times M \longrightarrow L$ is any R -balanced map. Then there is a unique group homomorphism $\phi : N \otimes_R M \longrightarrow L$ such that φ factors through ι , i.e., $\varphi = \phi \circ \iota$ as in (1).*

Equivalently, the correspondence $\varphi \leftrightarrow \phi$ in the commutative diagram

$$\begin{array}{ccc} N \times M & \xrightarrow{\iota} & N \otimes_R M \\ & \searrow \varphi & \downarrow \phi \\ & & L \end{array}$$

establishes a bijection

$$\{R\text{-balanced maps } \varphi : N \times M \longrightarrow L\} \leftrightarrow \{\text{group homomorphism } \phi : N \otimes_R M \longrightarrow L\}.$$

Let M an R -bimodule, then $M \otimes M$ is an R -bimodule again.

If R is a field, R -module N is canonically a bimodule. [13]

2.2. Tensor algebra.

DEFINITION 2.4. Let M a K -vector space. We define the *tensor powers* of M by: $T^0 M := K$ and for $d \geq 1$

$$T^d M := M \otimes \dots \otimes M =: M^{\otimes d}.$$

In particular, and by definition, $(T^d M)^*$ is the K -vector space of d -linear forms in M .

Moreover, we define the *tensor algebra* of M by

$$TM := \bigoplus_{d \geq 0} T^d M = K \oplus M \oplus T^2 M \oplus T^3 M \oplus \dots$$

where the product in TM is given by

$$T^d M \times T^e M \longrightarrow T^{d+e} M$$

and its unity is $1 \in K = T^0 M$. Thus, TM is a graded K -algebra.

The tensor algebra is not commutative if $\dim_K M \geq 2$.

THEOREM 2.5. *Universal property of the Tensor Algebra:*

For all K -linear map $f : M \rightarrow A$, where A is a K -algebra with unity, there is a unique algebra's morphism $\hat{f} : TM \rightarrow A$, such that $f = \hat{f} \circ i$, with $i : M \hookrightarrow TM$. In its means, the following diagram is commutative:

$$\begin{array}{ccc} M & \xrightarrow{f} & A \\ & \searrow i & \downarrow \hat{f} \\ & & TM \end{array}$$

(A curved arrow from M to A is labeled $\hat{f}$)

[13]

2.3. Semidirect product.

DEFINITION 2.6. (1) The pair of homomorphisms $X \xrightarrow{\alpha} Y \xrightarrow{\beta} Z$ is said to be *exact* (at Y) if $\text{Im}(\alpha) = \ker(\beta)$.

(2) A sequence $\cdots \rightarrow X_{n-1} \rightarrow X_n \rightarrow X_{n+1} \rightarrow \cdots$ of homomorphisms is said to be *an exact sequence* if it is exact at every X_i between a pair of homomorphisms.

We can also use this terminology to express the fact that for these maps β is injective and α is surjective:

PROPOSITION 2.7. *Let A, B and C be R -modules over some ring R Then*

- (1) *The sequence $0 \rightarrow A \xrightarrow{\beta} B$ is exact (at A) if and only if β is injective.*
- (2) *The sequence $B \xrightarrow{\alpha} C \rightarrow 0$ is exact (at C) if and only if α is surjective.*

COROLLARY 2.8. *The sequence $0 \rightarrow A \xrightarrow{\alpha} B \xrightarrow{\beta} C \rightarrow 0$ is exact if and only if β is injective, α is surjective, and $\text{Im}(\alpha) = \ker(\beta)$, i.e. B is an extension of C by A . This is also called a short exact sequence.*

If A, B and C are groups written multiplicatively, the analogous sequence will be written

$$1 \rightarrow A \xrightarrow{\alpha} B \xrightarrow{\beta} C \rightarrow 1,$$

where 1 denotes the trivial group. The previous results are valid with the obvious notational changes.

Note that any exact sequence can be written as a succession of short exact sequences, since to say $X \xrightarrow{\alpha} Y \xrightarrow{\beta} Z$ is exact at Y is the same as saying that the sequence

$$0 \rightarrow \alpha(X) \rightarrow Y \rightarrow Y/\ker(\beta) \rightarrow 0$$

is a short exact sequence.

THEOREM 2.9. *Let H and K be groups, and let φ be a homomorphism from K into $\text{Aut}(H)$. Let $\cdot$ denote the (left) action of K on H determined by φ . Let G be the set of ordered pairs (h, k) with $h \in H$ and $k \in K$ and define the following multiplication on G :*

$$(h_1, k_1)(h_2, k_2) = (h_1 k_1 \cdot h_2, k_1 k_2)$$

- (1) *This multiplication makes G into a group of order $|G| = |H||K|$.*
- (2) *The sets $\{(h, 1) \mid h \in H\}$ and $\{(1, k) \mid k \in K\}$ are subgroups of G and the maps $h \mapsto (h, 1)$ for $h \in H$ and $k \mapsto (1, k)$ for $k \in K$ are isomorphisms of these subgroups with the groups H and K respectively.*

DEFINITION 2.10. Let H and K be groups, and let φ be a homomorphism from K into $\text{Aut}(H)$. The group described in the previous theorem is called the *semidirect product* of H and K with respect to φ and will be denoted by $H \rtimes_{\varphi} K$ (where there is no danger of confusion, we shall simply write $H \rtimes K$).

- DEFINITION 2.11.** (1) Let R be a ring and let $0 \rightarrow A \xrightarrow{\alpha} B \xrightarrow{\beta} C \rightarrow 0$ be a short exact sequence of R -modules. The sequence is said to be *split* if there is an R -module complement to $\alpha(A)$ in B . In this case, up to isomorphism, $B = A \oplus C$ (more precisely, $B = \alpha(A) \oplus C'$ for some submodule C' , and C' is mapped isomorphically onto C by β : $\beta(C' \cong C)$).
- (2) If $1 \rightarrow A \xrightarrow{\alpha} B \xrightarrow{\beta} C \rightarrow 1$ is a short exact sequence of groups, then the sequence is said to be *split* if there is a subgroup complement to $\alpha(A)$ in B . In this case, up to isomorphism, $B = A \rtimes C$ (more precisely, $B = \alpha(A) \rtimes C'$ for some subgroup C' , and C' is mapped isomorphically onto C by β : $\beta(C' \cong C)$).

In either case, the extension B is said to be a *split extension* of C by A .

There is a variant of this construction in which a group G acts by automorphisms on an algebra A over a commutative ring K . In this case we write $A \rtimes G$ for $A \otimes_R KG$ with multiplication defined by

$$(a_1 \otimes g_1) \cdot (a_2 \otimes g_2) = a_1 g_1(a_2) \otimes g_1 g_2.$$

3. Dimensions.

3.1. Gelfand-Kirillov dimension. Let K be a field. By algebra, we will mean K -algebra unless otherwise specified.

Let A be a finitely generated algebra over K . If V is a subspace of A , we denote by V^n the subspace spanned by all products of elements of V of length n . By generating subspace V , we will mean a finite-dimensional subspace of A which generates A as algebra, and which

contains 1. Then V^n is spanned by the products of length $\leq n$ of elements of V , and because V generates, $A = \bigcup_n V^n$.

DEFINITION 3.1. The growth function associated to a generating subspace is

$$f_V(n) := \dim_K V^n.$$

An equivalent definition is to take a finite subset X of A which generates A and which contains 1. Let X^n denote the set of products of length n of elements of X . As before, this includes all shorter products. Let V^n be the vector space spanned by X^n . Then we set $f_X(n) = \dim V^n$, which is the number of linearly independent products of length $\leq n$ of elements of X .

DEFINITION 3.2. An algebra A is said to have polynomial growth if there are positive real numbers c, r such that

$$f_V(n) \leq cn^r$$

for all n . And A has exponential growth if there is a real number $s > 1$ so that $f_V(n) \geq s^n$ for infinitely many n .

This is independent of the choice of the generating subspace V .

DEFINITION 3.3. The Gelfand-Kirillov dimension of an algebra with polynomial growth is the infimum of the real numbers r such that $f_V(n) \leq cn^r$ holds for some c :

$$\text{GKdim}(A) = \inf\{r \mid f_V(n) \leq cn^r\}$$

If A does not have polynomial growth, then $\text{GKdim}(A) = \infty$.

Again, the GK dimension is independent of the choice of the subspace V .

PROPOSITION 3.4. $\text{GKdim}(A) = \inf\{r \mid f_V(n) \leq p(n)\}$ for some polynomial p of degree r . (see VI [1])

In contexts in which the generating set is given naturally, for example if the algebra is graded, we may also want to keep track of the constant coefficient c . In that case it would be more natural to ask for the smallest c such that (1.2) holds it for large n . Such a modification would not change r .

PROPOSITION 3.5. $\text{GKdim}(A) = \inf\{r \mid f_V(n) \leq n^r \text{ for large } n\}$. (see VI in [1])

Example: The commutative polynomial ring $A = K[x_1, \dots, x_d]$ has Gelfand-Kirillov dimension d . If we let V be the space spanned by $\{1, x_1, \dots, x_d\}$, then V^n is the space of polynomials of

degree $\leq n$. The dimension of this space is $\binom{n+d}{d}$, which is a polynomial of degree d in n .

If $d > 1$, the free ring $A = K\langle x_1, \dots, x_d \rangle$ has exponential growth, hence infinite GK dimension, because the number of noncommutative monomials of degree n is d^n .

The following proposition shows that the Gelfand-Kirillov dimension is independent of the choice of the generating space V .

PROPOSITION 3.6. *Let A be a finitely generated K -algebra, let V, W be generating subspaces, and set $f = f_V, g = f_W$. If $f(n) \leq cn^r$ for all n , then there is a c' such that $g(n) \leq c'n^r$ for all n . (see Proposition VI.1.5 in [1])*

The following is a lower bound on the growth of f_V .

PROPOSITION 3.7. *Suppose that A is finitely generated but not finite dimensional over K . Let $f = f_V$ be a growth function for A . Then f is a strictly increasing function, hence $f(n) \geq n$. (see Proposition VI.1.6 in [1])*

COROLLARY 3.8. *Let A be a finitely generated K -algebra. Then $\text{GKdim}(A) = 0$ if and only if A has finite dimension over K . If A is infinite dimensional, then $\text{GKdim}(A) \geq 1$. (see Corollary VI.1.7 in [1])*

The next three propositions derive elementary properties of the GK dimension.

PROPOSITION 3.9. *Let $A[t]$ denote the algebra of polynomials in t with coefficients in A , where t is a central variable. Then $\text{GKdim}(A[t]) = \text{GKdim}(A) + 1$. (see Proposition VI.1.8 in [1])*

PROPOSITION 3.10. *If γ is a regular central element of a finitely generated algebra A , then $\text{GKdim}(A) = \text{GKdim}(A[\gamma^{-1}])$. The same is true if γ is a normalizing element. (see Proposition VI.1.9 in [1])*

PROPOSITION 3.11. *Let A be a finitely generated graded algebra. Let $r = \text{gk}(A)$, and let*

$$s = \inf \{s \mid \dim A_n \leq cn^s\}$$

- (i) $r \leq s + 1$.
- (ii) *If there is an integer $d > 0$, such that $\dim A_n \leq \dim A_{n+d}$ for all sufficiently large n , then $r = s + 1$.*
- (iii) *The condition of (ii) is satisfied in each of the following cases:*
 - (a) $\dim A_n$ is monotonic,
 - (b) $\dim A_k d$ is a polynomial function of k , or
 - (c) A contains a homogeneous left or right regular element of degree d .

(see Proposition VI.1.10 in [1])

THEOREM 3.12. (Warfield) *For any real number $r \geq 2$, there is a finitely generated graded algebra R with $\text{gk}(R) = r$. (see Theorem, of Warfield, as VI.2.1 in [1] or as 2.9 in [27])*

THEOREM 3.13. (Bergman) *There is no algebra A whose GK dimension is in the open interval $1 < r < 2$. If A is a finitely generated algebra of GK dimension 1, then every growth function $f = f_V$ for A is bounded by a linear function, i.e., $f(n) \leq cn$ for some c . (see Theorem, of Bergman, as VI.4.1 in [1] or as 2.5 in [27])*

3.2. Classical Krull dimension.

DEFINITION 3.14. A K -algebra A has *classical Krull dimension* $\text{cl.Kdim}(A) = m$, if there exists a chain of prime ideals

$$P_0 \supsetneq P_1 \supsetneq \dots \supsetneq P_{m-1} \supsetneq P_m$$

of length m , and there is no such chain of greater length. If A has chains of prime ideals of arbitrary length, then $\text{cl.Kdim}(A) = \infty$

The previous definition can be extended for the classical Krull dimension of an arbitrary right Noetherian ring. (see 4.4 in 6.4.8 of [31]).

In Chapter 3, we will study algebras for which the classical Krull dimension coincides with the Krull dimension defined in 2.2 of 6.2.5 in [31].

PROPOSITION 3.15. *Let A be a commutative K -algebra. Then*

- (a) *If A is finitely generated, then $\text{GKdim}(A) = \text{cl.Kdim}(A)$.*
- (b) *$\text{GKdim}(A)$ is either infinite or a nonnegative integer.*

(see VI in [1] or 4.5 in [27])

4. Categories, functors and natural transformations.

4.1. Categories.

DEFINITION 4.1. A *directed graph* or *diagram scheme* is a set O of objects, a set A of arrows and two functions

$$A \xrightarrow{\text{dom}} O \quad \text{and} \quad A \xrightarrow{\text{cod}} O$$

DEFINITION 4.2. In this graph, the following set, called the *product over O* , is the set of composable pairs of arrows:

$$A \times_O A = \{\langle g, f \rangle \mid g, f \in A \text{ and } \text{dom} g = \text{cod} f\}$$

DEFINITION 4.3. A *category* is a graph with two additional functions called *identity* and *composition*.

$$\begin{array}{ll} O \xrightarrow{id} A, & A \times_O A \xrightarrow{\circ} A \\ c \mapsto id_c & \langle g, f \rangle \mapsto g \circ f \end{array}$$

such that

$$dom(id_a) = a = cod(id_a), \quad dom(g \circ f) = dom f, \quad cod(g \circ f) = cod(g),$$

for all object $a \in O$ and all composable pairs of arrows $\langle g, f \rangle \in A \times_O A$, and such that the following axioms hold:

Associativity: For given objects and arrows in the configuration

$$a \xrightarrow{f} b \xrightarrow{g} c \xrightarrow{k} d$$

one always has the equality

$$k \circ (g \circ f) = (k \circ g) \circ f$$

Unit law: For all arrows $f: a \rightarrow b$ and $g: b \rightarrow c$ composition with the identity arrow 1_b gives

$$1_b \circ f = f \quad \text{and} \quad g \circ 1_b = g$$

In treating a category C , we usually drop the letters A and C , and write

$$c \in C \quad \text{and} \quad f \in C$$

for " c is an object of C " and " f is an arrow of C ", respectively. We also write

$$Hom(b, c) = \{f \mid f \in C, dom f = b, cod f = c\}$$

for the set of arrows from b to c .

DEFINITION 4.4. A *subcategory* S of a category C is a collection of some objects and some arrows of C , which includes with each arrow f both the object $dom f$ and the object $cod f$, with each object s its identity arrow 1_s and with each pair of composable arrows $s \rightarrow s' \rightarrow s''$ their composite.

These conditions ensure that these collections of objects and arrows themselves constitute a category S .

4.2. Functors.

DEFINITION 4.5. A *functor* is a morphism of categories. In detail, for categories C and B a functor $T : C \rightarrow B$ with domain C and codomain B consists of two suitably related functions: The *object function* T , which assigns to each object c of C an object Tc of B , and the *arrow function* (also written T), which assigns to each arrow $f : c \rightarrow c'$ of C an arrow $Tf : Tc \rightarrow Tc'$ of B , in such a way that

$$T(1_c) = 1_{Tc}, \quad T(g \circ f) = Tg \circ Tf,$$

the latter whenever the composite $g \circ f$ is defined in C .

A functor, like a category, can be described in the "arrows-only" fashion: It is a function T from arrows f of C to arrows Tf of B , carrying each identity of C to an identity of B and each composable pair $\langle g, f \rangle$ in C to a composable pair $\langle Tg, Tf \rangle$ in B , with $Tg \circ Tf = T(g \circ f)$

Functors may be composed. Explicitly:

DEFINITION 4.6. Given functors

$$C \xrightarrow{T} B \xrightarrow{S} A$$

between categories A , B and C , the composite functions

$$c \mapsto S(Tc) \quad f \mapsto S(Tf)$$

on objects c and arrows f of C define a functor $S \circ T : C \rightarrow A$, called the *composite* (in that order) of S with T . This composition is associative. For each category B there is a functor $I_B : B \rightarrow B$, which acts as an identity for this composition.

DEFINITION 4.7. An *isomorphism* $T : C \rightarrow B$ of categories is a functor T from C to B which is a bijection, both on objects and on arrows.

Alternatively, but equivalently, a functor $T : C \rightarrow B$ is an isomorphism if and only if there is a functor $S : B \rightarrow C$ for which both composites $S \circ T$ and $T \circ S$ are identity functors; then S is the *two-sided inverse* $S = T^{-1}$.

The previous definition of isomorphism of categories is very strong. There is a concept more useful than this, the equivalence between categories, as we will see later.

DEFINITION 4.8. A functor $T : C \rightarrow B$ is *full* when to every pair c, c' of objects of C and to every arrow $g : Tc \rightarrow Tc'$ of B , there is an arrow $f : c \rightarrow c'$ of C with $g = Tf$.

Clearly, the composite of two full functors is a full functor.

DEFINITION 4.9. A functor $T : C \rightarrow B$ is *faithful* (or an embedding) when to every pair c, c' of objects of C and to every pair $f_1, f_2 : c \rightarrow c'$ of parallel arrows of C the equality $Tf_1 = Tf_2 : Tc \rightarrow Tc'$ implies $f_1 = f_2$.

Again, composites of faithful functors are faithful.

Then T is full when every such function is surjective, and faithful when every such function is injective. For a functor which is both, full and faithful, every such function is a bijection, but this need not mean that the functor itself is an isomorphism of categories, for there may be objects of B not in the image of T .

DEFINITION 4.10. Moreover, the injection (inclusion) map $S \rightarrow C$ which sends each object and each arrow of S to itself, in C , is a functor, called the *inclusion functor*.

This inclusion functor is automatically faithful.

DEFINITION 4.11. We say that S is a *full subcategory* of C when the inclusion functor $S \rightarrow C$ is full.

A full subcategory, given C , is thus determined by giving just the set of its objects, since the arrows between any two of these objects s, s' are all morphisms $s \rightarrow s'$ in C .

4.3. Natural transformations.

DEFINITION 4.12. Given two functors $S, T : C \rightarrow B$, a *natural transformation* $\tau : S \rightarrow T$ is a function which assigns to each object c of C an arrow $\tau_c = \tau c : Sc \rightarrow Tc$ of B in such a way that every arrow $f : c \rightarrow c'$ in C yields a diagram

$$\begin{array}{ccccc}
 & c & & Sc & \xrightarrow{\tau c'} & Tc \\
 & \downarrow f & & \downarrow Sf & & \downarrow Tf \\
 & c' & & Sc' & \xrightarrow{\tau c} & Tc'
 \end{array}$$

which is commutative. When this holds, we also say that $\tau_c : Sc \rightarrow Tc$ is *natural* in c .

If we think of the functor S as giving a picture in B of (all the objects and arrows of) C , then a natural transformation τ is a set of arrows mapping (or, translating) the picture S to the picture T , with all square (and parallelograms!) like that above commutative:

$$\begin{array}{ccccc}
 a & & Sa & \xrightarrow{\tau a} & Ta \\
 \downarrow h & \searrow f & \downarrow Sf & & \downarrow Tf \\
 & & Sb & \xrightarrow{\tau b} & Tb \\
 & \swarrow g & \downarrow Sg & & \downarrow Tg \\
 c & & Sc & \xrightarrow{\tau c} & Tc
 \end{array}$$

DEFINITION 4.13. We call $\tau a, \tau b, \tau c, \dots$, the *components* of the natural transformation τ .

A natural transformation is often called a *morphism of functors*.

DEFINITION 4.14. A natural transformation τ with every component τc invertible in B is called a *natural equivalence* or better a *natural isomorphism*; in symbols, $\tau : S \cong T$. In this case, the inverse $(\tau c)^{-1}$ in B are the components of a natural isomorphism $\tau^{-1} : T \rightarrow S$.

DEFINITION 4.15. More generally, an *equivalence* between categories C and D is defined to be a pair of functors $S : C \rightarrow D$, $T : D \rightarrow C$ together with natural isomorphism $I_C \cong T \circ S$, $I_D \cong S \circ T$.

DEFINITION 4.16. To each category C we also associate the *opposite* category C^{op} . The objects of C^{op} are the objects of C , the arrows of C^{op} are arrows f^{op} , is one-one correspondence $f \mapsto f^{op}$ with the arrows f of C . For each arrow $f : a \rightarrow b$ of C , the domain and codomain of the corresponding f^{op} are as in $f^{op} : b \rightarrow a$ (the direction is reversed).

The composite $f^{op} g^{op} = (gf)^{op}$ is defined in C^{op} exactly when the composite gf is defined in C . this clearly makes C^{op} a category.

4.4. Adjunctions.

DEFINITION 4.17. Let $\mathcal{C}$ and $\mathcal{D}$ be two categories, and let $F : \mathcal{C} \rightarrow \mathcal{D}$ and $G : \mathcal{D} \rightarrow \mathcal{C}$ be two functors. We say that F is a *left adjoint* of G or that G is a *right adjoint* of F , or that (F, G) is an *adjoint pair*, written $F \dashv G$, if there is a natural isomorphism of bifunctors

$$\varphi : \text{Hom}_{\mathcal{D}}(F(-), -) \rightarrow \text{Hom}_{\mathcal{C}}(-, G(-)),$$

i.e, if A is an object of $\mathcal{C}$ and B is an object of $\mathcal{D}$, there are bijections

$$\varphi_{A,B} : \text{Hom}_{\mathcal{D}}(F(A), B) \rightarrow \text{Hom}_{\mathcal{C}}(A, G(B)),$$

that are natural in A and B , that is, if we have morphisms $f : A \rightarrow A'$ in $\mathcal{C}$ and $g : B \rightarrow B'$ in $\mathcal{D}$, the diagram

$$\begin{array}{ccccc} \text{Hom}_{\mathcal{D}}(F(A'), B) & \xrightarrow{F(f)^*} & \text{Hom}_{\mathcal{D}}(F(A), B) & \xrightarrow{g^*} & \text{Hom}_{\mathcal{D}}(F(A), B') \\ \downarrow \varphi_{A',B} & & \downarrow \varphi_{A,B} & & \downarrow \varphi_{A,B'} \\ \text{Hom}_{\mathcal{C}}(A', G(B)) & \xrightarrow{f^*} & \text{Hom}_{\mathcal{C}}(A, G(B)) & \xrightarrow{G(g)^*} & \text{Hom}_{\mathcal{C}}(A, G(B')) \end{array}$$

is commutative.

We say that φ , or its inverse, is an *adjunction transform*, and the triple (F, G, φ) is called an *adjunction*. An usual notation is $\varphi : F \dashv G$.

4.4.1. *Tensor-Hom adjunction.* Let R and S be two rings. If N is a (R, S) -module and P is a right S -module, then $\text{Hom}_S(N, P)$ is a right R -module, with the following structure:

Let $f \in \text{Hom}_S(N, P)$, and $r \in R$ we define the action $(f \cdot r)(n) = f(rn)$, $\forall n \in N$.

Also, we can verify that if M is a right R -module, then $M \otimes_R N$ is a right S -module.

THEOREM 4.18. *Let N be a (R, S) -bimodule, then the functor tensor over R to the right with N , $-\otimes_R N$, is left adjoint of take morphism over S with domain N , $\text{Hom}_S(N, -)$. That is called *Tensor-Hom adjunction*.*

This means:

If M is a right R -module and P is and right S -module

$$\begin{array}{ccc} \exists \eta_{M,P} : \text{Hom}_S(M \otimes_R N, P) & \xrightarrow{\sim} & \text{Hom}_R(M, \text{Hom}_S(N, P)) \\ \varphi & \mapsto & M \longrightarrow \text{Hom}_S(N, P) \\ & & m \mapsto [n \mapsto \varphi(m \otimes n)] \\ M \otimes_R N \longrightarrow P & \longleftarrow & \psi \\ m \otimes n \mapsto (\psi(m))(n) & & \end{array}$$

This is the Tensor-Hom adjunction.

So, we have the adjunction for right modules,

$$\text{Hom}_S(M \otimes_R N, P) \cong \text{Hom}_R(M, \text{Hom}_S(N, P)),$$

and for left modules,

$$\text{Hom}_S(M \otimes_R N, P) \cong \text{Hom}_R(N, \text{Hom}_S(M, P)).$$

5. Tor.

DEFINITION 5.1. If M and N are R -modules, and

$$\dots \xrightarrow{\delta_{i+2}} F_{i+1} \xrightarrow{\delta_{i+1}} F_i \xrightarrow{\delta_i} F_{i-1} \xrightarrow{\delta_{i-1}} \dots \xrightarrow{\delta_1} F_0 \xrightarrow{\delta_0} M \rightarrow 0$$

a free resolution of M , it is an exact sequence of R -modules, and F_i is a free R -module for all i . We define $\text{Tor}_i^R(M, N)$ as the homology group of the sequence

$$\dots \xrightarrow{\delta_{i+2} \otimes 1} F_{i+1} \otimes N \xrightarrow{\delta_{i+1} \otimes 1} F_i \otimes N \xrightarrow{\delta_i \otimes 1} F_{i-1} \otimes N \xrightarrow{\delta_{i-1} \otimes 1} \dots \xrightarrow{\delta_1 \otimes 1} F_0 \otimes N \xrightarrow{\delta_0 \otimes 1} M \otimes N \rightarrow 0.$$

That is, $\text{Tor}_i^R(M, N) := \text{Ker}(\delta_i \otimes 1) / \text{Img}(\delta_{i+1} \otimes 1)$. This homology is independent of the resolution chosen.

PROPOSITION 5.2. (1) $\text{Tor}_0^R(M, N) = M \otimes_R N$.

(2) If M or N is free, then $\text{Tor}_i^R(M, N) = 0$ for $i > 0$.

- (3) If M or N is flat over R , then $\mathrm{Tor}_i^R(M, N) = 0$ for $i > 0$. (see section 6.2 of Chapter 6 of [14])

For any short exact sequence $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ of R -modules, and any R -module N , there is a *long exact sequence of Tor*:

$$\begin{aligned} \dots \rightarrow \mathrm{Tor}_i^R(M', N) \rightarrow \mathrm{Tor}_i^R(M, N) \rightarrow \mathrm{Tor}_i^R(M'', N) \rightarrow \\ \dots \rightarrow \mathrm{Tor}_1^R(M', N) \rightarrow \mathrm{Tor}_1^R(M, N) \rightarrow \mathrm{Tor}_1^R(M'', N) \\ \rightarrow M' \otimes_R N \rightarrow M \otimes_R N \rightarrow M'' \otimes_R N \rightarrow 0 \end{aligned}$$

6. Some results of Algebraic Geometry.

In this preliminary section, we will mention some necessary results in order to understand the strategy in Section 8 in chapter 3. These results concern the fiber product of affine schemes, flatness of modules, flatness of morphisms of affine schemes, and linearly reductive groups. For further details on the concepts and structures mentioned here, refer to [23] (for the structure of affine scheme and fiber product of affine schemes), [28] (for the categorical definition of pullback or fiber product), and [3] (for linearly reductive groups).

6.1. Fiber product of affine schemes.

DEFINITION 6.1. Let A , B and C be rings. Let $\mathrm{Spec}(A)$, $\mathrm{Spec}(B)$, $\mathrm{Spec}(C)$ be affine schemes and let

$$f : \mathrm{Spec}(A) \rightarrow \mathrm{Spec}(C) \text{ and } g : \mathrm{Spec}(B) \rightarrow \mathrm{Spec}(C)$$

be morphisms of affine schemes. These morphisms induce ring homomorphism $\tilde{f} : C \rightarrow A$ and $\tilde{g} : C \rightarrow B$. The *fiber product* between the affine schemes $\mathrm{Spec}(A)$ and $\mathrm{Spec}(B)$ over the scheme $\mathrm{Spec}(C)$ is defined by

$$\mathrm{Spec} A \times_C \mathrm{Spec} B = \mathrm{Spec}(A) \times_{\mathrm{Spec}(C)} \mathrm{Spec}(B) := \mathrm{Spec}(A \otimes_C B).$$

Diagrams arising from fiber products are frequently called *Cartesian diagram*; that is, the diagram

$$\begin{array}{ccc} Z & \xrightarrow{\quad} & \mathrm{Spec} A \\ \downarrow & \boxtimes & \downarrow f \\ \mathrm{Spec} B & \xrightarrow{\quad g \quad} & \mathrm{Spec} C \end{array}$$

is said to be a *Cartesian diagram* if there is an isomorphism

$$Z \cong \mathrm{Spec} A \times_C \mathrm{Spec} B.$$

(see Theorem 3.3 in Chapter 2 of [23].)

Note that A is a C -module, considering the action $c \cdot a = \tilde{f}(c)a$, for all $c \in C$ and all $a \in A$. Similarly, B is a C -module, considering the action $c \cdot b = \tilde{g}(c)b$, for all $c \in C$ and all $b \in B$.

The fiber product of affine schemes exists (see Theorem 3.3 in Chapter II of [23]). It follows immediately, from the universal property of the tensor product, that the fiber product of affine schemes is unique up to a unique isomorphism. So, the fiber product of schemes is the product (also called pullback) in the category of schemes.

The map $Z \rightarrow \text{Spec} A$ is called the *base change* or the *pullback* of the morphism $g : \text{Spec} B \rightarrow \text{Spec} C$ via the morphism $f : \text{Spec} A \rightarrow \text{Spec} C$.

Example: An example of a fiber product, or pullback, is a base change. Let us see a base change in the category of affine schemes, which is of interest to us in the section 8 of Chapter 3:

Let A be a ring, let $\text{Spec} A$ be an affine scheme over K , and let R be a K -algebra. The base change of $\text{Spec} A$ to an affine scheme over R is the fiber product $Z = \text{Spec} A \times_{\text{Spec} K} \text{Spec} R$, which is a pullback of the morphisms $f : \text{Spec} A \rightarrow \text{Spec} K$ and $g : \text{Spec} R \rightarrow \text{Spec} K$, induced by the inclusions of K into the K -algebras R and A .

DEFINITION 6.2. Let A and B be rings. Given $X = \text{Spec} A$ an affine scheme. $\mathcal{O}(X)$ is the ring of all regular functions over X . If $p \in X$ is a point, we define the *local ring* of p over X , $\mathcal{O}_{p,Y} =: \mathcal{O}_p$ by

$$\mathcal{O}_p = \left\{ \langle U, f \rangle \mid p \in U \subseteq_{\text{open}}, f : U \rightarrow K \text{ regular} \right\} / \sim$$

where $\langle U, f \rangle \sim \langle V, g \rangle$ if and only if $f|_{U \cap V} = g|_{U \cap V}$ is an equivalence relation.

$\mathcal{O}_p$ is a local ring, it means that it has only one maximal ideal. The unique maximal ideal of $\mathcal{O}_p$ is

$$\mathfrak{m}_p = \{ \langle U, f \rangle \in \mathcal{O}_p \mid f(p) = 0 \}.$$

The quotient $\mathcal{O}_p / \mathfrak{m}_p$ is the *residual field* of p . The residual field is isomorphic to K .

DEFINITION 6.3. Let $f : X = \text{Spec} A \rightarrow Y = \text{Spec} B$ be a morphism of affine schemes, and let $p \in Y$ be a point. Let $K(p)$ be the residue field of p , and let $\text{Spec} k(p) \rightarrow \text{Spec} B$ be the natural morphism. Then we define the *fibre* of the morphism f over the point p to be the scheme:

$$X_p = X \times_Y \text{Spec} k(p)$$

where the product on the right hand is the fiber product of affine schemes.

The fibre X_p is a scheme over $k(p)$, its underlying topological space is homeomorphic to the subset $f^{-1}(p)$ of X . (see 3 in Chapter II of [23])

6.2. Flatness.

DEFINITION 6.4. Let R be a ring. Let M be an R -module, M is said *flat* if for every monomorphism $N' \rightarrow N$ of R -modules, the induced map $M \otimes_R N' \rightarrow M \otimes_R N$ is again a monomorphism.

Since $M \otimes -$ is a right-exact functor, it means that tensor products always preserve right-exact sequences; this definition of flat is the same as saying that $M \otimes -$ is an exact functor. (see section 2.2 in Chapter 1 of [14])

PROPOSITION 6.5. Let M_i be R -modules for i in a index set I . M_i is flat if and only if $\bigoplus_{i \in I} M_i$ is flat.

DEFINITION 6.6. Let $\phi : A \rightarrow B$ be a morphism of rings. ϕ is said *flat* if B is flat as an A -module.

PROPOSITION 6.7. It follows, from the definition of a R -module flat, that if M is flat as R -module then $S \otimes_R M$ is flat as S -module. (i.e. if $R \rightarrow M$ is a flat morphism of ring, then $S \rightarrow S \otimes_R M$ is also flat).

DEFINITION 6.8. Let A and B be two rings and let $X = \text{Spec} A$ and $Y = \text{Spec} B$ be two affine schemes. The morphism of affine schemes $\psi : X \rightarrow Y$ is said *flat* if $\mathcal{O}_{X,x}$ is flat as an $\mathcal{O}_{Y,\psi(x)}$ -module, $\forall x \in X$.

PROPOSITION 6.9. A morphism between regular smooth affine schemes is flat if and only if its fibers have dimension constant. (see Theorem 18.16 of [14])

PROPOSITION 6.10. A morphism of rings $A \rightarrow B$ is flat, if and only if $\mathcal{O}_{\text{Spec} B, p} \rightarrow \mathcal{O}_{\text{Spec} A, p}$ is flat $\forall p \in \text{Spec} A$. (see proposition 9.1A (d) in Chapter III of [23])

PROPOSITION 6.11. The flatness of a morphism of affine schemes is stable under pullbacks:

Let A, B and C be rings, and let $f : \text{Spec} A \rightarrow \text{Spec} C$, $g : \text{Spec} B \rightarrow \text{Spec} C$ two morphisms of affine schemes. Consider the Cartesian square:

$$\begin{array}{ccc} Z & \xrightarrow{\hat{g}} & \text{Spec } A \\ \hat{f} \downarrow & \boxtimes & \downarrow f \\ \text{Spec } B & \xrightarrow{g} & \text{Spec } C \end{array}$$

In this context, if f is flat then $\hat{f}$ is also flat. And the same happens for g and $\hat{g}$.

PROOF. Since f is flat as a morphism of affine schemes, by the definition 6.8 and the Proposition 6.10, is the same as saying that A is flat as C -module. Then, by Proposition 6.7,

$A \otimes_C B$ is flat as B -module. And Again, by the definition 6.8 and the Proposition 6.10, this is equivalent to say that $\hat{f}$ is flat. $\square$

REMARK 6.12. A base change is a pullback (see the example 6.1 in this section). So, the flatness of morphisms between affine schemes is stable under base change.

6.3. Linearly reductive groups. We will give definitions and results about affine algebraic groups. These groups are the same as linear algebraic groups, which are closed subgroups of the general linear groups.

DEFINITION 6.13. Suppose that G is a group which also has the structure of an affine variety. Let $m : G \times G \rightarrow G$ be the multiplication map $(g, h) \mapsto gh$ and let $i : G \rightarrow G$ be the map that sends an element $g \in G$ to its inverse $g^{-1} \in G$.

The group G is called an *affine algebraic group* if m and i are morphisms of affine varieties.

DEFINITION 6.14. A group homomorphism $\varphi : G \rightarrow H$ between two affine algebraic groups is a *homomorphism of algebraic groups* if it is also a morphism of affine varieties. Such a homomorphism is an *isomorphism of algebraic groups* if φ is an isomorphism of affine varieties.

PROPOSITION 6.15. *If G is an affine algebraic group, then there exists a finite dimensional vector space W such that G is isomorphic, as an affine algebraic group, to a Zariski closed subgroup of $\mathrm{GL}(W)$. (see Proposition 9.18 in [11])*

Because of the proposition, affine algebraic groups are also called *linear algebraic groups*.

Now, consider the action of an affine algebraic group G on itself on the left. Then G also acts on $K[G]$ as follows. If $h \in G$ and $f \in K[G]$, then we have

$$(h \cdot f)(g) = f(h^{-1}g)$$

for all $g \in G$.

DEFINITION 6.16. Suppose that G is an affine algebraic group. A linear map

$$\mathcal{R} : K[G] \rightarrow K$$

is called a *Reynolds operator* if it satisfies the following 2 axioms:

- (1) $\mathcal{R}(1) = 1$
- (2) $\mathcal{R}(h \cdot f) = \mathcal{R}(f)$ for all $h \in G$ and all $f \in K[G]$.

An algebraic group G is called *linearly reductive* if there exists a Reynolds operator

$$\mathcal{R} : K[G] \rightarrow K.$$

PROPOSITION 6.17. *Let G be a linearly reductive group over K , and let A and B be K -linear representations of G , then $A = A^G \oplus A'$, where A' is a subrepresentation of A . Moreover, if there is a surjection $A \twoheadrightarrow B$, there is an induced surjection $A^G \twoheadrightarrow B^G$. (see [3])*

7. Representations

7.1. Representation of groups.

DEFINITION 7.1. Let G be a group. A *linear representation* of G is a pair (V, ρ) , where V is a vector space over a field K and

$$\rho : G \rightarrow \mathrm{GL}(V)$$

is a group homomorphism.

DEFINITION 7.2. We define the *degree* of (V, ρ) as the dimension of V as a K -vector space. The representation (V, ρ) is *finite dimensional* or *of finite degree* if $\dim_K V$ is finite. We often abuse the language saying that ρ is a *representation of G in V* .

DEFINITION 7.3. Given two linear representations (V, ρ) and (W, θ) of G , a *homomorphism of representations* from (V, ρ) to (W, θ) is a K -linear map $\varphi : V \rightarrow W$ such that, for all $g \in G$, the diagram

$$\begin{array}{ccc} V & \xrightarrow{\varphi} & W \\ \rho(g) \downarrow & & \downarrow \theta(g) \\ V & \xrightarrow{\varphi} & W \end{array}$$

commutes.

DEFINITION 7.4. If (V, ρ) is a linear representation of G , we say that a K -vector subspace W of V is a *subrepresentation* of (V, ρ) or that is *G -invariant* or *invariant under G* if $\rho(g)(W) \subseteq W$ for all $g \in G$.

If V is a representation (we usually omit ρ and say that V is a representation of G and we omit the term *linear* and call a *linear representation* simply a *representation*) of G and W is a subrepresentation, then we define a group homomorphism

$$G \rightarrow \mathrm{GL}(W), g \mapsto [w \mapsto gw].$$

Thus W is also a representation of G .

7.1.1. Some examples of group representations.

Example 1. *The trivial representation.* If G is any group and K is any field, then there is a natural representation of G on K given by sending g to the identity map on K .

Example 2. *The permutation representation.* Let G be a group acting on a set X and let K be a field. Let KX be the free K -module with bases X . Then we can define a representation of G on KX by extending linearly the action of each element of G on X . That is, if we write

$$v = \sum_{x \in X} a_x x \in KX$$

where $a_x \in K$ and $a_x = 0$ for all but finitely many $x \in X$, then

$$gv = \sum_{x \in X} a_x gx = \sum_{x \in X} a_{g^{-1}x} x.$$

Example 3. *The regular representation.* Every group G acts on itself by left multiplication. The permutation representation associated to this action is called the *regular representation* of G . More precisely, we have KG , the free K -module with bases the elements of G . The action of an element $h \in G$ on an element

$$v = \sum_{g \in G} a_g g \in KG, \quad a_g \in K$$

is given by

$$hv = \sum_{g \in G} a_g hg = \sum_{g \in G} a_{h^{-1}g} g.$$

PROPOSITION 7.5. *Let G be a group and let $\varphi : V \rightarrow W$ be a homomorphism of representations of G . Then the kernel $\ker(\varphi)$ of φ are subrepresentations of V and W , respectively.*

7.1.2. **Modules over the group algebra.** Let G be a group, K be a field, and KG be the free K -module with basis G . The multiplication on G extends by bilinearity to KG . Explicitly, we define

$$\left(\sum_{h \in G} a_h h \right) \left(\sum_{g \in G} b_g g \right) = \sum_{g, h \in G} a_g b_h gh = \sum_{g \in G} \left(\sum_{h \in G} a_{gh^{-1}b_h} \right) g.$$

This multiplication on KG is associative and the identity element $1 \in G$ is a multiplicative identity on KG . Thus KG is an associative K -algebra with 1.

DEFINITION 7.6. We call the algebra KG the *group algebra* of G over K .

Now let (V, ρ) be a representation of G . We define a (left) KG -module V_ρ as follows. As a set, $V_\rho = V$, and if $\alpha = \sum_{g \in G} a_g g \in KG$ and $v \in V$, then

$$\alpha v = \sum_{g \in G} a_g \rho(g)(v).$$

Conversely, given a KG -module M we define a representation (V_M, ρ_M) of G as follows. As a vector space, $V_M = M$ and

$$\rho(g)(v) = gv, \quad g \in G, v \in V_M.$$

It is clear that $\rho : G \rightarrow \text{GL}(V_M)$ is a well defined group homomorphism, thus (V_M, ρ_M) is indeed a representation of G .

This construction gives an isomorphism of categories between the category of K -linear representations of G ($\mathbf{Rep}_K(G)$) and the category of left KG -modules ($KG\text{-}\mathbf{Mod}$).

7.1.3. Construction of representations. Let G be a group and (ρ, V) and (θ, W) be two representations of G . The direct sum $V \oplus W$ is also a representation of G , with the action of G given by

$$g(v, w) = (gv, gw), \quad g \in G, v \in V, w \in W.$$

More precisely, we define a group homomorphism $\rho \oplus \theta : G \rightarrow \text{GL}(V \oplus W)$ by

$$(\rho \oplus \theta)(g)(v, w) = (\rho(g)(v), \theta(g)(w)).$$

This is called the *direct sum* of the representations V and W . More generally, given a family of representations $(V_i)_{i \in I}$ indexed by a set I , we can define the direct sum representation $\bigoplus_{i \in I} V_i$ where G acts component-wise.

Now let G and H be two groups. Let (V, ρ) be a representation of G and (W, θ) be a representation of H . Let $V \otimes_K W$ be the tensor product of V and W over K . For each $g \in G$ and $h \in H$, the map

$$V \times W \rightarrow V \otimes_K W, (v, w) \mapsto \rho(g)(v) \otimes \theta(h)(w)$$

is clearly K -bilinear, thus induces a K -linear map

$$(\rho \boxtimes \theta)(g, h) : V \otimes_K W \rightarrow V \otimes_K W.$$

It is easy to see that $(\rho \boxtimes \theta)(g) \in \text{GL}(V \otimes_K W)$. Thus, we obtain a group homomorphism

$$\rho \boxtimes \theta : G \times H \rightarrow \text{GL}(V \otimes_K W),$$

and hence a representation of $G \times H$ on $V \otimes_K W$.

DEFINITION 7.7. With the notation as above, we call $(V \otimes_K W, \rho \boxtimes \theta)$ the *(external) tensor product* of the representation V and W and denote it by $V \boxtimes W$.

If $G = H$ then we have that $\rho \boxtimes \theta$ is a representation of $G \times G$. Let $\Delta : G \rightarrow G \times G$ be the *diagonal map* $g \mapsto (g, g)$. Then the composite

$$G \xrightarrow{\Delta} G \times G \xrightarrow{\rho \boxtimes \theta} \text{GL}(V \otimes_K W)$$

gives a representation of G , which we denote by $\rho \otimes \theta$.

DEFINITION 7.8. The representation $(V \otimes_K W, \rho \otimes \rho)$ is called the *(internal) tensor product* of the representations V and W . We denote it by $V \otimes W$.

DEFINITION 7.9. Let G be a group, and let V and W be two representations of G over a field K . The K -vector space $\text{Hom}_K(V, W)$ of linear maps from V to W has a structure of representation of G given by

$$(g\varphi)(x) = g\varphi(g^{-1}x), \quad g \in G, \varphi \in \text{Hom}_K(V, W), x \in V.$$

We call this the *Hom-representation*.

In particular, if $W = K$ is the trivial representation, then

$$V^* = \text{Hom}_K(V, K)$$

is a representation of G , called the *dual representation* or *contragredient representation* of V .

PROPOSITION 7.10. Let G be a group and V and W two representations of G over a field K . Assume that W is finite dimensional. Then

$$\text{Hom}_K(V, W) \text{ and } V^* \otimes W$$

are naturally isomorphic as representations of G .

Let V be a representation of a group G . We write $V^{\otimes n}$ for the n -fold tensor product of n copies of V . Then $V^{\otimes n}$ is a representation of G . Let U be the subspace of $V^{\otimes n}$ spanned by the vectors

$$v_1 \otimes \dots \otimes v_n - v_{\sigma(1)} \otimes \dots \otimes v_{\sigma(n)}, \quad v_1, \dots, v_n \in V, \sigma \in S_n.$$

Note that U is a subrepresentation of $V^{\otimes n}$, thus $V^{\otimes n}/U$ is also a representation of G .

DEFINITION 7.11. The n th symmetric power of a representation V is the representation

$$\text{Sym}^n(V) := V^{\otimes n}/U$$

Keeping the above notation, let W be the subspace of $V^{\otimes n}$ spanned by the vectors

$$v_1 \otimes \dots \otimes v_n - \text{sign}(\sigma) v_{\sigma(1)} \otimes \dots \otimes v_{\sigma(n)}, \quad v_1, \dots, v_n \in V, \sigma \in S_n.$$

Then W is also a subrepresentation of $V^{\otimes n}$.

DEFINITION 7.12. The n th exterior power of a representation V is the representation

$$\Lambda^n V = V^{\otimes n}/W.$$

7.1.4. Irreducibility and indecomposability.

DEFINITION 7.13. If V is a representation of a group G , then V is a KG -module, and we say that V is an *irreducible representation* if it is a simple KG -module (i.e. that is non-zero and has no non-zero proper submodules).

PROPOSITION 7.14. *If G is a finite group, any non zero representation of G contains a non zero finite dimensional subrepresentation.*

Moreover, any irreducible representation of G is finite dimensional.

DEFINITION 7.15. Let G be a group and V be a representation of G . V is said *indecomposable* if V can not be written as a direct sum of two non zero subrepresentations.

THEOREM 7.16. Maschke. *Let G be a finite group and K be a field whose characteristic does not divide the order of G . Let (V, ρ) be a finite dimensional representation of G over K and W be a subrepresentation of V . Then there is a subrepresentation U of V such that $V = U \oplus W$.*

COROLLARY 7.17. *Every irreducible representation is indecomposable.*
(The converse is not always true.)

THEOREM 7.18. (Schur's lemma). *Let G be a finite group and K be a field.*

- (1) *If V and W are two irreducible representation of G over K and $f : V \rightarrow W$ is a morphism of representations, then $f = 0$ or f is an isomorphism.*
- (2) *If V is an irreducible representation of G and K is algebraically closed, then any morphism of representation $V \rightarrow V$ is a homothety. In other words,*

$$\text{End}_{\text{Rep}_K(G)}(V) = KI \cong K.$$

THEOREM 7.19. *Let V be a finite dimensional representation of a finite group G over an algebraically closed field K . Assume that the order of G does not divide the characteristic of K . Then*

$$V \cong V_1^{\oplus a_1} \oplus \dots \oplus V_r^{\oplus a_r}$$

where V_i are irreducible representations of G , unique up to isomorphism, and the multiplicities a_i are uniquely determined by V .

DEFINITION 7.20. The number a_i in the previous theorem is called the multiplicity of V_i in V .

7.1.5. **Character Theory.** Let G be a group and $K = \mathbb{C}$ be the field of complex numbers. Every representation is assumed to be finite dimensional.

If V is a representation of G we say that a bilinear form $B : V \times V \rightarrow \mathbb{C}$ is invariant under G if

$$B(gv, gw) = B(v, w), \text{ for all } g \in G, v, w \in V.$$

DEFINITION 7.21. Let V be a representation of G . The *character* of V is the map

$$\chi_V : G \rightarrow \mathbb{C}, g \mapsto \text{tr}(g|V).$$

PROPOSITION 7.22. Let V be a representation of G . There is an inner product on V that is invariant under G , defined by

$$\langle v, w \rangle_W = \frac{1}{|G|} \sum_{g \in G} (gv, gw),$$

where $(\cdot, \cdot)$ is any inner product on V .

COROLLARY 7.23. If V is a representation of G , then $g|V$ is diagonalizable for all $g \in G$. Moreover, the eigenvalues of $g|V$ are roots of unity.

COROLLARY 7.24. Let V and W be two isomorphic representations of G , then $\chi_V = \chi_W$.

PROPOSITION 7.25. Let V be a representation of G .

- $\chi_V(1) = \dim V$.
- $\chi_V(g^{-1}) = \overline{\chi_V(g)}$ for all $g \in G$.

PROPOSITION 7.26. Let V and W be two representations of G .

- (1) $\chi_{V \oplus W} = \chi_V + \chi_W$.
- (2) $\chi_{V \otimes W} = \chi_V \chi_W$.
- (3) $\chi_{V^*} = \overline{\chi_V}$.
- (4) $\chi_{\text{Hom}_{\mathbb{C}}(V, W)} = \overline{\chi_V} \chi_W$.
- (5) If W is a subrepresentation of V , then $\chi_{V/W} = \chi_V - \chi_W$.
- (6) $\chi_{\text{Sym}^2(V)}(g) = \frac{1}{2}[\chi_V(g)^2 + \chi_V(g^2)]$, for all $g \in G$.
- (7) $\chi_{\Lambda^2(V)}(g) = \frac{1}{2}[\chi_V(g)^2 - \chi_V(g^2)]$, for all $g \in G$.

DEFINITION 7.27. A character is said to be irreducible if it is the character of an irreducible representation.

7.1.6. **The group algebra.** Let G be a finite group. Denote by $\mathbb{C}G$ the group algebra of G . Given a representation (V, ρ) of G , the group homomorphism

$$\rho : G \rightarrow \text{GL}(V)$$

extends uniquely to a $\mathbb{C}$ -algebra homomorphism

$$\tilde{\rho} : \mathbb{C}G \rightarrow \text{End}_{\mathbb{C}}(V)$$

by the formula

$$\tilde{\rho} \left(\sum_{g \in G} \alpha_g g \right) = \sum_{g \in G} \alpha_g \rho(g)$$

Note that this is the same as giving V the structure of a left $\mathbb{C}G$ -module as made previously.

Now let $(V_1, \rho_1), \dots, (V_r, \rho_r)$ be a complete family of pairwise non-isomorphic irreducible representations of G and let

$$\tilde{\rho} = \bigoplus_{j=1}^r \tilde{\rho}_j : \mathbb{C}G \rightarrow \bigoplus_{j=1}^r \text{End}_{\mathbb{C}}(V_j)$$

Then

PROPOSITION 7.28. *$\tilde{\rho}$ is an isomorphism of $\mathbb{C}$ -algebras.*

By choosing a basis for each V_i we can identify

$$\text{End}_{\mathbb{C}}(V_j) \cong \mathbf{M}_{n_j}(\mathbb{C})$$

where $n_j = \dim V_j$, that is, the space of $n_j \times n_j$ complex matrices. Thus, we have the following

THEOREM 7.29 (Wedderburn-Artin). *The group algebra $\mathbb{C}G$ is isomorphic to the algebra*

$$\bigoplus_{j=1}^r \mathbf{M}_{n_j}(\mathbb{C})$$

PROPOSITION 7.30. *Any irreducible representation of G is isomorphic to a minimal left ideal of $\mathbb{C}G$.*

PROOF. Let V be an irreducible representation of G . Then V occurs in $\mathbb{C}G$ with multiplicity $\dim V$, so there is a subrepresentation (that is, a left $\mathbb{C}G$ -submodule) I of $\mathbb{C}G$ isomorphic to V as a $\mathbb{C}G$ -module. Thus I is a left-ideal. By the irreducibility of I , the only ideals contained in I are I itself and 0 , so that I is a minimal left ideal of $\mathbb{C}G$.

Recall that minimal left ideals in any ring R are generated by primitive idempotents, that is, elements $e \in R$ such that $e^2 = e$ and such that e is not a sum of two or more idempotents. Thus, the irreducible representations of G are generated as left ideals of the group algebra by primitive left idempotents. There is no known formula for such idempotents. Nevertheless, there are idempotents generating the isotypic components of $\mathbb{C}G$, namely

$$\eta_j = \frac{\dim V_j}{|G|} \sum_{g \in G} \overline{\chi_{V_j}(g)} g, \quad j = 1, \dots, r$$

which follows from the second projection formula. $\square$

7.1.7. The regular representation.

LEMMA 7.31. *Let X be a set endowed with a G -action and let $V = \mathbb{C}V$ be the permutation representation associated to X . For $g \in G$ write*

$$\text{Fix}_X(g) = \{x \in X \mid gx = x\}.$$

Then

$$\chi_V(g) = |\text{Fix}_X(g)|$$

for all $g \in G$.

PROPOSITION 7.32. *Let $\mathbb{C}V$ be the regular representation of G and let χ be its character. Then*

$$\chi(g) = |G| \text{ if } g = 1$$

and

$$\chi(g) = 0 \text{ if } g \neq 1$$

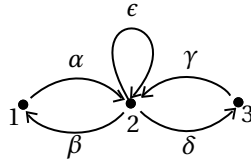
PROPOSITION 7.33. *Let $(V_i)_{i \in I}$ be a complete family of pairwise non-isomorphic irreducible representations of G . Let $n_i = \dim V_i$. Then*

$$\mathbb{C}G \cong_{\text{rep}} \bigoplus_{i \in I} V_i^{n_i}$$

as representations of G . In particular, the number of isomorphism classes of representations of G is finite.

Observe that the previous proposition implies that $|G| = \sum_{i \in I} n_i^2$. [33]

7.2. Representations of quivers. A *quiver* is an oriented graph. For example



is a quiver.

In this work, we consider only finite quivers, namely quivers with finitely many vertices and arrows.

A *quiver* Q is endowed with a vertex set I . We write $a \in Q$ to mean that a is an arrow. Its *head* or *target* is at the vertex $h(a)$ and its *tail* or *source* is at $t(a)$. Thus, $a : i \rightarrow j$ means that $h(a) = j$ and $t(a) = i$.

In the preceding example $I = \{1, 2, 3\}$ and $Q = \{\alpha, \beta, \gamma, \delta, \epsilon\}$

The *underlying graph* of a quiver Q is the graph obtained from Q by forgetting the orientation of the arrow.

A quiver Q' is a *subquiver* of a quiver Q , if $I' \subset I$ and $Q' \subset Q$.

An *oriented cycle* is a subgraph with vertices $I_0 = \{s_1, \dots, s_r\} \subset I$ and arrows $Q_0 = \{a_1, \dots, a_r\} \subset Q$ such that $t(a_i) = s_i$, $h(a_i) = s_{i+1}$ if $i < r$ and $t(a_r) = s_r$, $h(a_r) = s_1$.

In our example $Q_0 = \{\alpha, \beta\}$ with vertex set $I_0 = \{1, 2\}$ is an oriented cycle.

A *loop* is an arrow whose head and tail coincide. In our example, there is only one loop, $\epsilon : 2 \longrightarrow 2$.

A *double quiver* $\overline{Q}$ of a quiver Q is obtained by adjoining a reverse arrow $a^* : j \longrightarrow i$ for each arrow $a : i \longrightarrow j$ in Q .

DEFINITION 7.34. Fix a field K . Let Q be a quiver. Consider a K -vector space

$$V = \bigoplus_{i \in I} V(i)$$

and a collection of K -linear maps

$$\rho = \{\rho_a : V(i) \longrightarrow V(j) \mid a \in Q, t(a) = i, h(a) = j\}$$

Then (V, ρ) is called a *representation of Q* . The *dimension vector* of the representation is d , where $d_i = \dim V(i)$

DEFINITION 7.35. Let (V, ρ) and (W, σ) to be two representations of Q . A *morphism of representations* $\varphi : (V, \rho) \longrightarrow (W, \sigma)$ is a set of linear maps $\{\varphi_i : V(i) \longrightarrow W(i) \mid i \in I\}$ such that

$$\varphi_j \circ \rho_a = \sigma_a \circ \varphi_i$$

for every $a \in Q$, whew $i = t(a)$, $j = h(a)$

DEFINITION 7.36. We say that two representations (V, ρ) and (W, σ) of Q are *isomorphic* if there is existing a morphism $\phi : (V, \rho) \longrightarrow (W, \sigma)$ such that ϕ_i is an isomorphism for every $i \in I$.

DEFINITION 7.37. The *direct sum* $(V \oplus W, \rho \oplus \sigma)$ of two representations (V, ρ) and (W, σ) of a quiver Q is defined in the obvious way.

DEFINITION 7.38. A representation (W, σ) is a *subrepresentation* of (V, ρ) if, for every $i \in I$, we are given an inclusion $W(i) \subseteq V(i)$ such that, for every $a \in Q$, with $t(a) = i$, the restriction of ρ_a to $W(i)$ coincides with σ_a .

DEFINITION 7.39. A non-zero representation (V, ρ) is called *irreducible* if it does not contain any non-trivial proper subrepresentation; it is called *indecomposable* if it cannot be written as the direct sum of two non-trivial subrepresentations.

DEFINITION 7.40. Let Q be a quiver. A *path* p is a sequence $a_1, \dots, a_k$ of arrows such that $t(a_i) = h(a_{i+1})$. Set $t(p) = t(a_k)$, $h(p) = h(a_1)$. The number k of arrows is called the *length* of p .

DEFINITION 7.41. Let $p_1 = a_1, \dots, a_k$ and $p_2 = \delta_1, \dots, \delta_l$ be two paths of Q . We define the product of p_1 and p_2 to be the path $p_1 p_2 = \delta_1, \dots, \delta_l, a_1, \dots, a_k$ if $h(a_1) = t(a_l)$ and zero otherwise.

We introduce an element e_i for each vertex $i \in I$ and define the product of e_i and e_j by the formula

$$e_i e_j = \delta_{ij} e_i$$

For a path p , we set

$$e_i p = \begin{cases} p, & \text{if } i = h(p) \\ 0, & \text{otherwise} \end{cases},$$

$$p e_i = \begin{cases} p, & \text{if } i = t(p) \\ 0, & \text{otherwise} \end{cases}$$

7.2.1. The path algebra.

DEFINITION 7.42. The *path algebra* $K(Q)$ of Q is the vector space of k -linear combinations of all paths of Q and elements $\{e_i\}_{i \in I}$, with the multiplication obtained by extending the product defined above by bilinearity.

Note that every e_i , $i \in I$, is an idempotent in $K(Q)$ and that $\sum_{i \in I} e_i = 1$.

We write $K\overline{Q}$ for the path algebra of $\overline{Q}$.

LEMMA 7.43. Let Q be a quiver.

- (1) The path algebra $K(Q)$ is generated by the idempotents $\{e_i \mid i \in I\}$ and the paths $\{a \mid a \in Q\}$ of length 1.
- (2) The algebra $K(Q)$ is finite-dimensional if and only if Q does not contain any oriented cycle.
- (3) If Q is the disjoint union of two quivers Q' and Q'' , then $k(Q)$ is isomorphic to the direct product $K(Q') \times K(Q'')$.
- (4) The path algebra has a natural $\mathbb{Z}$ -grading

$$K(Q) = \bigoplus_{n=0}^{\infty} K(Q)_{(n)},$$

where $K(Q)_{(0)}$ is the span of the idempotents e_i for all $i \in I$ and $K(Q)_{(n)}$ is the span of all paths of length n .

- (5) For every vertex $i \in I$ the element e_i is a primitive idempotent of $K(Q)$, and hence $K(Q)e_i$ is an indecomposable projective $K(Q)$ -module.

THEOREM 7.44. *The category of representations of Q over a field K is equivalent to the category of $K(Q)$ -modules.*

PROOF. Given a representation (V, ρ) of a quiver Q , $V = \bigoplus_{i \in I} V(i)$ one can equip V with a structure of $K(Q)$ -module in the following way

- (1) The idempotent e_i acts on $V(j)$ by $\delta_{ij} \text{Id}_{V(j)}$.
- (2) For $a \in Q$ and $v \in V(i)$ we set $av = \rho_a(v)$ if $t(a) = i$ and zero otherwise.
- (3) We extend this action for the whole $K(Q)$ using a Lemma 7.43 (1).

Conversely, every $K(Q)$ -module V gives rise to a representation ρ of Q when one sets $V(i) = e_i V$. $\square$

We denote by $\text{Rep}_K Q$ the category representations of Q . In particular, it follows that $\text{Rep}_K Q$ is an abelian K -category.

DEFINITION 7.45. Let $x \in \mathbb{Z}^I$, we define the *support* of x , $\text{supp} x$, to be the set of vertices $i \in I$ such that $x_i \neq 0$.

LEMMA 7.46. *Let Q be a quiver, X be an indecomposable representation of Q . Then the support of X is connected. (see [7])*

7.2.2. Standard resolution. Let R be a unital ring.

DEFINITION 7.47. An R -module P is *projective* if for any surjective morphism $\varphi : M \rightarrow N$ of R -modules and any morphism $\psi : P \rightarrow N$ there exists a morphism $f : P \rightarrow M$ such that $\psi = \varphi \circ f$.

DEFINITION 7.48. Let M be an R -module. A complex $(P_\bullet, d)$ of R -modules

$$\dots \xrightarrow{d_{i+1}} P_i \xrightarrow{d_i} \dots \xrightarrow{d_2} P_1 \xrightarrow{d_1} P_0 \rightarrow 0$$

such that P_i is projective for all $i \geq 0$, the homology groups are $H_0(P_\bullet) := \text{Ker} d_0 / \text{Im} d_1 = M$ and $H_i(P_\bullet) := \text{Ker} d_i / \text{Im} d_{i+1} = 0$ for all $i \geq 1$, is called a *projective resolution* of M . It is sometimes useful to see a projective resolution as the exact complex

$$\dots \rightarrow P_i \rightarrow \dots \rightarrow P_1 \rightarrow P_0 \xrightarrow{l} M \rightarrow 0,$$

where $l : P_0 \rightarrow M$ is the lift of the identity map between $H_0(P_\bullet)$ and M .

Here, d is an R -morphism from $P_\bullet$ to $P_\bullet$ of degree -1 , it means that d_i maps P_i to P_{i-1} for all i , such that $d_i^2 = 0$ for all i .

PROPOSITION 7.49. *Every R -module has a projective resolution. (see Proposition 6.13 in Chapter 5 of [20])*

A remarkable property of path algebras is the fact that every module has a projective resolution of length at most 2. In order to understand this fact, we need some definitions and results of Homologic Algebra, for more details see [32] and [30].

DEFINITION 7.50. Let R be a unital ring. An R -module P is *projective* if for any surjective morphism $\varphi : M \rightarrow N$ of R -modules and any morphism $\psi : P \rightarrow N$ there exists a morphism $f : P \rightarrow M$ such that $\psi = \varphi \circ f$. (See 6.1 in Chapter 5 of [20]).

LEMMA 7.51. *Let P be an R -module, the following conditions are equivalent*

- (1) *P is projective;*
- (2) *There exists a free module F such that F is isomorphic to $P \oplus P'$;*
- (3) *Any exact sequence of R -modules*

$$0 \rightarrow N \rightarrow M \rightarrow P \rightarrow 0$$

splits.

(see Lemma 6.2 in chapter 5 of [20])

DEFINITION 7.52. Let M be an R -module. A complex $(P_\bullet, d)$ of R -modules

$$\dots \xrightarrow{d_{i+1}} P_i \xrightarrow{d_i} \dots \xrightarrow{d_2} P_1 \xrightarrow{d_1} P_0 \xrightarrow{d_0} 0$$

such that P_i is projective for all $i \geq 0$, $H_0(P_\bullet) := (\text{Ker} d_0) / (\text{Im} d_{i+1}) = M$ and $H_i(P_\bullet) := (\text{Ker} d_i) / (\text{Im} d_{i+1}) = 0$ for all $i \geq 1$, is called a *projective resolution* of M .

DEFINITION 7.53. Let M and N be two R -modules and $P_\bullet$ be a projective resolution of M . Consider the complex of abelian groups

$$0 \rightarrow \text{Hom}_R(P_0, N) \xrightarrow{\delta_0} \text{Hom}_R(P_1, N) \xrightarrow{\delta_1} \dots$$

where δ is defined by $\delta(\varphi)(x) := \varphi(d_{i+1}x)$ for all $\varphi \in \text{Hom}_R(P_i, N)$ and $x \in P_{i+1}$. We define the *i -th extension group* $\text{Ext}_R^i(M, N)$ as the *i -th cohomology group* of this complex. It means $\text{Ext}_R^i(M, N) := \text{Ker} \delta_i / \text{Im} \delta_{i-1}$.

LEMMA 7.54. $\text{Ext}_R^0 = \text{Hom}_R(M, N)$. (see section 6.4 in Chapter 5 of [20])

LEMMA 7.55. *Let $(P_\bullet, d)$ a projective resolution for M . We can associate the following short exact sequence with $[\gamma] \in \text{Ext}_R^1(M, N)$*

$$0 \rightarrow P_1 / \text{Ker} \gamma \rightarrow P_0 / d_1(\text{Ker} \gamma) \rightarrow M \rightarrow 0,$$

where $\gamma \in \text{Hom}_R(P_1, N)$.

Moreover, this exact sequence splits if and only if $[\gamma] = 0$. (see section 6.4 in Chapter 5 of [20])

THEOREM 7.56. Let Q be a quiver, V be a KQ -module. Recall that $V = \bigoplus_{i \in I} V(i)$. Then the following sequence of KQ -modules is exact:

$$0 \rightarrow \bigoplus_{a \in Q} KQe_{h(a)} \otimes V_{t(a)} \xrightarrow{f} \bigoplus_{i \in I} KQe_i \otimes V_i \xrightarrow{g} V \rightarrow 0,$$

where

$$f(be_{h(a)} \otimes v) = ae_{h(a)}a \otimes v - ae_{h(a)} \otimes av$$

for all $b \in KQ$, $a \in Q$, $v \in V_{t(a)}$, and

$$g(be_i \otimes v) = bv$$

for any $i \in I$, $v \in V_i$. Hence it is a projective resolution of V .

The structures of KQ -module considered in the statement are defined by the action of KQ on the left-hand side of every tensor product. (see Theorem 3.1 of Chapter 7 in [20])

7.2.3. Extension group for modules over the path algebra.

DEFINITION 7.57. Let X and Y be two KQ -modules. We define a linear map

$$d : \bigoplus_{i \in I} \text{Hom}_K(X_i, Y_i) \rightarrow \bigoplus_{\gamma \in Q} \text{Hom}_K(X_{t(\gamma)}, Y_{h(\gamma)}) \quad (7.1)$$

$$(d\varphi)_\gamma(x) = \varphi_{h(\gamma)}(\gamma \cdot x) - \gamma \cdot \varphi_{t(\gamma)}(x),$$

where $\gamma \in Q$, $x \in X_{t(\gamma)}$ and $\varphi = (\varphi_i) \in \bigoplus_{i \in I} \text{Hom}_K(X_i, Y_i)$.

PROPOSITION 7.58. $\text{Ker} d = \text{Hom}_{KQ}(X, Y) = \text{Ext}_{KQ}^0(X, Y)$ and $\text{coker} d = \text{Ext}_{KQ}^1(X, Y)$. (see section 3 in Chapter 7 of [20])

PROPOSITION 7.59. In the category of representations of Q , $\text{Ext}_{KQ}^1(X, Y) = 0$ for all $i \geq 2$. (see section 3 in chapter 5 of [20])

COROLLARY 7.60. Let

$$0 \rightarrow X \rightarrow Y \rightarrow Z \rightarrow X \rightarrow 0$$

be a short exact sequence of representations of Q , then the maps

$$\text{Ext}_{KQ}^1(V, Z) \rightarrow \text{Ext}_{KQ}^1(V, X) \text{ and } \text{Ext}_{KQ}^1(Z, V) \rightarrow \text{Ext}_{KQ}^1(Y, V)$$

are surjective for every representation V of Q . (see section 3 in chapter 5 of [20])

LEMMA 7.61. If X and Y are indecomposable finite-dimensional KQ -modules and if $\text{Ext}_{KQ}^1(X, Y) = 0$, then every non-zero $\varphi \in \text{Hom}_{KQ}(X, Y)$ is either surjective or injective. (see section 3 in chapter 5 of [20])

7.2.4. *Canonical bilinear form.*

DEFINITION 7.62. Let $(C_\bullet, d)$ be a complex of finite-dimensional K -vector spaces, we assume that $C_i = \{0\}$ for almost every i . The *Euler characteristic* of the complex is the number $\sum_{i \in \mathbb{Z}} (-1)^i \dim_K C_i = \sum_{i \in \mathbb{Z}} (-1)^i \dim_K H_i(C_\bullet)$.

DEFINITION 7.63. Let X be a finite-dimensional KQ -module. We define the bilinear form on $\mathbb{Z}^I$ by the formula

$$\langle \dim X, \dim Y \rangle = \sum_{i \in I} \dim X_i \dim Y_i - \sum_{a \in Q} \dim X_{t(a)} \dim Y_{h(a)} = \dim \text{Hom}_{KQ}(X, Y) - \dim \text{Ext}_{KQ}^1(X, Y),$$

where the second equality follows from calculating Euler characteristic in 7.1.

This bilinear form defined over $\mathbb{Z}^I$ is called the *Ringel form* for the quiver Q and $q(x) = \langle x, x \rangle$, for $x \in \mathbb{Z}^I$ is called the *Tits form* for Q .

For a graph Γ , with set of vertices Γ and set of edges Γ_1 we can generalize the Ringel form as follows:

$$(x, y) := \sum_{i \in \Gamma_0} (2 - l(i)) x_i y_i - \sum_{(i, j) \in \Gamma_1} x_i y_j,$$

where $x, y \in \mathbb{Z}^{\Gamma_0}$ and $l(i)$ is the number of loops at i .

7.2.5. *Coxeter-Dynkin and affine graphs.*

DEFINITION 7.64. A connected graph Γ (it no have orientation) is called *Coxeter-Dynkin* if its Ringel form is positive definite and *affine* is its Tits form is positive semidefinite but not positive definite. If Γ is neither Coxeter-Dynkin or affine, then we say that it is of *indefinite type*.

PROPOSITION 7.65. *Let Γ be a Coxeter-Dynkin or affine graph. Any proper connected subgraph of Γ is a Coxeter-Dynkin graph.*

The following Table illustrates the Classification of the Coxeter-Dynkin and affine graphs (below n is the number of vertices).

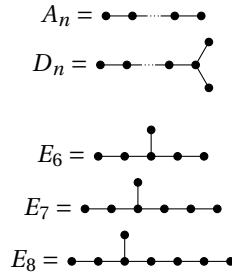
The affine graph, except the non-oriented loop A_0^1 ,



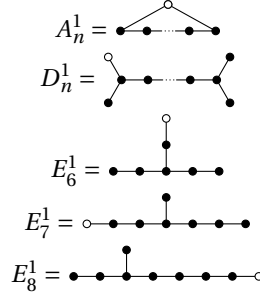
are obtained from the Coxeter-Dynkin graphs by adding a vertex. And A_1^1 is the graph



Coxeter-Dynkin graphs

(Also called finite or
Dynkin of type ADE):

Affine graphs

(Also called Dynkin extended or
Extended Dynkin of type ADE):**7.2.6. Gabriel Theorem.**

DEFINITION 7.66. A quiver is of *finite type* if it has finitely many isomorphism classes of indecomposable representations.

PROPOSITION 7.67. A quiver is of *finite type* if and only if all its connected components are of *finite type*. (see section 7 in chapter 7 of [20])

THEOREM 7.68 (Gabriel). Let Q be a connected quiver and Γ be its underlying graph. Then

- (1) The quiver Q is of *finite type* if and only if Γ is a Coxeter-Dynkin graph.
- (2) Assume that Γ is a Coxeter-Dynkin graph and (X, ρ) is an indecomposable representation of Q . Then $q(\dim X) \leq 1$ and $\dim X_i \geq 0$ for all $i \in I$.
- (3) If Γ is a Coxeter-Dynkin graph, then for every $x \in \mathbb{Z}^I$, such that $q(x) \leq 1$ and $x_i \geq 0$ for all $i \in I$, there is exactly one indecomposable representation of Q of dimension x . (see section 7 in chapter 7 of [20])

REMARK 7.69. For a quiver Q of finite representation type, and a dimension vector β , the linearly reductive group $\text{GL}(\beta)$ acts on $\text{Rep}(Q, \beta)$. Since there are finitely many indecomposable representations, there are only finitely many isomorphism classes in $\text{Rep}(Q, \beta)$, so there are only finitely many orbits. One of the orbits has to be dense. (see chapter 7 in [20])

PROPOSITION 7.70. The orbit of $V \in \text{Rep}(Q, \alpha)$ is closed if and only if V is a direct sum of simple representations.

In particular, Let $V \in \text{Rep}(Q, \alpha)$ be a nonzero representation.

- V is semi-stable with respect to $\text{GL}(\alpha)$ if and only if it is semisimple.
- V is stable if and only if it is simple.

(see Propositions 9.7.7 and 9.7.8 in [11])

[20]

Non-commutative deformations of Kleinian singularities.

1. The algebras $\mathcal{S}^\lambda$ and $\mathcal{O}^\lambda$.

Let K be an algebraically closed field of characteristic zero and let $\Gamma \leq \mathrm{SL}_2(K)$ be a non-trivial finite subgroup.

In this section, we introduce relevant algebras, which are deformations of the skew group algebra $K\langle x, y \rangle \otimes_K K\Gamma$.

DEFINITION 1.1. For $\lambda \in Z(K\Gamma)$ let

$$\mathcal{S}^\lambda := (K\langle x, y \rangle \rtimes \Gamma) / (xy - yx - \lambda) = \mathbb{K}\langle x, y \rangle \otimes_{\mathbb{K}} K\Gamma / (xy - yx - \lambda).$$

We then define $\mathcal{O}^\lambda := e\mathcal{S}^\lambda e$, where $e = \frac{1}{|\Gamma|} \sum_{\gamma \in \Gamma} \gamma$.

We also need to consider a global version of these algebras:

DEFINITION 1.2. Let R be a commutative K -algebra and let $\lambda \in Z(R\Gamma)$. The action of Γ extends to $R\langle x, y \rangle$, with the action of $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma$ on R being trivial, $\gamma \cdot x = ax + cy$ and $\gamma \cdot y = bx + dy$. We define

$$\mathcal{S}^{R,\lambda} = (R\langle x, y \rangle \rtimes \Gamma) / (xy - yx - \lambda)$$

and

$$\mathcal{O}^{R,\lambda} = e\mathcal{S}^{R,\lambda} e,$$

where $e = \frac{1}{|\Gamma|} \sum_{\gamma \in \Gamma} \gamma$

The following is a special case of the Theorem 2.1 in [18].

LEMMA 1.3. $\mathrm{gr}\mathcal{S}^{R,\lambda} \cong R[x, y] \rtimes \Gamma$ and $\mathrm{gr}\mathcal{O}^{R,\lambda} \cong R[x, y]^\Gamma$

LEMMA 1.4. $\mathcal{S}^\lambda$ and $\mathcal{O}^\lambda$ are Noetherian rings.

PROOF. Since K is Noetherian, $K\Gamma$ is also Noetherian, because it is a finitely generated K -algebra. Then $K[x, y] \otimes_K K\Gamma$ is Noetherian, since $K\Gamma$ is Noetherian and $K[x, y]$ is finitely generated over K . By the Theorem 1.3, $\mathrm{gr}\mathcal{S}^\lambda \cong K[x, y] \otimes_K K\Gamma$. Then, since $\mathcal{S}^\lambda$ is filtered and $\mathrm{gr}\mathcal{S}^\lambda$ is Noetherian, $\mathcal{S}^\lambda$ is Noetherian (see Lemma 1.6.9 in [31]).

The same argument holds for $\mathcal{O}^\lambda$, because $K[x, y]$ is Noetherian and $\mathcal{O}^\lambda$ has a filtration inherited from $\mathcal{S}^\lambda$. $\square$

1.1. Example using the cyclic group of order 2. Part 1. In this example, we consider $K = \mathbb{C}$ as the base field.

Let $\Gamma = C_2 = \langle \sigma | \sigma^2 = 1_\Gamma \rangle$ be the cyclic group of order 2. For simplicity, use $1 = 1_\Gamma$ for the identity of the group C_2 . Let $\zeta = e^{2\pi i/2} = e^{\pi i} = -1$, the irreducible representations for C_2 are (ρ_0, V_0) and (ρ_1, V_1) , where

$$\rho_0 : C_2 \rightarrow \mathbb{C} = V_0, 1 \mapsto 1, \sigma \mapsto \zeta^0 = 1$$

and

$$\rho_1 : C_2 \rightarrow \mathbb{C} = V_1, 1 \mapsto 1 = \zeta^0, \sigma \mapsto \zeta^1 = -1.$$

Note that they are the only irreducible representations, because $|C_2| = 2 = (\dim V_0)^2 + (\dim V_1)^2$. Let χ_0 and χ_1 be the characters corresponding to the irreducible representations (ρ_0, V_0) and (ρ_1, V_1) , respectively, the Table 1 is the table of characters for C_2 .

C_2	1	σ
χ_0	$1 = \dim V_0$	1
χ_1	$1 = \dim V_1$	-1

TABLE 1.

Note that $\sigma \in Z(C_2)$, then $\sigma \in Z(\mathbb{C}(C_2))$. And the average element of C_2 is $e = \frac{1+\sigma}{2}$. So, with $\lambda = \sigma$ we can construct the algebras $\mathcal{S}^\lambda$ and $\mathcal{O}^\lambda$ over $\mathbb{C}$:

$$\mathcal{S}^\lambda := \frac{\mathbb{C}\langle x, y \rangle \otimes_{\mathbb{C}} \mathbb{C}C_2}{(xy - yx - \sigma)},$$

$$\mathcal{O}^\lambda := \frac{1+\sigma}{2} \frac{\mathbb{C}\langle x, y \rangle \otimes_{\mathbb{C}} \mathbb{C}C_2}{(xy - yx - \sigma)} \frac{1+\sigma}{2}$$

In order to exemplify the Lemma 1.3, we will find the graded associated algebras for our examples of $\mathcal{S}^\lambda$ and $\mathcal{O}^\lambda$, and we will verify that $\text{gr}\mathcal{S}^\lambda \cong \mathbb{C}[x, y] \otimes \mathbb{C}C_2$ and $\text{gr}\mathcal{O}^\lambda \cong \mathbb{C}[x, y]^{C_2}$.

The invariant polynomials under the action of C_2 are $\mathbb{C}[x, y]^{C_2} = \text{span}\{x^2, y^2, xy\}$:

Indeed, since $C_2 \leq \text{SL}_2(\mathbb{C})$, σ , the generator of C_2 can be seen as $\sigma = \begin{pmatrix} \zeta & 0 \\ 0 & \zeta^{-1} \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$. We shall see that x and y are semi-invariant under C_2 :

$$\begin{aligned} \sigma \cdot x &= \begin{pmatrix} \zeta & 0 \\ 0 & \zeta^{-1} \end{pmatrix} \cdot x = \zeta^{-1}x = -x \\ \sigma \cdot y &= \begin{pmatrix} \zeta & 0 \\ 0 & \zeta^{-1} \end{pmatrix} \cdot y = \zeta y = -y \end{aligned}$$

So, $x^2, y^2, xy \in \mathbb{C}[x, y]^{C_2}$. It is easy to see that $\{x^2, y^2, xy\}$ is linearly independent. Suppose the null polynomial is written as a linear combination of x^2 , y^2 and xy . Then, by equality

of polynomials, the scalars are equal to zero. Now, let $f \in \mathbb{C}[x, y]^{C_2}$, i.e. $\sigma \cdot f = f$. Since f is a polynomial function from $\mathbb{C}^2$ to $\mathbb{C}$, we can write f by

$$f(x, y) = \sum_{0 \leq i \leq k, 0 \leq j \leq l} a_{ij} x^i y^j,$$

for some non-negative integers k and l . Then,

$$\begin{aligned} \sigma \cdot f(x, y) &= \sum_{0 \leq i \leq k, 0 \leq j \leq l} a_{ij} \sigma \cdot x^i \sigma \cdot y^j \\ f(x, y) &= \sum_{0 \leq i \leq k, 0 \leq j \leq l} a_{ij} (-1)^{j-i} x^i \sigma \cdot y^j \\ &\Rightarrow j = i \bmod 2 \end{aligned}$$

$\Rightarrow f \in \text{span}\{x^2, y^2, xy\}$. Thus, $\mathbb{C}[x, y]^{C_2} = \text{span}\{x^2, y^2, xy\}$.

Now, a graduation for $\mathbb{C}\langle x, y \rangle$ is $\{V^{\otimes i}\}_{i \geq 0}$, where $V = \mathbb{C}x \oplus \mathbb{C}y$ and $V^{\otimes 0} = \mathbb{C}$. Then a graduation for $\mathbb{C}\langle x, y \rangle \otimes \mathbb{C}C_2$ is $\{V^{\otimes i} \otimes \mathbb{C}C_2\}_{i \geq 0}$. Thus, a filtration for $\mathbb{C}\langle x, y \rangle \otimes \mathbb{C}C_2$ is given by $\{\bigoplus_{i=0}^d V^{\otimes i} \otimes \mathbb{C}C_2\}_{d \geq 0}$. And let $\pi : \mathbb{C}\langle x, y \rangle \otimes \mathbb{C}C_2 \rightarrow S^\lambda$ be the canonical projection, a filtration for S^λ is $\{\bigoplus_{i=0}^d \pi(V^{\otimes i} \otimes \mathbb{C}C_2)\}_{d \geq 0}$. Therefore, the graded associated algebra of S^λ is

$$\text{gr}S^\lambda = \bigoplus_{d \geq 0} \frac{\bigoplus_{i=0}^d \pi(V^{\otimes i} \otimes \mathbb{C}C_2)}{\bigoplus_{i=0}^{d-1} \pi(V^{\otimes i} \otimes \mathbb{C}C_2)}$$

We have a surjection

$$\mathbb{C}[x, y] \otimes \mathbb{C}C_2 \xrightarrow{\varphi} \text{gr}S^\lambda, x \otimes 1 \mapsto \overline{x \otimes 1}, y \otimes 1 \mapsto \overline{y \otimes 1}, 1 \otimes \sigma \mapsto \overline{1 \otimes \sigma}.$$

And also we have that

$$\varphi|_{\mathbb{C}[x, y]^{C_2} \otimes \mathbb{C}} : \mathbb{C}[x, y]^{C_2} \rightarrow \text{gr}S^\lambda.$$

It is not easy to see that ϕ and this restriction of ϕ are also injective.

2. Deformed preprojective algebras.

In this section K is an arbitrary field and Q is an arbitrary finite quiver with vertex set I . We define the preprojective algebra of a quiver and its deformations.

DEFINITION 2.1. We define the *Ringel form* for Q to be the bilinear form on $\mathbb{Z}^I$ with

$$\langle \alpha, \beta \rangle = \sum_{i \in I} \alpha_i \beta_i - \sum_{a \in Q} \alpha_{t(a)} \beta_{h(a)},$$

for $\alpha, \beta \in \mathbb{Z}^I$. And the *symmetric bilinear form* is

$$(\alpha, \beta) = \langle \alpha, \beta \rangle - \langle \beta, \alpha \rangle$$

Elements of K^I are called *weights* and if λ is a weight then we define the *deformed preprojective algebra* to be

$$\Pi^\lambda = K\overline{Q}/(c - \lambda),$$

where c is the *commutator element* of $K\overline{Q}$

$$c = \sum_{a \in Q} [a, a^*] = \sum_{a \in Q} aa^* - a^*a,$$

and λ is identified with the element $\sum_{i \in I} \lambda_i e_i$ of $K\overline{Q}$.

If the quiver is understood, we just write Π^λ .

LEMMA 2.2. *The ideal $(c - \lambda)$ is the same as the ideal, with a generator*

$$g_i = \sum_{\substack{a \in Q \\ h(a)=i}} aa^* - \sum_{\substack{a \in Q \\ t(a)=i}} a^*a - \lambda_i e_i$$

for each $i \in I$.

PROOF. Note that

$$\begin{aligned} e_i(c - \lambda)e_i &= e_i c e_i - e_i \lambda e_i = e_i \sum_{a \in Q} (aa^* - a^*a)e_i - e_i \sum_{j \in I} \lambda_j e_j e_i \\ &= \sum_{a \in Q} e_i aa^* e_i - \sum_{a \in Q} e_i a^* a e_i - e_i \lambda_i e_i = \sum_{\substack{a \in Q \\ h(a)=i}} aa^* - \sum_{\substack{a \in Q \\ t(a)=i}} a^*a - \lambda_i e_i \end{aligned}$$

Since $e_i(c - \lambda)e_i \in (c - \lambda)$, so the elements g_i belongs to the ideal $(c - \lambda)$. But also $c - \lambda$ is the sum of them, so the ideal they generate contains $c - \lambda$. $\square$

This deformed preprojective algebra is a generalization of the *preprojective algebra* $\Pi(Q) = \Pi^0(Q)$.

There is a natural grading (see section 1) of $K\overline{Q}$ over the length of the path, with the trivial paths e_i in the degree zero and the arrows a, a^* in the degree one (see section 7.2). This induces a grading of Π . It also gives rise to a filtration of $K\overline{Q}$ which induces one of Π^λ

More general, if R is a commutative ring and $\lambda \in R^I$, we define $\Pi^{R,\lambda}(Q) = R\overline{Q}/(c - \lambda)$

LEMMA 2.3. *Up to isomorphism, $\Pi^{R,\lambda}(Q)$ does not depend on the orientation of Q .*

PROOF. Assume that a new quiver Q' is obtained from Q by reversing the arrow $a \in Q$. Note that $\overline{Q} = \overline{Q'}$. Then there is an isomorphism $\Pi^{R,\lambda}(Q) \longrightarrow \Pi^{R,\lambda}(Q')$ which send a to a^* and a^* to $-a$. $\square$

LEMMA 2.4. *There is a natural surjective homomorphism $\Pi^0 \longrightarrow \text{gr} \Pi^\lambda$.*

PROOF. Give the ideal $(c - \lambda)$ the induced filtration from the filtration for $K\overline{Q}$ (see section 1.4 in the chapter 2). Then, by Lemma 1.13 there is an exact sequence

$$0 \longrightarrow \text{gr}(c - \lambda) \longrightarrow \text{gr}K\overline{Q} \longrightarrow \text{gr}(\Pi^\lambda(Q)) \longrightarrow 0.$$

Then by the First Isomorphism Theorem $\text{gr}K\overline{Q}/\text{gr}(c - \lambda) \cong \text{gr}(\Pi^\lambda(Q))$. Since $\text{gr}K\overline{Q} = K\overline{Q}$, $c \in \text{gr}(c - \lambda)$ and then $(c) \subseteq \text{gr}(c - \lambda)$. Thus,

$$\text{gr}(\Pi^\lambda(Q)) \cong K\overline{Q}/\text{gr}(c - \lambda) \leftarrow K\overline{Q}/(c) =: \Pi^0(Q)$$

and there is a surjection $\Pi^0 \longrightarrow \text{gr}\Pi^\lambda$. $\square$

LEMMA 2.5. *If Q is Dynkin (or a disjoint union of Dynkin quivers) then $\Pi(Q)$ is finite dimensional.*

PROOF. For an arbitrary quiver, the preprojective algebra $\Pi(Q) := \Pi^0(Q)$ decomposes into the direct sum of the indecomposable preprojective KQ -modules. (see Lemma 3.4 in [2] and Theorems 0.1, 0.2 in [8]). In particular, if Q is a Dynkin quiver then Π^0 is finite-dimensional, by Gabriel's Theorem (see section 7.2 in the chapter 2) and the previously mentioned result. Since $\text{gr}\Pi^\lambda(Q)$ is a quotient of $\Pi(Q)$, also $\Pi^\lambda(Q)$ is finite-dimensional if Q is a Dynkin quiver. $\square$

3. Morita equivalence.

Let K be an algebraically closed field of characteristic zero and Γ a non-trivial finite subgroup of $SL_2(K)$.

In this section, we will prove the Theorem 0.1, that S^λ is Morita equivalent to $\Pi^\lambda(Q)$ and $\mathcal{O}^\lambda \cong e_0\Pi^\lambda(Q)e_0$, where Q is any orientation of the McKay graph of Γ (see *The McKay graph* in the introductory chapter 1). This is the first significant result that relates deformations $\mathcal{O}^\lambda$ and Π^λ .

For technical reasons, we need to prove it for the global versions over a commutative K -algebra R : S^λ , $\mathcal{O}^\lambda$ and $\Pi^\lambda(Q)$. These global versions are:

$$S^{R,\Lambda} = (R\langle x, y \rangle \rtimes \Gamma) / (xy - yx - \Lambda),$$

$$\mathcal{O}^{R,\Lambda} = eS^{R,\Lambda}e$$

where $e = \frac{1}{|\Gamma|} \sum_{\gamma \in \Gamma} \gamma$ (see definition 1.2 in section 1) and

$$\Pi^{R,\Lambda} = R\overline{Q}/(c - \Lambda)$$

(see up to lemma 2.3 in section 2).

Let V the natural 2-dimensional representation of Γ , it means, the inclusion of Γ is $\text{SL}_2(K)$. Let $\{N_0, \dots, N_n\}$ the irreducible representations of Γ with N_0 the trivial representation, and $I = \{0, 1, \dots, n\}$.

If η_i ($i \in I$) are the central idempotents in $K\Gamma$ corresponding to the irreducible representations N_i (see section 7.1 in preliminary chapter 2), then the following map is a bijection:

$$R^I \rightarrow Z(K\Gamma), \Lambda \mapsto \sum_{i \in I} \frac{\Lambda_i}{\delta_i} \eta_i$$

and we use this to identify R^I with $Z(R\Gamma)$. In the case $R = K$ we have the identification: $Z(K\Gamma) = K^I$, with $\lambda \in Z(K\Gamma)$ corresponding to the vector whose i -th component λ_i is the trace of $\lambda \in Z(K\Gamma)$ on the regular representation $K\Gamma$.

Let $e \in K\Gamma$ be the average of the group elements, and for each $i \in I$ let f_i be an idempotent in $K\Gamma$ with $K\Gamma f_i \cong N_i$. The f_i are orthogonal, so $f = f_0 + f_1 + \dots + f_n$ is idempotent. Also $f_0 = e$, so $r = fef$.

Now, if x, y is the standard basis of the natural representation of Γ , V , the symplectic bilinear form $\langle, \rangle$ with

$$\langle x, y \rangle = -1, \langle y, x \rangle = 1, \langle x, x \rangle = \langle y, y \rangle = 0,$$

is Γ -invariant, so defines a homomorphism $\langle, \rangle : V \otimes V \rightarrow K$. We have also a homomorphism

$$\zeta : K \rightarrow V \otimes V, \zeta(1) = x \otimes y - y \otimes x.$$

LEMMA 3.1. *If L and M are $K\Gamma$ -modules then there are inverse bijections*

$$\text{Hom}_{K\Gamma}(L, V \otimes M) \xrightleftharpoons[\theta \mapsto \theta^\#]{\phi \mapsto \phi^\flat} \text{Hom}_{K\Gamma}(V \otimes L, M)$$

where $\theta^\# = (\langle, \rangle \otimes 1)(1 \otimes \theta)$ and $\phi^\flat = (1 \otimes \phi)(\zeta \otimes 1)$.

PROOF. It follows from the Tensor-Hom adjunction (see 4.4 in the Preliminary Chapter 2). And also we can obtain that $(\theta^\#)^\flat = \theta$ and $(\phi^\flat)^\# = \phi$ by a direct calculation:

$$\begin{aligned} (\theta^\#)^\flat &= (1_V \otimes \theta^\#)(\zeta \otimes 1_L) = (1_V \otimes [(\langle, \rangle \otimes 1_M)(1_V \otimes \theta)])(\zeta \otimes 1_L) \\ &= (1_V \otimes \langle, \rangle \otimes 1_M)(1_V \otimes 1_V \otimes \theta)(\zeta \otimes 1_L) \end{aligned}$$

$\Rightarrow (\theta^\#)^\flat : L \rightarrow V \otimes M$ and

$$\begin{aligned} (\theta^\#)^\flat(l) &= (1_V \otimes \langle, \rangle \otimes 1_M)(1_V \otimes 1_V \otimes \theta)(\zeta \otimes 1_L)(l) \\ &= (1_V \otimes \langle, \rangle \otimes 1_M)(1_V \otimes 1_V \otimes \theta)(\zeta \otimes 1_L)(1 \otimes l) = (1 \otimes \langle, \rangle \otimes 1)(1_V \otimes 1_V \otimes \theta)((x \otimes y - y \otimes x) \otimes l) \\ &= (1_V \otimes \langle, \rangle \otimes 1_M)((x \otimes y - y \otimes x)\theta(l)). \end{aligned}$$

Let $\theta(l) = x \otimes m_1 + y \otimes m_2 \in V \otimes M$, with $m_1, m_2 \in M$. Then

$$\begin{aligned}
 (\theta^\#)^b(l) &= (1_V \otimes \langle, \rangle \otimes 1_M)((x \otimes y - y \otimes x)\theta(l)) \\
 &= (1_V \otimes \langle, \rangle \otimes 1_M)(x \otimes y \otimes x \otimes m_1 + x \otimes y \otimes y \otimes m_2 - y \otimes x \otimes x \otimes m_1 - y \otimes x \otimes y \otimes m_2) \\
 &= x \otimes \langle y, x \rangle \otimes m_1 + x \otimes \langle y, y \rangle \otimes m_2 - y \otimes \langle x, x \rangle \otimes m_1 - y \otimes \langle x, y \rangle \otimes m_2 \\
 &= x \otimes 1 \otimes m_1 + 0 + 0 - (-1)y \otimes 1 \otimes m_2 \leftrightarrow x \otimes m_1 + y \otimes m_2 = \theta(l)
 \end{aligned}$$

On the other hand

$$\begin{aligned}
 (\phi^b)^\# &= (\langle, \rangle \otimes 1_M)(1_V \otimes \phi^b) = (\langle, \rangle \otimes 1)(1_V \otimes [(1_V \otimes \phi)(\zeta \otimes 1_L)]) \\
 &= (\langle, \rangle \otimes 1_M)(1_V \otimes 1_V \otimes \phi)(1_V \otimes \zeta \otimes 1_L) \\
 &\Rightarrow (\phi^b)^\# : V \otimes L \rightarrow M
 \end{aligned} \tag{3.1}$$

and let $l_1, l_2 \in L$,

$$\begin{aligned}
 (\phi^b)^\#(x \otimes l_1 + y \otimes l_2) &= (\langle, \rangle \otimes 1_M)(1_V \otimes 1_V \otimes \phi)(1_V \otimes \zeta \otimes 1_L)(x \otimes l_1 + y \otimes l_2) \\
 &= (\langle, \rangle \otimes 1_M)(1_V \otimes 1_V \otimes \phi)(1_V \otimes \zeta \otimes 1_L)(x \otimes 1 \otimes l_1 + y \otimes 1 \otimes l_2) \\
 &= (\langle, \rangle \otimes 1_M)(1_V \otimes 1_V \otimes \phi)(x \otimes (x \otimes y - y \otimes x) \otimes l_1 + y \otimes (x \otimes y - y \otimes x) \otimes l_2) \\
 &= (\langle, \rangle \otimes 1_M)(1_V \otimes 1_V \otimes \phi)(x \otimes x \otimes y \otimes l_1 - x \otimes y \otimes x \otimes l_1 + y \otimes x \otimes y \otimes l_2 - y \otimes y \otimes x \otimes l_2) \\
 &= (\langle, \rangle \otimes 1_M)(x \otimes x \otimes \phi(y \otimes l_1) - x \otimes y \otimes \phi(x \otimes l_1) + y \otimes x \otimes \phi(y \otimes l_2) - y \otimes y \otimes \phi(x \otimes l_2)) \\
 &= \langle x, x \rangle \otimes \phi(x \otimes l_1) - \langle x, y \rangle \otimes \phi(y \otimes l_1) + \langle y, x \rangle \otimes \phi(x \otimes l_2) - \langle y, y \rangle \otimes \phi(y \otimes l_2) \\
 &= 0 - (-1) \otimes \phi(x \otimes l_1) + 1 \otimes \phi(y \otimes l_2) + 0 = 1 \otimes \phi(x \otimes l_1) + 1 \otimes \phi(y \otimes l_2) \\
 &\leftrightarrow \phi(x \otimes l_1 + y \otimes l_2).
 \end{aligned}$$

□

LEMMA 3.2. *To each arrow $a : i \rightarrow j$ in Q one can associate homomorphisms*

$$\theta_a : N_i \longrightarrow V \otimes N_j \text{ and } \phi_a : N_j \longrightarrow V \otimes N_i$$

in such a way that, for each vertex i ,

$$\sum_{\substack{a \in Q \\ h(a)=i}} (a \otimes \theta_a) \phi_a - \sum_{\substack{a \in Q \\ t(a)=i}} (1 \otimes \phi_a) \theta_a = \delta_i(\zeta \otimes 1)$$

as maps $N_i \longrightarrow V \otimes V \otimes N_i$; and the θ_a, ϕ_a ($a \in Q$) combine to give a basis for each of the spaces $\text{Hom}(N_i, V \otimes N_j)$.

PROOF. We will analyze separately the cases where Q is type A_n^1 and where Q is type D_n^1 , E_6^1 , E_7^1 or E_8^1 .

Case 1: Suppose that Q is of type A_n^1 (see 7.2.5 in section 7.2 in preliminary chapter 2). Let u, v be a basis of the natural representation V with respect to which the action of Γ is diagonal. By scaling u one can ensure that $u \otimes v - v \otimes u = x \otimes y - y \otimes x$. Thus, the generator g of cyclic group $\Gamma = \langle g \rangle$ of order $n+1$, acting via $g \cdot u = e^{\frac{2\pi i}{n+1}} u$ and $g \cdot v = e^{-\frac{2\pi i}{n+1}} v$ (here i is the imaginary unity).

We have $n+1$ irreducible representations of Γ , with g acting as multiplication by $(e^{\frac{2\pi i}{n+1}})^j$, with $j \in \{0, 1, \dots, n+1\}$. Then, by proposition 7.33 in section 7.1 of the preliminary chapter 2, those $n+1$ are the only irreducible representations of Γ , called N_j in our notation.

Now, the arrows a_i in Q connect vertices i and $i+1$ (0 is the vertex $n+1$). According to the orientation:

If $a_i : i \rightarrow i+1$ let $\theta_{a_i}(z) = v \otimes z$ and $\phi_{a_i}(z) = -u \otimes z$. Then we have

$$\begin{aligned} \sum_{\substack{a \in Q \\ ha=i}} (1 \otimes \theta_a) \phi_a(z) - \sum_{\substack{a \in Q \\ ta=i}} (1 \otimes \phi_a) \theta_a(z) &= (1 \otimes \theta_{a_{i-1}}) \phi_{a_{i-1}}(z) - (1 \otimes \phi_{a_i}) \theta_{a_i}(z) \\ &= (1 \otimes \theta_{a_{i-1}})(-u \otimes z) - (1 \otimes \phi_{a_i}) \theta_{a_i}(z) \\ &= -u \otimes \theta_{a_{i-1}}(z) - v \otimes \phi_{a_i}(z) \\ &= -u \otimes v \otimes z - v \otimes (-u) \otimes z = 1(\zeta \otimes 1)(1 \otimes z) \end{aligned}$$

If $a_i : i+1 \rightarrow i$, let $\theta_{a_i}(z) = u \otimes z$ and $\phi_{a_i}(z) = v \otimes z$. Then we have

$$\begin{aligned} \sum_{\substack{a \in Q \\ ha=i}} (1 \otimes \theta_a) \phi_a(z) - \sum_{\substack{a \in Q \\ ta=i}} (1 \otimes \phi_a) \theta_a(z) &= (1 \otimes \theta_{a_i}) \phi_{a_i}(z) - (1 \otimes \phi_{a_{i-1}}) \theta_{a_{i-1}}(z) \\ &= (1 \otimes \theta_{a_i})(v \otimes z) - (1 \otimes \phi_{a_{i-1}})(u \otimes z) \\ &= v \otimes \theta_{a_i}(z) - u \otimes \phi_{a_{i-1}}(z) \\ &= v \otimes u \otimes z - u \otimes v \otimes z = 1(\zeta \otimes 1)(1 \otimes z) \end{aligned}$$

Case 2: Now suppose that Q is of type D_n^1 , E_6^1 , E_7^1 or E_8^1 (see 7.2.5 in section 7.2 in preliminary chapter 2). In this case Q is a tree. For each arrow $a \in Q$ choose a non-zero homomorphism

$$\theta_a : N_{t(a)} \rightarrow V \otimes N_{h(a)},$$

and a non-zero homomorphism

$$\phi_a : N_{h(a)} \rightarrow V \otimes N_{t(a)}.$$

Note, that $V \otimes N_i$ are multiplicity-free, it means that the irreducible representations N_i appears in $V \otimes N_i$ once at most. And recall that $\phi_a^\# : V \otimes N_{h(a)} \rightarrow N_{t(a)}$ and $\theta_a^\# : V \otimes N_{t(a)} \rightarrow N_{h(a)}$. Then, the map $\phi_a^\# \theta_a : N_{t(a)} \rightarrow N_{t(a)}$ is a non-zero endomorphism of $N_{t(a)}$ and the map $\theta_a^\# \phi_a : N_{h(a)} \rightarrow N_{h(a)}$ a non-zero endomorphism of $N_{h(a)}$.

Now, by rescaling θ_a , one can simultaneously rescale these two endomorphism. Since, Q is a tree, and $N_{t(a)}$ and $N_{h(a)}$ are irreducible representation, by Schur's lemma (see 7.18), $\phi_a^\# \theta_a$ and $\theta_a^\# \theta_a$ are scalars, it means, that there are scalars m_i for each vertex i such that the following equations hold

$$\phi_a^\# \theta_a = m_{h(a)} 1_{N_{t(a)}}, \quad (3.2)$$

$$\theta_a^\# \phi_a = -m_{t(a)} 1_{N_{h(a)}}, \quad (3.3)$$

for every arrow $a \in Q$.

One proceeds as follows: First fix the value of m_i for one vertex. At this moment, the piece of Q consisting of vertex i is known. Since Q is of type D_n^1 , E_6^1 , E_7^1 or E_8^1 , there is a arrow a connecting the vertex i in the piece to a vertex j outside. Without lost of generality, assume that $a : i \rightarrow j$. Then the equation 3.3 are determined by $m_i = m_{ta}$ by rescaling θ_a . Therefore, are fixed θ_a and ϕ_a and now, we obtain $m_j = m_{ha}$ from the equation 3.2. We can recursively define non-zero scalars m_i such that 3.2 and 3.3 holds for all $i \in I$ and all arrow $a \in Q$.

Thus, $\theta_a \phi_a^\#$ is m_{ha} times the projection of $V \otimes N_{ha}$ onto N_{ta} and $\phi_a \theta_a^\#$ is m_{ta} times the projection of $V \otimes N_{ta}$ onto N_{ha} .

Let $\pi_{ta} : V \otimes N_{ha} \rightarrow N_{ta}$ and $\pi_{ha} : V \otimes N_{ta} \rightarrow N_{ha}$ be the canonical projections. Then for every $i \in I$:

$$\begin{aligned} \sum_{\substack{a \in Q \\ ha=i}} \theta_a \phi_a^\# - \sum_{\substack{a \in Q \\ ta=i}} \phi_a \theta_a^\# &= \sum_{\substack{a \in Q \\ ha=i}} m_{ha} \pi_{ta} - \sum_{\substack{a \in Q \\ ta=i}} (-1) m_{ta} \pi_{ha} \\ &= m_i \left(\sum_{\substack{a \in Q \\ ha=i}} \pi_{ta} + \sum_{\substack{a \in Q \\ ta=i}} \pi_{ha} \right) = m_i 1_{V \otimes N_i}. \end{aligned}$$

Let's call this map $\psi := m_i 1_{V \otimes N_i}$. Then considering $(1_V \otimes \psi)(\zeta \otimes 1_{N_i}) : N_i \rightarrow V \otimes V \otimes N_i$ and using Lemma 3.1 one obtains:

$$\begin{aligned} \sum_{\substack{a \in Q \\ ha=i}} (1_V \otimes \theta_a) \phi_a - \sum_{\substack{a \in Q \\ ta=i}} (1_V \otimes \phi_a) \theta_a &= \sum_{\substack{a \in Q \\ ha=i}} (1_V \otimes \theta_a) (1_V \otimes \phi_a^\#) (\zeta \otimes 1_{N_{ha}}) - \sum_{\substack{a \in Q \\ ta=i}} (1_V \otimes \phi_a) (1_V \otimes \theta_a^\#) (\zeta \otimes 1_{N_{ta}}) \\ &= \left(\sum_{\substack{a \in Q \\ ha=i}} (1_V \otimes \theta_a) (1_V \otimes \phi_a^\#) - \sum_{\substack{a \in Q \\ ta=i}} (1_V \otimes \phi_a) (1_V \otimes \theta_a^\#) \right) (\zeta \otimes 1_{N_i}) \\ &= \left(\sum_{\substack{a \in Q \\ ha=i}} (1_V \otimes \theta_a \phi_a^\#) - \sum_{\substack{a \in Q \\ ta=i}} (1_V \otimes \phi_a \theta_a^\#) \right) (\zeta \otimes 1_{N_i}) \end{aligned}$$

$$\begin{aligned}
&= \left(1_V \otimes \left(\sum_{\substack{a \in Q \\ ha=i}} \theta_a \phi_a^\# - \sum_{\substack{a \in Q \\ ta=i}} \phi_a \theta_a^\# \right)\right) (\zeta \otimes 1_{N_i}) \\
&= (1_V \otimes m_1 1_{V \otimes N_i}) (\zeta \otimes 1_{N_i}) = (1_V \otimes 1_V \otimes 1_{N_i}) m_i (\zeta \otimes 1_{N_i}) \\
&= m_i (\zeta \otimes 1_{N_i}). \tag{3.4}
\end{aligned}$$

Now, composing this with the map $\langle, \rangle \otimes 1_{N_i} : V \otimes V \otimes N_i \rightarrow N_i$, the resulting endomorphism of N_i is multiplication by $-2m_i$: Let $n \in N_i$, we can identify n with $1 \otimes n \in K \otimes N_i$, then

$$\begin{aligned}
(\langle, \rangle \otimes 1_{N_i}) m_i (\zeta \otimes 1_{N_i}) (1 \otimes n) &= (\langle, \rangle \otimes 1_{N_i}) (m_i x \otimes y \otimes n - m_i y \otimes x \otimes n) \\
&= -2m_i \otimes n \leftrightarrow -2m_i n
\end{aligned}$$

From 3.4 we have,

$$\begin{aligned}
&(\langle, \rangle \otimes 1_{N_i}) \left(\sum_{\substack{a \in Q \\ ha=i}} (1_V \otimes \theta_a) \phi_a - \sum_{\substack{a \in Q \\ ta=i}} (1_V \otimes \phi_a) \theta_a \right) = (\langle, \rangle \otimes 1_{N_i}) m_i (\zeta \otimes 1_{N_i}) \\
&\sum_{\substack{a \in Q \\ ha=i}} (\langle, \rangle \otimes 1_{N_i}) (1_V \otimes \theta_a) \phi_a - \sum_{\substack{a \in Q \\ ta=i}} (\langle, \rangle \otimes 1_{N_i}) (1_V \otimes \phi_a) \theta_a = -2m_i 1_{N_i} \\
&\sum_{\substack{a \in Q \\ ha=i}} \theta_a^\# \phi_a - \sum_{\substack{a \in Q \\ ta=i}} \phi_a^\# \theta_a = -2m_i 1_{N_i} \\
&- \sum_{\substack{a \in Q \\ ha=i}} m_{ta} 1_{N_i} - \sum_{\substack{a \in Q \\ ta=i}} m_{ha} 1_{N_i} = -2m_i 1_{N_i} \\
&\Leftrightarrow - \sum_{\substack{a \in Q \\ ha=i}} m_{ta} - \sum_{\substack{a \in Q \\ ta=i}} m_{ha} = -2m_i
\end{aligned}$$

Now, using the symmetric bilinear form $(\cdot, \cdot)$ defined in section 2 from the Ringel form, consider the vector $(m_i)_{i \in I} = m \in K^I$ and the coordinate vector e^i , then

$$\begin{aligned}
(m, e^i) &= \langle m, e^i \rangle + \langle e^i, m \rangle = \sum_{j \in I} m_j e_j^i - \sum_{a \in Q} m_{ta} e_{ha}^i + \sum_{j \in I} m_j e_j^i - \sum_{a \in Q} m_{ha} e_{ta}^i \\
&= 2m_i - \sum_{\substack{a \in Q \\ ha=i}} m_{ta} - \sum_{\substack{a \in Q \\ ta=i}} m_{ha} = 2m_i - 2m_i = 0
\end{aligned}$$

So, $(m, e^i) = 0$ for all i , then, by definition, m belongs to the radical of the form $(\cdot, \cdot)$. Since Q is an affine quiver, this forces m to be a multiple of δ , because δ is the basis of the radical of $(\cdot, \cdot)$ (see [26]). The assertion follows by rescaling all θ_a . $\square$

In order to understand the following lemma and the main result of this section (Theorem 3.8) and its consequences, we need the definition of the Morita equivalence:

DEFINITION 3.3. A left R -module M is called a *generator* for R if $MM^* = R$. (see definition 5.3, for right R -modules, in section 3.5.3 of [31])

DEFINITION 3.4. A left R -module M is called a *progenerator* if and only if M is projective and generator for R . (see definition 5.4, for right R -modules, in section 3.5.3 of [31])

DEFINITION 3.5. Two rings R and S are *Morita equivalent* if there is a progenerator M such that $S \cong \text{End}_R(M)$. We write $R \stackrel{M}{\sim} S$. (see definition 5.5, for right R -modules, in chapter 3 of [31])

REMARK 3.6. Morita equivalence arose first in terms of category equivalence. Indeed, one can define $R \stackrel{M}{\sim} S$ to mean there is a category equivalence between their categories of left modules; $F: R\text{-mod} \rightarrow S\text{-mod}$. Given such an F it follows that $F(R)_S$ is a progenerator with $\text{End}(F(R)_S) \cong R$. It turns out that the two definitions agree. (see 7.5 in chapter 3 of [31])

LEMMA 3.7. *To each arrow $a: i \rightarrow j$ in Q one can associate elements*

$$\theta_a \in f_i C_1 f_j \text{ and } \phi_a \in f_j C_1 f_i$$

in such a way that, for each vertex i ,

$$\sum_{\substack{a \in Q \\ h(a)=i}} \theta_a \phi_a - \sum_{\substack{a \in Q \\ t(a)=i}} \phi_a \theta_a = \delta_i f_j (xy - yx)$$

and the ϕ_a, θ_a ($a \in Q$) combine to give a basis for each of the spaces $f_i C_1 f_j$.

PROOF. Let $C = K\langle x, y \rangle \rtimes \Gamma = K\langle x, y \rangle \otimes_K \Gamma$, graded with x, y is degree 1 and elements of Γ in degree 0. Let $V = \text{span}\{x, y\}$ the natural representation of Γ considered above. Since $\{V^{\otimes d}\}$ is a graduation for $K\langle x, y \rangle$, then $\{V^{\otimes d} \otimes \Gamma\}$ is a graduation for C . Thus C is the tensor algebra over Γ of the bimodule $V \otimes \Gamma$ (see 2.2 in the preliminary chapter 2). Thus $C_0 = \Gamma$ and $C_1 = V \otimes \Gamma$. Remark that we consider $V \otimes \Gamma$ as a bimodule with a diagonal left action of Γ over $V \otimes \Gamma$ and the right action of Γ over $V \otimes \Gamma$ given by multiplication only on the second term in the tensor product.

Since the idempotent f is such that $fN_i \neq 0$ for all i , induces a Morita equivalence (see Example IV of 3.6.3 in [31]), it follows that fCf is the tensor algebra over $f\Gamma f = fC_0f$ of $f(V \otimes \Gamma)f = fC_1f$.

Now,

$$\begin{aligned} fC_0f = f\Gamma f &\stackrel{(a)}{\cong} \bigoplus_{i \in I} fN_i \cong \bigoplus_{i \in I} f_i N_i \stackrel{(b)}{\cong} \bigoplus_{i \in I} f_i \Gamma f_i \stackrel{(c)}{\cong} \bigoplus_{i \in I} \text{Hom}_{\Gamma}(f_i \Gamma, f_i \Gamma) \\ &\stackrel{(d)}{\cong} \bigoplus_{i \in I} \text{Hom}_{\Gamma}(N_i, N_i) \stackrel{(e)}{\cong} \bigoplus_{i \in I} K \end{aligned}$$

For (a), (b) and (d): Use that $f\Gamma f_i \cong N_i$ and $f = f_0 + \dots + f_n$.

For (c): Use a more general fact, let R be a ring, $M \in R\text{-mod}$ and $f^2 = f \in R$ we have the inverse bijections:

$$\begin{aligned} fM &\rightarrow \text{Hom}_R(Rf, M), \\ fm &\mapsto [rf \mapsto rffm = rfm], \end{aligned}$$

and

$$\begin{aligned} \text{Hom}_R(Rf, M) &\rightarrow fM, \\ \varphi &\mapsto \varphi(f) = \varphi(f^2) = f\varphi(f). \end{aligned}$$

In the case of the identification (c), $K\Gamma$ is the ring, $f_i \in K\Gamma$ is the idempotent and $K\Gamma f_i \in K\Gamma\text{-mod}$.

For (e): Apply the Schur's Lemma (see 7.18 in section 7.1 of the preliminary chapter 2), remark that K is algebraically closed.

Thus, $fC_0f \cong K^I$.

And, we have the following identifications:

$$f_i C_1 f_j \underset{(1)}{\cong} \text{Hom}_{K\Gamma}(K\Gamma f_i, C_1 f_j) \cong \text{Hom}_{K\Gamma}(K\Gamma f_i, V \otimes K\Gamma f_j) \underset{(2)}{\cong} \text{Hom}_{K\Gamma}(N_i, V \otimes N_j).$$

For (1): Use the same general fact that in (c). In this case, the ring is $K\Gamma$, $f_i^2 = f_i \in K\Gamma$ and $C_1 f_j = V \otimes K\Gamma f_j \in K\Gamma\text{-mod}$, so:

$$f_i C_1 f_j \cong \text{Hom}_{K\Gamma}(K\Gamma f_i, C_1 f_j).$$

For (2): Recall that $K\Gamma f_i \cong N_i$ and the right action of $K\Gamma$ insures that $C_1 f_j = (V \otimes K\Gamma) f_j = V \otimes (K\Gamma) f_j$.

Thus, we can restate the Lemma 3.2 as in the statement. $\square$

THEOREM 3.8. *There is an isomorphism $fS^{R,\Lambda}f \cong \Pi^{R,\Lambda}(Q)$. Under this isomorphism e corresponds to the trivial path e_0 .*

PROOF. There is an isomorphism:

$$\begin{aligned} K\overline{Q} &\rightarrow fCf, \\ e_i &\mapsto f_i, \\ [a : i \rightarrow j] &\mapsto \theta_a \in f_i C_1 f_j, \\ [a^* : j \rightarrow i] &\mapsto \phi_a \in f_j C_1 f_i. \end{aligned}$$

Let C^R be the induced R -algebra $R\langle x, y \rangle \otimes_R R\Gamma$, then we have an isomorphism $R\overline{Q} \rightarrow fC^Rf$.

Observe that $a(xy - yx) - a\Lambda = a(xy - yx) - \Lambda a = (xy - yx)a - \lambda a$ for all $a \in K\Gamma$. The first equality comes from the fact that $\Lambda \in Z(R\Gamma) \subset Z(K\Gamma)$, and the second equality it follows from

that $x \otimes y - y \otimes x$ belongs to a trivial submodule of $V \otimes V$ by using ζ , and because, let $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $g \cdot x = ax + dy$ and $g \cdot y = cx + dy$, then

$$g(xy - yx)g^{-1} = gxg^{-1}gyg^{-1} - gyg^{-1}gxg^{-1} = (ax + by)(cx + dy) - (cx + dy)(ax + by) = xy - yx.$$

Now, let $I = (xy - yx - \Lambda)$ be an ideal in C^R , then $J = I \cap fC^R f$ is generated as an ideal in $fC^R f$ by $f(xy - yx - \Lambda)f = ff(xy - yx - \Lambda) = f(xy - yx - \Lambda)$. Since f corresponds to a Morita equivalence (see Example IV of 3.6.3 in [31]), so there are $a_i, b_i \in K\Gamma$ with $\sum_{i \in I} a_i f b_i = 1$ (see Lemma 5.2 in 3.5.2 of [31]). Then the ideal in C^R generated by $f(xy - yx - \Lambda)$ contains

$$\sum_{i \in I} a_i (xy - yx - \Lambda) f b_i = (xy - yx - \Lambda) \sum_{i \in I} a_i f b_i = (xy - yx - \Lambda).$$

Then $f_i(xy - yx - \Lambda) \in J$. And

$$J = (f_0(xy - yx - \Lambda) + \dots + f_n(xy - yx - \Lambda))$$

which implies that $J \subseteq (f_0(xy - yx - \Lambda), \dots, f_n(xy - yx - \Lambda))$. Thus, J is generated by the elements $f_i(xy - yx - \Lambda)$. Identifying Λ with an element in R^I , $f_i(xy - yx - \Lambda) \leftrightarrow f_i(xy - yx - \Lambda_i / \delta_i)$.

Now, from the Lemma 3.7:

$$\begin{aligned} f_i(xy - yx - \Lambda_i / \delta_i) &= f_i(xy - yx) - f_i \Lambda_i / \delta_i = \frac{1}{\delta_i} \left(\sum_{\substack{a \in Q \\ ha=i}} \theta_a \phi_a - \sum_{\substack{a \in Q \\ ta=i}} \phi_a \theta_a \right) - f_i \Lambda_i / \delta_i \\ &= \frac{1}{\delta_i} \left(\sum_{\substack{a \in Q \\ ha=i}} \theta_a \phi_a - \sum_{\substack{a \in Q \\ ta=i}} \phi_a \theta_a - f_i \Lambda_i \right), \end{aligned}$$

which is the defining relations for $\Pi^{R, \Lambda}(Q)$ via the isomorphism $R\bar{Q} \cong fCf$. $\square$

There is an immediate corollary of the previous theorem:

COROLLARY 3.9. $\mathcal{S}^{\mathcal{R}, \Lambda}$ is Morita equivalent to $\Pi^{R, \Lambda}(Q)$ and $\mathcal{O}^{R, \Lambda} \cong e_0 \Pi^{R, \Lambda}(Q) e_0$.

PROOF. First, since f induce a Morita equivalence (see Example IV of 3.6.3 in [31]), $\mathcal{S}^{\Lambda} \overset{M}{\sim} \Pi^{\Lambda}(Q)$.

Now,

$$\mathcal{O}^{\Lambda} = e \mathcal{S}^{\Lambda} e = f e f \mathcal{S}^{\Lambda} f e f \overset{3.8}{\cong} f e \Pi^{\Lambda}(Q) e f \cong f f_0 \Pi^{\Lambda}(Q) f_0 f = f_0 \Pi^{\Lambda}(Q) f_0 \cong e_0 \Pi^{\Lambda}(Q) e_0.$$

$\square$

We have a consequence of this corollary for the deformed preprojective algebras:

COROLLARY 3.10. If Q is an extended Dynkin quiver with vertex set I and $\lambda \in K^I$ then the natural map $\Pi(Q) \rightarrow \text{gr} \Pi^{\lambda}(Q)$ is an isomorphism. and $\Pi^{\lambda}(Q)$ is of GK dimension 2.

PROOF. Consider an extended Dynkin quiver, then it arises by orienting the McKay graph of some Kleinian singularity. By the Corollary 3.9 $\Pi^\lambda(Q)$ is Morita equivalent to $\mathcal{S}^\lambda$, so $\Pi^\lambda(Q)$ is also a ring of GK dimension 2. Also, the map $\Pi(Q) \rightarrow \text{gr}\Pi^\lambda(Q)$ is surjective, by lemma 2.4. So this surjection must also be injective since every proper quotient of $\Pi(Q)$ has GK dimension strictly less than 2 (see Theorem 3.13 in section 3 of the preliminary chapter 2, for more details see Theorem, of Bergman, as VI.4.1 in [1] or as 2.5 in [27]).

Remark that in the case of the quiver consisting of one vertex and one loop, $\Pi(Q)$ is the polynomial algebra in 2 variables $K[x, y]$ and $\Pi^\lambda(Q)$ is the Weyl algebra of differential operator in 2 variables with parameter λ , $\frac{K[x, y]}{(xy - yx - \lambda)}$. $\square$

LEMMA 3.11. *Let Q be an extended Dynkin quivers, then $\Pi^\lambda(Q)$ is a Noetherian ring.*

PROOF. We know from the Lemma 1.4 in section 1, that $\mathcal{S}^\lambda$ is a Noetherian ring. Now, since being Noetherian is a Morita invariant property (see Proposition 3.5.10 in [31]), and by Corollary 3.9 $\mathcal{S}^\lambda \stackrel{M}{\sim} \Pi^\lambda(Q)$, then Π^λ is Noetherian. $\square$

3.1. Example using the cyclic group of order 2. Part 2. Consider the field $K = \mathbb{C}$ and the finite group C_2 of $\text{SL}_2(\mathbb{C})$, just as in the example 1.1 in section 1.

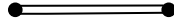
Let (ρ, V) be the 2-dimsional natural representation of C_2 given by $V = \mathbb{C}^2$ and

$$\rho : C_2 \rightarrow \text{SL}_2(\mathbb{C}), 1 \mapsto \begin{pmatrix} \zeta^0 & 0 \\ 0 & \zeta^0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \sigma \mapsto \begin{pmatrix} \zeta^1 & 0 \\ 0 & \zeta^1 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}.$$

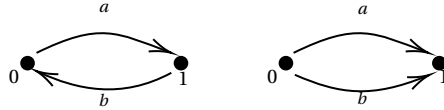
Let χ be the character of (ρ, V) , then $\chi(1) = 2$, $\chi(\sigma) = -2$. Let $\chi_{V_i \otimes V} = \chi_i \chi$ be the character of the representation $(\rho_i \otimes \rho, V_i \otimes V)$, $i = 0, 1$. We concluded that

$$\rho_0 \otimes \rho = \rho_1 \oplus \rho_1 \text{ and } \rho_1 \otimes \rho = \rho_0 \oplus \rho_0.$$

Then, the extended Dynkin quiver for C_2 is A_1^1 :



There are two possible orientations for A_1^1 :



A cycle, to the left, and the Kronecker quiver, to the right. We will consider the Kronecker quiver, which we will call Q . The problem of classification of the finite-dimensional representations of Q , up to isomorphism, is also the problem of classification of linear transformations $S, T : V_0 \rightarrow V_1$, up to left multiplication by $X \in \text{GL}(V_0)$ and right multiplication by $Y \in \text{GL}(V_1)$. In the section 7.1 of [11] the classification of the indecomposable representations of Q can be seen.

Now, we construct the deformed preprojective algebra of Q for some $\lambda \in \mathbb{C}^2$. Take, for example $\lambda = (1, -1)$, then

$$\Pi^\lambda(Q) = \mathbb{C}\overline{Q}/(-a^*a - b^*b - \lambda_0 e_0, aa^* + bb^* - \lambda_1 e_1) = \mathbb{C}\overline{Q}/(-a^*a - b^*b - e_0, aa^* + bb^* + e_1).$$

Now, $\lambda = (1, -1) \in \mathbb{C}^2$ is identified with $\frac{\lambda_0}{\delta_0}\eta_0 + \frac{\lambda_1}{\delta_1}\eta_1 \in Z(\mathbb{C}C_2)$.

Let's calculate the fundamental root δ for Q , for which one of its components is 1, take this component as $\delta_0 = 1$.

$$0 = (\delta, \delta) = 2\langle \delta, \delta, \rangle = 2\left(\sum_{i=0,1} \delta_i \delta_i - \sum_{f=a,b} \delta_{t(f)} \delta_{h(f)}\right) = 2(\delta_0^2 + \delta_1^2) - 2(\delta_0 \delta_1 + \delta_0 \delta_1) = 2 + 2\delta_1^2 - 4\delta_1$$

$$0 = (\delta_1 - 1)^2$$

$$\Rightarrow \delta_1 = 1.$$

So, the fundamental root for Q is $\delta = (1, 1)$.

By the other hand, the primitive idempotent elements of $\mathbb{C}C_2$, η_i , are:

$$\eta_0 = \frac{\dim V_0}{|C_2|} \sum_{\gamma \in \Gamma} \overline{\chi_0(\gamma)} \gamma = \frac{1}{2}(1 \cdot 1_\Gamma + 1 \cdot \sigma) = \frac{1}{2}(1 + \sigma) = e$$

and

$$\eta_1 = \frac{\dim V_1}{|C_2|} \sum_{\gamma \in \Gamma} \overline{\chi_1(\gamma)} \gamma = \frac{1}{2}(1 \cdot 1_\Gamma - 1 \cdot \sigma) = \frac{1}{2}(1 - \sigma).$$

Then, $\lambda = (1, -1)$ is identify with $\frac{1}{1} \frac{1}{2}(1 + \sigma) + \frac{-1}{1} \frac{1}{2}(1 - \sigma) = \frac{1+\sigma-1+\sigma}{2} = \sigma$.

4. Representations of deformed preprojective algebras.

Let K be an arbitrary field and Q is an arbitrary quiver with vertex set I .

In this section, we begin to study the representations of the deformed preprojective algebra $\Pi^\lambda(Q)$ of a quiver Q .

DEFINITION 4.1.

- The variety of representation of Q with dimension vector $\alpha \in \mathbb{Z}_{\geq 0}^I$ is given by the elements of the vector space

$$\text{Rep}(Q, \alpha) := \bigoplus_{a \in Q} \text{Hom}(K^{\alpha_{t(a)}}, K^{\alpha_{h(a)}}).$$

- For representations of $\overline{Q}$ with dimension vector α we write

$$\text{Rep}(\overline{Q}, \alpha) := \text{Rep}(Q, \alpha) \oplus \text{Rep}(Q^{op}, \alpha) = \bigoplus_{a:i \rightarrow j} \text{Hom}(K^{\alpha_i}, K^{\alpha_j}) \oplus \text{Hom}(K^{\alpha_j}, K^{\alpha_i}),$$

when Q^{op} is the opposite quiver to Q .

- Also, we define the following algebra

$$\text{End}(\alpha) := \bigoplus_{i \in I} \text{End}(K^{\alpha_i}).$$

LEMMA 4.2. *If $\lambda \cdot \alpha \neq 0$ there is no representation of Π^λ of dimension vector α .*

PROOF. Assume that there is a representation X of Π^λ of dimension vector α . Then X satisfies the condition

$$0 = c - \lambda \iff 0 = \sum_{\substack{a \in Q \\ h(a)=i}} x_a x_{a^*} - \sum_{\substack{a \in Q \\ t(a)=i}} x_{a^*} x_a - \lambda_i e_i \text{ for all } i \in I,$$

which implies

$$0 = \text{tr} \left(\sum_{\substack{a \in Q \\ h(a)=i}} x_a x_{a^*} - \sum_{\substack{a \in Q \\ t(a)=i}} x_{a^*} x_a - \lambda_i e_i \right) = \text{tr} \left(\sum_{\substack{a \in Q \\ h(a)=i}} x_a x_{a^*} - \sum_{\substack{a \in Q \\ t(a)=i}} x_{a^*} x_a \right) - \text{tr}(\lambda_i e_i) = \text{tr}(\lambda_i e_i) = \lambda_i \cdot \alpha_i$$

$$\iff 0 = \lambda \cdot \alpha.$$

But, this is impossible because $\lambda \cdot \alpha \neq 0$. $\square$

LEMMA 4.3. *If $x = (x_a)_{a \in Q} \in \text{Rep}(Q, \alpha)$ and X is the corresponding KQ -module, then there is an exact sequence*

$$0 \rightarrow D\text{Ext}^1(X, X) \rightarrow \text{Rep}(Q^{op}, \alpha) \xrightarrow{c} \text{End}(\alpha) \rightarrow D\text{End}(X) \rightarrow 0,$$

where c sends $(y_{a^*}) \in \text{Rep}(Q^{op}, \alpha)$ to $\sum_{a \in Q} [x_a, y_{a^*}]$ and D is duality with the field.

PROOF. We consider the map f defined by

$$f : \text{End}(\alpha) \rightarrow \text{Rep}(Q, \alpha)$$

$$(\theta_i)_{i \in I} \mapsto (\theta_{h(a)} x_a - x_a \theta_{t(a)})_{a \in Q}.$$

Note that $\ker(f) = \text{End}(X)$ and using the standard projective resolution of X (see Proposition 7.58 in section 7.2 of preliminary chapter 2, for more details see section 3 in chapter 7 of [20]), $\text{coker}(f) \cong \text{Ext}^1(X, X)$.

Thus, the following sequence is exact

$$0 \rightarrow \text{End}(X) \xrightarrow{\iota} \text{End}(\alpha) \xrightarrow{f} \text{Rep}(Q, \alpha) \xrightarrow{\zeta} \text{Ext}^1(X, X) \rightarrow 0,$$

where ι is the inclusion map and ζ is given by the isomorphism $\text{coker}(f) \cong \text{Ext}^1(X, X)$. Now, the dual of this sequence is the following exact sequence

$$0 \leftarrow D\text{End}(X) \leftarrow D\text{End}(\alpha) \leftarrow D\text{Rep}(Q, \alpha) \leftarrow D\text{Ext}^1(X, X) \leftarrow 0,$$

and because

$$D\text{Rep}(Q, \alpha) = D \bigoplus_{a \in Q} \text{Hom}(K^{\alpha_{t(a)}}, K^{\alpha_{h(a)}}) = \bigoplus_{a \in Q} \text{Hom}(K^{\alpha_{h(a)}}, K^{\alpha_{t(a)}})$$

$$= \bigoplus_{a^* \in Q} \text{Hom}(K^{\alpha_{t(a^*)}}, K^{\alpha_{h(a^*)}}) = \text{Rep}(Q^{op}, \alpha)$$

and $D\text{End}(\alpha) = \text{End}(\alpha)$ via the trace form, we obtain the required exact sequence. $\square$

THEOREM 4.4. *If $\lambda \cdot \alpha = 0$ and there is a representation of Q of dimension vector α and endomorphism algebra K , then there is a representation of Π^λ of dimension vector α .*

PROOF. If $x \in \text{Rep}(Q, \alpha)$ has an endomorphism algebra K , then the map $c : \text{Rep}(Q^{op}, \alpha) \rightarrow \text{End}(\alpha)$, of the previous lemma, has one-dimensional cokernel, because its cokernel is isomorphic to $D\text{End}(X)$. Since the image of c is contained in $\text{End}(\alpha)_0$ and $\dim(\text{End}(\alpha)_0) = \dim(\text{End}(\alpha)) - 1$, c has image $\text{End}(\alpha)_0$.

Now, if λ is identified with a central element of the ring $\text{End}(\alpha)$, then the condition $\lambda \cdot \alpha = 0$ implies that $\lambda \in \text{End}(\alpha)_0$. Thus $c^{-1}(\lambda)$ is not empty. Now, any element of this fiber extends x to a representation of Π^λ . $\square$

In particular, we can apply the previous results for extended Dynkin quivers and the dimension vector δ . See, for example, the following lemma (in [12]).

LEMMA 4.5. *If Q is an extended Dynkin quiver and δ is the minimal positive vector with $(\delta, \delta) = 0$ then there is a representation of Q of dimension vector δ whose endomorphism algebra is K . In addition, if K is algebraically closed then the representations with endomorphism algebra K form a dense open subset in $\text{Rep}(Q, \delta)$, and its complement is the union of only finitely many $\text{GL}(\delta)$ -orbits.*

4.1. Example using the cyclic group of order 2. Part 3. According to what was established in parts 1 and 2 of our example (see 1.1 in section 1 and 3.1 in section 3), we will continue using $K = \mathbb{C}$ as the base field, $\Gamma = C_2$ and Q the Kronecker quiver.

In order to exemplify the Lemma 4.2, let $\lambda \in \mathbb{C}^2$ such that $\lambda \cdot \alpha = 0$, for some root α and let M be a representation of Q with dimension vector α and let's find a representation of $\Pi^\lambda(Q)$ of dimension vector α .

Let $\alpha = \delta + \phi \in \mathbb{Z}^2$, where $\delta = (1, 1)$ is the fundamental root for Q (see the Part 2 of our example 3.1) and $\phi = (1, 0)$ is a simple root for Q (see section 6). So, $\alpha = (1, 1) + (1, 0) = (2, 1)$. Let $\lambda \in \mathbb{C}^2$ such that $\lambda \cdot \alpha = 0$, then $\lambda_1 = -2\lambda_0$. Take, for example $\lambda = (1, -2)$, then

$$\Pi^\lambda(Q) = \mathbb{C}\overline{Q}/(-a^*a - b^*b - e_0, aa^* + bb^* + 2e_1).$$

Now, we exhibit a representation of $\overline{Q}$ with dimension vector $\alpha = (2, 1)$, which is also a representation for Π^λ .

Let $M \in \text{Rep}(\overline{Q})$ given by the C -vector spaces $M(0) = \mathbb{C}^2$, $M(1) = \mathbb{C}$ and the linear transformations:

$$M(a) : M(0) \rightarrow M(1), (v_0, v_1) \mapsto v_0,$$

$$M(b) : M(0) \rightarrow M(1), (v_0, v_1) \mapsto v_1,$$

$$M(a^*) : M(1) \rightarrow M(0), w \mapsto (-w, 0),$$

$$M(b^*) : M(1) \rightarrow M(0), w \mapsto (0, -w).$$

Remark that M is also a representation of Π^λ if and only if the ideal $(-a^*a - b^*b - e_0, aa^* + bb^* + 2e_1)$ annihilates M . Indeed, let $v + w \in M(0) \oplus M(1)$, where $v = (v_0, v_1)$, we have

$$\begin{aligned} (-a^*a - b^*b - e_0) \cdot (v + w) &= -a^*av - a^*aw - b^*bv - b^*bw - e_0v - e_0w \\ &= -a^*av - b^*bv - e_0v = -a^*v_0 - b^*v_1 - (v_0, v_1) \\ &= -(-v_0, 0) - (0, -v_1) - (v_0, v_1) = 0 \end{aligned}$$

and

$$\begin{aligned} (aa^* + bb^* + 2e_1) \cdot (v + w) &= aa^*v + aa^*w + bb^*v + bb^*w + 2e_1v + 2e_1w \\ &= aa^*w + bb^*w + 2e_1w = a(-w, 0) + (0, -w) + 2w \\ &= -w - w + 2w = 0 \end{aligned}$$

5. Reflection functors.

In this section we will define certain *reflection functors* which are the key tools for studying deformed preprojective algebras.

Let be K an arbitrary field and let be Q an arbitrary quiver with vertex set I .

DEFINITION 5.1. We say that a vertex i is *loopfree* if there are no arrows $a : i \rightarrow i$.

DEFINITION 5.2. In the case of a vertex i loopfree, the *simple reflection* $s_i : \mathbb{Z}^I \rightarrow \mathbb{Z}^I$ is defined by:

$$s_i(\alpha) = \alpha - (\alpha, \epsilon_i)\epsilon_i$$

where ϵ is the coordinate vector at i .

DEFINITION 5.3. Let be s_i the simple reflection of a loopfree vertex i , there is a *dual reflection* $r_i : K^I \rightarrow K^I$ with the property that

$$r_i\lambda \cdot \alpha = \lambda \cdot s_i\alpha$$

for all $\lambda \in K^I$ and all $\alpha \in \mathbb{Z}^I$. Explicitly, $(r_i\lambda)_j = \lambda_j - (\epsilon_i, \epsilon_j)\lambda_i$.

DEFINITION 5.4. The *Weyl group* W is the group of automorphisms of $\mathbb{Z}^I$ generated by the simple reflections for loopfree vertices.

It acts on K^I , with

$$(w\lambda) \cdot \alpha = \lambda \cdot (w^{-1}\alpha)$$

for all $w \in W$, $\lambda \in K^I$ and $\alpha \in \mathbb{Z}^I$.

DEFINITION 5.5. The *length* of an element $w \in W$, denoted by $l(w)$, is the smallest number n so that w is a product of n reflections by simple roots. In particular, a simple reflection has length one.

It follows immediately from the definition that $l(w^{-1}) = l(w)$ and that $l(ww'^{-1}) \leq l(w) + l(w')$, for all $w, w' \in W$.

THEOREM 5.6. *If i is a loopfree vertex and $\lambda_i \neq 0$, then there is an equivalence*

$$E_i : \Pi^\lambda(Q)\text{-mod} \longrightarrow \Pi^{r_i \lambda}(Q)\text{-mod}$$

which acts as the simple reflection s_i on dimension vectors.

PROOF. By Lemma 2.3 $\Pi^\lambda(Q)$ does not depend on the orientation of Q , then we can assume that no arrows have tail at i , in this case i is called a *sink* for Q .

First: We consider a $\Pi^\lambda(Q)$ -module V as a representation of $\overline{Q}$ with vector space V_j at vertex j and linear map V_a for each arrow $a \in \overline{Q}$, satisfying the relation

$$\sum_{\substack{a \in Q \\ h(a)=j}} V_a V_{a^*} - \sum_{\substack{a \in Q \\ t(a)=j}} V_{a^*} V_a - \lambda_j 1_{V_j}$$

for each vertex j .

Given such a representation, let

$$V_\oplus = \bigoplus_{\substack{a \in Q \\ h(a)=i}} V_{t(a)},$$

and write $\mu_a : V_{t(a)} \hookrightarrow V_\oplus$ and $\pi_a : V_\oplus \twoheadrightarrow V_{t(a)}$ for the inclusion and the projection between $V_{t(a)}$ and $V_\oplus$. Define the maps $\mu : V_i \longrightarrow V_\oplus$ and $\pi : V_\oplus \longrightarrow V_i$ as follow

$$\mu = \sum_{\substack{a \in Q \\ h(a)=i}} \mu_a V_{a^*}, \quad \pi = \frac{1}{\lambda_i} \sum_{\substack{a \in Q \\ h(a)=i}} V_a \pi_a.$$

Then $\pi\mu = 1_{V_i}$. In effect,

$$\pi\mu = \frac{1}{\lambda_i} \sum_{\substack{a \in Q \\ h(a)=i}} V_a \pi_a \left(\sum_{\substack{a \in Q \\ h(a)=i}} \mu_a V_{a^*} \right) = \frac{1}{\lambda_i} \sum_{\substack{a \in Q \\ h(a)=i}} V_a \pi_a \mu_a V_{a^*} = \frac{1}{\lambda_i} \sum_{\substack{a \in Q \\ h(a)=i}} V_a V_{a^*}.$$

Note that $\sum_{\substack{a \in Q \\ t(a)=i}} V_{a^*} V_a = 0$ because no arrows have tail at i . Then

$$\pi\mu = \frac{1}{\lambda_i} \sum_{\substack{a \in Q \\ h(a)=i}} V_a V_{a^*} - \frac{1}{\lambda_i} \sum_{\substack{a \in Q \\ t(a)=i}} V_{a^*} V_a = 1_{V_i}.$$

And also $\mu\pi$ is an idempotent endomorphism of $V_\oplus$:

$$(\mu\pi)^2 = \mu\pi\mu\pi = \mu 1_{V_i}\pi = \mu\pi.$$

Second: Starting from V , we will construct another representation or Q , V' . Consider the representation V' of Q as follows:

$$V'_j = V_j, \text{ for } j \neq i$$

and for $j = i$

$$V'_i = \text{Img}(1 - \mu\pi) \stackrel{*}{=} \ker \pi,$$

with the linear applications

$$V'_a = V_a \text{ and } V'_{a^*} = V_{a^*}, \text{ if } h(a) \neq i$$

while if $h(a) = i$

$$\begin{aligned} V'_a &= -\lambda_i(1 - \mu\pi)\mu_a : V'_{t(a)} \rightarrow V'_i \\ V'_{a^*} &= \pi_a|_{V'_i} : V'_i \rightarrow V'_{t(a)}. \end{aligned}$$

In (*) we claim that $\text{Img}(1 - \mu\pi) = \ker \pi$. Indeed, if $v \in \ker \pi$,

$v = v - \mu\pi(v) = (1 - \mu\pi)(v)$ which implies that $v \in \text{Img}(1 - \mu\pi)$. On the other hand, if

$v \in \text{Img}(1 - \mu\pi)$, there is $w \in V_\oplus$ such that $v = (1 - \mu\pi)w = w - \mu\pi(w)$. Then

$\pi(v) = \pi(w) - \pi\mu\pi(w) = 0$, because $\pi\mu = 1_{V_i}$, then $v \in \ker \pi$.

Third: We shall prove that $V' \in \Pi^{r_i\lambda}(Q) - \text{mod}$.

For the vertex i :

$$\begin{aligned} \sum_{\substack{a \in Q \\ h(a)=i}} V'_a V'_{a^*} &= \sum_{\substack{a \in Q \\ h(a)=i}} -\lambda_i(1 - \mu\pi)\mu_a \pi_a|_{V'_i} = -\lambda_i(1 - \mu\pi) \sum_{\substack{a \in Q \\ h(a)=i}} \mu_a \pi_a|_{V'_i} \\ &= -\lambda_i(1 - \mu\pi)|_{V'_i} = -\lambda_i 1_{V'_i} + \lambda_i \mu\pi|_{V'_i} \\ &= -\lambda_i 1_{V'_i} + \sum_{\substack{a \in Q \\ h(a)=i}} \mu_a V_{a^*} V_a \pi_a|_{V'_i} = -\lambda_i 1_{V'_i} \end{aligned}$$

$$\begin{aligned} \text{And } (r_i\lambda)_i &= \lambda_i - (\epsilon_i, \epsilon_i)\lambda_i = \lambda_i - 2\langle \epsilon_i, \epsilon_i \rangle = \lambda_i - 2 \left(\sum_{k \in I} \epsilon_i(k)\epsilon_i(k) - \sum_{a \in Q} \epsilon_i(t(a))\epsilon_i(h(a)) \right) \lambda_i \\ &= \lambda_i - 2(1 - 0)\lambda_i = -\lambda_i. \end{aligned}$$

Note that, for $i \neq j$ two vertices in I

$$\langle \epsilon_i, \epsilon_j \rangle = \sum_{k \in I} \epsilon_i(k)\epsilon_j(k) - \sum_{a \in Q} \epsilon_i(t(a))\epsilon_j(h(a)) = - \sum_{a \in Q} \epsilon_i(t(a))\epsilon_j(h(a))$$

is the number of arrows in Q with tail in i and head in j . And because i is a sink for Q , $\langle \epsilon_i, \epsilon_j \rangle = 0$.

Therefore, for $i \neq j$, $\langle \epsilon_i, \epsilon_j \rangle = \langle \epsilon_i, \epsilon_j \rangle + \langle \epsilon_j, \epsilon_i \rangle = \langle \epsilon_j, \epsilon_i \rangle$ is the number of arrows with tail in j and head in i .

Now, if $a : i \rightarrow j$ is an arrow in Q then

$$V'_{a^*} V'_a = \pi_a(-\lambda_i(1 - \mu\pi)\mu_a) = -\lambda_i\pi_a\mu_a + \lambda_i\pi_a\mu\pi\mu_a = -\lambda_i 1_{V_{t(a)}} + V_{a^*} V_a,$$

thus, if $j \neq i$ then

$$\begin{aligned} \sum_{\substack{a \in Q \\ h(a)=j}} V'_a V'_{a^*} - \sum_{\substack{a \in Q \\ t(a)=j}} V'_{a^*} V'_a &= \sum_{\substack{a \in Q \\ h(a)=j}} V_a V_{a^*} - \sum_{\substack{a \in Q \\ t(a)=j \\ h(a) \neq i}} V_{a^*} V_a - \sum_{\substack{a \in Q \\ t(a)=j \\ h(a)=i}} V'_{a^*} V'_a \\ &= \sum_{\substack{a \in Q \\ h(a)=j}} V_a V_{a^*} - \sum_{\substack{a \in Q \\ t(a)=j \\ h(a) \neq i}} V_{a^*} V_a - \sum_{\substack{a \in Q \\ t(a)=j \\ h(a)=i}} (-\lambda_i 1_{V_{t(a)}} + V_{a^*} V_a) \\ &= \sum_{\substack{a \in Q \\ h(a)=j}} V_a V_{a^*} - \sum_{\substack{a \in Q \\ t(a)=j}} V_{a^*} V_a + \lambda_i \sum_{\substack{a \in Q \\ t(a)=j \\ h(a)=i}} 1_{V_j} \\ &= \sum_{\substack{a \in Q \\ h(a)=j}} V_a V_{a^*} - \sum_{\substack{a \in Q \\ t(a)=j}} V_{a^*} V_a - (\epsilon_j, \epsilon_i) \lambda_i 1_{V_j} \\ &= \lambda_i 1_{V_j} - (\epsilon_j, \epsilon_i) \lambda_i 1_{V_j} \\ &= (r_i \lambda)_j 1_{V_j}. \end{aligned}$$

Fourth: The assignment $V \mapsto V'$ extends to a functor $E_i : \Pi^\lambda Q - \text{mod} \rightarrow \Pi^{r_i \lambda} Q - \text{mod}$, where i is a sink for Q .

We have defined how E_i acts on objects in $\Pi^\lambda Q - \text{mod}$. Now, we need to define how it acts on morphisms in $\text{Rep}(Q)$:

Let $\phi : V \rightarrow W$ be a morphism in $\Pi^\lambda Q - \text{mod}$ and let i be a vertex in Q which is a sink. We define $(E_i \phi)_j = \phi_j : V_j = V'_j \rightarrow W_j = W'_j$ for $j \neq i$.

For the vertex i we know that the following diagram is commutative

$$\begin{array}{ccc} V_{t(a)} & \xrightarrow{V_a} & V_i \\ \downarrow \phi_{t(a)} & & \downarrow \phi_i \\ W_{t(a)} & \xrightarrow{W_a} & W_i \end{array}$$

from ϕ is a morphism of representation.

Let μ_a^V be the inclusion μ_a for V , π_a^V be the projection π_a for V , μ^V be the map μ for V and π be the map π for V ; respectively for W . We define $\phi_{\hat{a}} : V_{\oplus} \rightarrow W_{\oplus}$ by $\phi_{\hat{a}} = \mu_a^W \phi_{t(a)} \pi_a^V$.

Then,

$$\begin{aligned}\phi_i \pi^V &= \frac{1}{\lambda_i} \sum_{\substack{a \in Q \\ h(a)=i}} \phi_i V_a \pi_a^V = \frac{1}{\lambda_i} \sum_{\substack{a \in Q \\ h(a)=i}} W_a \phi_{t(a)} \pi_a^V = \frac{1}{\lambda_i} \sum_{\substack{a \in Q \\ h(a)=i}} W_a \pi_a^W \mu_a^W \phi_{t(a)} \pi_a^V \\ &= \pi^W \phi_{\hat{a}}\end{aligned}$$

and the following diagram is also commutative,

$$\begin{array}{ccc} V_{\oplus} & \xrightarrow{\pi^V} & V_i \\ \downarrow \phi_{\hat{a}} & & \downarrow \phi_i \\ W_{\oplus} & \xrightarrow{\pi^W} & W_i \end{array}$$

Finally, we define $(E_i \phi)_i : V'_i \rightarrow W'_i$ as the unique linear application such that the following diagram is commutative,

$$\begin{array}{ccc} V'_i & \xrightarrow{\iota^V} & V_{\oplus} \\ \vdots \downarrow (E_i \phi)_i & & \downarrow \phi_{\hat{a}} \\ W'_i & \xrightarrow{\iota^W} & W_{\oplus} \end{array}$$

which exists from the Universal Property of the Kernel, since $V'_i = \ker \pi^V$ (see 1 in Chapter III of [28]).

Now, we prove that E_i is, in fact, a functor.

- a) Let $1_V : V \rightarrow V$ a trivial morphism in $\text{Rep}(Q)$. Let see that $E_i 1_V = 1_{V'}$.

For $j \neq i$, $(E_i 1_V)_j = (1_V)_j = 1_{V_j}$.

For i , by definition, $(E_i 1_V)_i : V'_i \rightarrow W'_i$ is the unique linear application such that the previous diagram for 1_V commutes.

Then, by the uniqueness of $(E_i 1_V)_i$, $(E_i 1_V)_i = 1_{V'_i} = (1_V)_i$.

- b) For the another axiom, let $\psi : U \rightarrow V$ and $\phi : V \rightarrow W$ be two morphism in $\text{Rep} Q$.

Let see that $E_i(\phi \circ \psi) = E_i(\phi) \circ E_i(\psi)$.

We have the following diagram with each small square commutes:

$$\begin{array}{ccccc}
U'_i & \xrightarrow{\iota^U} & U_\oplus & \xrightarrow{\pi^U} & U_i \\
\vdots \downarrow (E_i\psi)_i & & \downarrow \psi_{\hat{a}} & & \downarrow \psi_i \\
V'_i & \xrightarrow{\iota^V} & V_\oplus & \xrightarrow{\pi^V} & V_i \\
\vdots \downarrow (E_i\phi)_i & & \downarrow \phi_{\hat{a}} & & \downarrow \phi_i \\
W'_i & \xrightarrow{\iota^W} & W_\oplus & \xrightarrow{\pi^W} & W_i
\end{array}$$

Since, $\pi^W(\phi_{\hat{a}}\psi_{\hat{a}}\iota^U) = \phi_i\pi^V\iota^U = \phi_i\psi_i\pi^U\iota^U = 0$, by the universal mapping property of the Kernel $W'_i = \ker\pi^W$, there is a unique linear application $\varphi : U'_i \rightarrow W'_i$ such $(\phi_{\hat{a}}\psi_{\hat{a}}\iota^U) = \iota^W\varphi$.

So, by definition of E_i this application is $\varphi = E_i(\phi \circ \psi)_i$. Thus, by the uniqueness $\varphi = E_i(\phi \circ \psi)_i = (E_i\phi)_i \circ (E_i\psi)_i$.

Thus, $E_i : \Pi^\lambda - \text{mod} \rightarrow \Pi^{r_i\lambda} - \text{mod}$ is a functor.

Fifth: We shall prove that E_i acts as s_i on dimension vectors.

If $\dim V = \alpha$. Recall that

$$\begin{aligned}
s_i(\alpha) &= \alpha - (\alpha, \epsilon_i)\epsilon_i = \alpha - \langle \alpha, \epsilon_i \rangle \epsilon_i - \langle \epsilon_i, \alpha \rangle \epsilon_i \\
&= \alpha - (2 \sum_{i \in I} \alpha(i)\epsilon_i(i) + \sum_{a \in Q} \alpha(t(a))\epsilon_i(h(a)) + \sum_{a \in Q} \epsilon_i(t(a))\alpha(h(a)))\epsilon_i \\
&= \alpha - 2\alpha(i)\epsilon_i + \sum_{\substack{a \in Q \\ h(a)=i}} \alpha(t(a))\epsilon_i,
\end{aligned}$$

then

$$s_i(\alpha)(j) = \alpha(j)$$

and

$$s_i(\alpha)(i) = \alpha(i) - 2\alpha(i) + \sum_{\substack{a \in Q \\ h(a)=i}} \alpha(t(a)) = \sum_{\substack{a \in Q \\ h(a)=i}} \alpha(t(a)) - \alpha(i).$$

By the other hand, the exact sequence

$$0 \rightarrow V'_i \xrightarrow{\iota^V} V_\oplus \xrightarrow{\pi^V} V_i \rightarrow 0$$

shows that

$$\dim V'_i = \dim V_\oplus - \dim V_i = \sum_{\substack{a \in Q \\ h(a)=i}} \alpha(t(a)) - \alpha(i).$$

Sixth: The reflection also defines a functor

$$E'_i : \Pi^{r_i \lambda} Q - \text{mod} \rightarrow \Pi^\lambda Q - \text{mod}.$$

Let V be a $\Pi^{r_i \lambda} - \text{mod}$. Just define $E'_i(V)$, in the same way as E_i . By analogous calculation, it is obtained that $E'_i(V)$ is a $\Pi^{r_i(r_i \lambda)} - \text{mod}$, but $r_i^2(\lambda) = \lambda$, so $E'_i(V)$ is a $\Pi^\lambda - \text{mod}$.

Seventh: There is a natural isomorphism $V \mapsto E'_i(E_i(V))$. It follows that E_i is an equivalence. $\square$

COROLLARY 5.7. (1) Π^λ is Morita equivalent to Π^μ for any $\mu \in W\lambda$.

(2) In fact, if $\mu = w\lambda$ and w has a minimal length with this property, then there is an equivalence

$$\Pi^\lambda - \text{modules} \longrightarrow \Pi^\mu - \text{modules},$$

which acts as w on dimension vectors.

PROOF. The first assertion is immediately from the Theorem 5.6 and the Remark 3.6.

We prove the second assertion by induction on the length of w , and hence reduce to the case when w is a simple reflection s_i ($l(s_i) = 1$). Assume that $\lambda_i = 0$, so $(r_i \lambda)_i = -\lambda_i = 0$ and $(w\lambda)_i \cdot \alpha = \lambda_i \cdot (w^{-1}\alpha) = 0$ for all $\alpha \in \mathbb{Z}^I$. Then, since $(r_i \lambda)_j = \lambda_j = (w\lambda)_j$, $w\lambda = r_i \lambda = \lambda$; contradicting the minimality of w . Thus $\lambda_i \neq 0$, therefore, the reflection functor exists. $\square$

5.1. An example using the cyclic group of order 2. Part 4. Consider the Kronecker quiver Q associated to the group C_2 , just as in the previous parts of our example, we will give the Weyl group for Q . The simple reflection for Q are:

$$s_0 : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z} \times \mathbb{Z}, (\alpha_0, \alpha_1) \mapsto \alpha - (\alpha, \epsilon^0) \epsilon^0$$

and

$$s_1 : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z} \times \mathbb{Z}, (\alpha_0, \alpha_1) \mapsto \alpha - (\alpha, \epsilon^1) \epsilon^1,$$

where $\epsilon^0 = (1, 0)$ and $\epsilon^1 = (0, 1)$. We have,

$$\begin{aligned} (\alpha, \epsilon^0) &= \langle \alpha, \epsilon^0 \rangle + \langle \epsilon^0, \alpha \rangle = \sum_{i=0,1} \alpha_i \epsilon_i^0 - \sum_{f=a,b} \alpha_{t(f)} \epsilon_{h(f)}^0 + \sum_{i=0,1} \epsilon_i^0 \alpha_i - \sum_{f=a,b} \epsilon_{t(f)}^0 \alpha_{h(f)} \\ &= 2\alpha_0 - 2\alpha_1 = 2(\alpha_0 - \alpha_1) \end{aligned}$$

and

$$\begin{aligned} (\alpha, \epsilon^1) &= \langle \alpha, \epsilon^1 \rangle + \langle \epsilon^1, \alpha \rangle = \sum_{i=0,1} \alpha_i \epsilon_i^1 - \sum_{f=a,b} \alpha_{t(f)} \epsilon_{h(f)}^1 + \sum_{i=0,1} \epsilon_i^1 \alpha_i - \sum_{f=a,b} \epsilon_{t(f)}^1 \alpha_{h(f)} \\ &= 2\alpha_1 - 2\alpha_0 = 2(\alpha_1 - \alpha_0) \end{aligned}$$

Then

$$s_0 : (\alpha_0, \alpha_1) \mapsto (\alpha_0 - 2\alpha_0 + 2\alpha_1, \alpha_1) = (2\alpha_1 - \alpha_0, \alpha_1)$$

and

$$s_1 : (\alpha_0, \alpha_1) \mapsto (\alpha_0, \alpha_1 - 2\alpha_1 + 2\alpha_0) = (\alpha_0, 2\alpha_0 - \alpha_1).$$

And let W be the Weyl group for Q , $W = \langle s_0, s_1 \rangle$.

6. Roots for a quiver.

In this section K is an arbitrary field and Q is an arbitrary quiver with vertex set I . We will introduce the root system associated to a quiver, and we will prove fundamental results of classifications of the representations of Π^λ .

DEFINITION 6.1. The *simple roots* for Q are the coordinate vectors of loopfree vertices.

DEFINITION 6.2. An element $\alpha \in \mathbb{Z}^I$ is a *real root* if it is an image of a simple root under the Weyl group W . It is standard that any real root is either *positive* or *negative*, i.e. either α or $-\alpha$ belongs to $\mathbb{N}^I$.

DEFINITION 6.3. Let be $\alpha \in \mathbb{Z}^I$ a root, we define a support of α as

$$\text{supp}(\alpha) = \{i \in I \mid \alpha_i \neq 0\}$$

DEFINITION 6.4. The *fundamental region* is

$$F = \{\alpha \in \mathbb{N}^I \mid \alpha \neq 0, \text{ support of } \alpha \text{ is connected, and } (\alpha, \epsilon_i) \leq 0 \text{ for all } i\}$$

$(\alpha, \epsilon_i) \leq 0$ is automatic if i is not loopfree.

DEFINITION 6.5. An *imaginary root* is an element of $\mathbb{Z}^I$ of the form $w\beta$ or $-w\beta$ with $w \in W$ and $\beta \in F$. The elements $w\beta$ all belong to $\mathbb{N}^I$, so these are the *positive imaginary roots*. [26]

A *root* is a real or imaginary root.

DEFINITION 6.6. We say that $\lambda \in K^I$ is *on a wall*, if $\lambda \cdot \alpha = 0$ for some real root α .

THEOREM 6.7. Π^λ has a representation of dimension vector ϵ_i if and only if $\lambda_i = 0$. It is unique if i is loopfree.

PROOF. $\implies$ Representations of Π^λ with dimension vector ϵ_i are the same things as representations of $K\overline{Q}$ with dimension vector ϵ_i on which $c - \lambda$ acts by 0.

Since $c = \sum_{a \in Q} x_a x_{a^*} - x_{a^*} x_a$ as an operator on such a representation, with $x_a = 0$ unless a is a loop based at i and $x_a x_{a^*} - x_{a^*} x_a = 0$ if a is a loop based at i , we see that c acts by 0 on such representations.

This implies that $\lambda_i = 0$.

$\Leftarrow$ On the other hand, let X be the representation of $K\overline{Q}$ with dimension vector ϵ_i , such that $x_a = 0 = x_{a^*}$ for all arrows $a \in Q$.

If $\lambda_i = 0$ then X descends to Π^λ , since both c and λ acts by 0. $\square$

THEOREM 6.8. *Suppose that $\lambda \in K^I$*

- (1) *If λ is on a wall, then Π^λ has a simple representation whose dimension vector is a real root.*
- (2) *If λ is not on a wall, then the dimension vector of any indecomposable finite-dimensional representation of Π^λ is an imaginary root.*

PROOF. (1) Since λ is on a wall, there is $\alpha \in R_{re}$ such that $\lambda \cdot \alpha = 0$. And $\alpha = v(\beta_i)$ for β_i a simple root, some $v \in W$ and some $i \in I$, by definition of real root.

By the action of W on K^I

$$0 = \lambda \cdot \alpha = \lambda \cdot v(\beta_i) = v^{-1}(\lambda) \cdot \beta_i$$

This implies that $(v^{-1}(\lambda))_i = 0$, for some loopfree vertex i .

By Theorem 6.7 $\Pi^{v^{-1}(\lambda)}$ has a representation of dimension vector ϵ_i , obviously simple. And, by Corollary 5.7 there is a simple Π^λ -module of dimension vector $w'\epsilon_i$ for some $w' \in W$.

- (2) Since λ is not on a wall, $(w \cdot \lambda)_i \neq 0$ for all $w \in W$ and for all $i \in I$. In particular, for $w = 1$, $\lambda_i \neq 0$ for all $i \in I$.

Let X an indecomposable representation of Π^λ of dimension vector d . By Theorem 5.6, for some loopfree vertex j , $E_j X$ is an indecomposable representation of $\Pi^{j\lambda}$ whose dimension vector is $s_j(d)$.

By Lemma 7.46 $\text{supp}(\dim(X)) = \text{supp}(d)$ is connected, so $\text{supp}(s_j(d))$ is connected. It follows that $\text{supp}(w(d))$ is connected for all $w \in W$.

We can apply any sequence of reflection functors, because $(w \cdot \lambda)_i \neq 0$ for all $w \in W$ and for all $i \in I$. We are looking to obtain $(w'(d), \epsilon_i) \leq 0$ for all $i \in I$ and for some $w' \in W$. We choose w' such that $\sum_{j=1}^n (w'd)_j$ is minimal. If $(w'd, \epsilon_i) > 0$, we apply the reflection functor E_i to obtain a smaller $\sum_{j=1}^n (s_i w'd)_j$, which is a contradiction. Thus $w'(d)$ belongs to the fundamental region, so d is an imaginary root. $\square$

PROPOSITION 6.9. *If Q is a Dynkin quiver, then it does not have imaginary roots.*

PROOF. If Q is a Dynkin quiver, then $(\cdot, \cdot)$ is positive definite (see section 7.2 in the chapter 2). Let $\alpha = \sum a_i \epsilon_i$, with $a_i \geq 0$, if $(\alpha, \epsilon_i) \geq 0$ for all $i \in I$, this implies that

$$0 < (\alpha, \alpha) \leq \sum a_i (\alpha, \epsilon_i) \leq 0,$$

which is a contradiction. Then the fundamental region F for Q is empty, which implies Q have no imaginary roots. $\square$

COROLLARY 6.10. *If Q is Dynkin (or is a disjoint union of Dynkin quivers) then $\Pi^\lambda(Q) \neq 0$ if and only if λ is on a wall.*

PROOF. First, note that by Lemma 2.5, $\Pi^\lambda(Q)$ is finite dimensional.

$\implies$ If λ is not on a wall, by Theorem 6.8 part 2, the dimension vector of any indecomposable finite-dimensional representation of $\Pi^\lambda(Q)$ is an imaginary root. Since Q is a Dynkin quiver, by Proposition 6.9, $\Pi^\lambda(Q)$ has no indecomposable finite-dimensional representations.

Thus, it must be zero, because it is finite-dimensional.

$\impliedby$ Now, if λ is on a wall, by Theorem 6.8 part 1, $\Pi^\lambda(Q)$ has a simple representation whose dimension vector a real root, so $\Pi^\lambda(Q) \neq 0$. $\square$

We shall prove another Corollary, but first we need two definitions for R -modules over a ring R .

COROLLARY 6.11. *Let Q be an arbitrary quiver, $\lambda \in K^I$ and $J \subset I$. Suppose that the Q' , the full subquiver of Q on J , is a disjoint union of Dynkin quivers. If $e = \sum_{i \in J} e_i$ then, the module $\Pi^\lambda(Q)e$ is a progenerator if and only if the restriction λ' of λ to J is not on a wall for Q' . In this case, $\Pi^\lambda(Q)$ is Morita equivalent to $e\Pi^\lambda(Q)e$.*

PROOF. In general, let R be a ring and let e be an idempotent element in R , we have the following three facts:

- i) Re is a projective left R -module.

Clearly Re is a left R -module, because $sre \in Re$ for all $s, r \in R$.

Now, by the idempotence of e , $e^2 = e$ which implies $(1 - e)^2 = 1 - e$ and $1 = e + 1 - e$. On the other hand, $x \in Re \cap R(1 - e) \implies ae = x = b(1 - e)$ for some $a, b \in R \implies x = xe = b(e - e^2) = 0 \implies Re \cap R(1 - e) = 0$. So $R = Re \oplus R(1 - e)$ and Re is a projective R -module.

Now, Re is progenerator if and only if Re is a generator.

- ii) $(Re)^* := \text{Hom}_R(Re, R) \cong eR$ via the map:

$$\begin{aligned} \text{Hom}_R(Re, R) &\longrightarrow eR \\ f &\mapsto f(e), \end{aligned}$$

which inverse is

$$\begin{aligned} eR &\longrightarrow \text{Hom}_R(Re, R) \\ er &\mapsto [se \mapsto ser] \end{aligned}$$

$$\implies Re(Re)^* = ReeR = ReR.$$

iii) $eRe \cong \text{End}_R Re$, via the map:

$$eRe \longrightarrow \text{End}_R Re$$

$$ere \mapsto [se \mapsto sere]$$

Obviously is injective and let $\varphi \in \text{End}_R Re$, $\varphi(e) = \varphi(e^2) = e\varphi(e) \in eRe$ and $\varphi(se) = s\varphi(e) = s\varphi(e^2) = se\varphi(e)$, which implies that the map is surjective.
 $\implies eRe = \text{End}_R Re$.

Then,

$$Re \text{ is a progenerator } \xLeftrightarrow{3.3} R = Re(Re)^* \xLeftrightarrow{(ii)} R = ReR \xrightarrow{3.5 \text{ and } (iii)} R \overset{M}{\sim} eRe$$

In particular, for $R = \Pi^\lambda(Q)$ and the idempotent $e = \sum_{i \notin J} e_i$ we have:

$$\Pi^\lambda(Q)e \text{ is a progenerator } \iff \Pi^\lambda(Q)e = \Pi^\lambda(Q)e\Pi^\lambda(Q).$$

Also, in this case $\Pi^\lambda(Q) \overset{M}{\sim} e\Pi^\lambda(Q)e$

On the other hand, there is a canonical isomorphism

$$\Pi^\lambda(Q)/\Pi^\lambda(Q)e\Pi^\lambda(Q) \cong \Pi^{\lambda'}(Q'),$$

because on the left hand we make the quotient on all class of path in $\Pi^\lambda(Q)$ which begin or end at a vertex of J .

Then,

$$\Pi^\lambda(Q)e = \Pi^\lambda(Q)e\Pi^\lambda(Q) \iff 0 = \Pi^\lambda(Q)/\Pi^\lambda(Q)e\Pi^\lambda(Q) = \Pi^{\lambda'}(Q').$$

And because Q' is a union of Dynkin quivers, by Corollary 6.10, λ' is not on a wall for Q' if and only if $\Pi^{\lambda'}(Q') = 0$.

So $\Pi^\lambda(Q)e$ is a progenerator $\iff \lambda'$ is not on a wall. $\square$

6.1. An example using the cyclic group of order 2. Part 5. Continuing with our example, using group C_2 and the associated Kronecker quiver Q , we give some examples of the real root for the quiver Q . In the part 4 of our example, we find the simple reflections which generate the Weyl group for Q :

$$s_0 : (\alpha_0, \alpha_1) \mapsto (\alpha_0 - 2\alpha_0 + 2\alpha_1, \alpha_1) = (2\alpha_1 - \alpha_0, \alpha_1)$$

and

$$s_1 : (\alpha_0, \alpha_1) \mapsto (\alpha_0, \alpha_1 - 2\alpha_1 + 2\alpha_0) = (\alpha_0, 2\alpha_0 - \alpha_1).$$

So,

$$s_0(1, 0) = (2 \cdot 0 - 1, 0) = (-1, 0), \quad s_0(0, 1) = (2 \cdot 1 - 0, 1) = (2, 1),$$

$$s_1(1, 0) = (1, 2 \cdot 1 - 0) = (1, 2) \text{ and } s_1(0, 1) = (0, 2 \cdot 0 - 1) = (0, -1)$$

are some real roots for Q .

7. The case when $\lambda \cdot \delta \neq 0$.

Consider $K = \mathbb{C}$ and fix a total ordering $<$ on K as follows:

$$z < z' \text{ if and only if either } \mathcal{R}z < \mathcal{R}z' \text{ or } \mathcal{R}z = \mathcal{R}z' \text{ and } \mathcal{I}z < \mathcal{I}z'.$$

In this section, using a notion of dominance, we study $\mathcal{O}^\lambda$ when $\lambda \cdot \delta \neq 0$. The next results are also true for any field of characteristic zero.

DEFINITION 7.1. We say that a weight $\lambda \in K^I$ is *dominant* if $\lambda_i \geq 0$, for all $i \in I$.

DEFINITION 7.2. If λ is dominant, or more generally a non-zero scalar multiple of a dominant weight, we define $I_\lambda = \{i \in I \mid \lambda_i = 0\}$.

We write Q_λ for a full subquiver of Q with vertex set I_λ .

If $i \in I_\lambda$ is a loopfree vertex, as in Theorem 6.7, we have a simple module for Π^λ with dimension vector ϵ_i , we denote it S_i .

THEOREM 7.3. *If λ is dominant then any finite-dimensional Π^λ -module has support contained in I_λ , so is a module for the algebra $\Pi(Q_\lambda)$. In particular:*

- (1) *If Q is a disjoint union of Dynkin quivers, then $\Pi^\lambda \cong \Pi(Q_\lambda)$.*
- (2) *If Q_λ is a disjoint union of Dynkin quivers, then the S_i , with $i \in I_\lambda$, are the finite-dimensional simple Π^λ -modules.*

PROOF. Let be M a finite-dimensional Π^λ -module with dimension vector α . Then, by Lemma 4.2, $\lambda \cdot \alpha = 0$, i.e. $\sum_{i \in I} \lambda_i \alpha_i = 0$. Now, since λ is dominant, all $\lambda_i \geq 0$ so the sum is also ≥ 0 , and if $i \in I$ such that $\alpha_i \neq 0$ then $\lambda_i = 0$.

- (1) If Q is a disjoint union of Dynkin quivers, then, by Lemma 2.5, $\Pi^\lambda(Q)$ is finite dimensional. We write

$$c_i = \sum_{\substack{a \in Q \\ h(a)=i}} aa^* - \sum_{\substack{a \in Q \\ t(a)=i}} a^*a - \lambda_i e_i,$$

$$c'_i = \sum_{\substack{b \in Q_\lambda \\ h(b)=i}} bb^* - \sum_{\substack{b \in Q_\lambda \\ t(b)=i}} b^*b.$$

Thus, $\Pi^\lambda(Q) = k(\overline{Q})/(c_i \mid i \in I)$ and $\Pi(Q_\lambda) = k(\overline{Q}_\lambda)/(c'_i \mid i \in I_\lambda)$

We define a map

$$\psi : \Pi(Q_\lambda) \longrightarrow \Pi^\lambda(Q)$$

given by $\psi(\overline{c}) = \overline{c}$.

This map is well-defined, because there is an obvious inclusion $k(\overline{Q}_\lambda) \hookrightarrow k(\overline{Q})$ and by this inclusion $(c'_i \mid i \in I_\lambda) \hookrightarrow (c_i \mid i \in I)$.

Let $\bar{p} \in \Pi^\lambda(Q)$ and let p its path class representative in $k(\bar{Q})$. There is a arrow $a \in Q$ in p such that $h(a) \notin I_\lambda$ and $t(a) \notin I_\lambda$.

$$\Rightarrow \bar{p} = \overline{\dots(e_i a \dots)} = 0 \Rightarrow \bar{p} = 0 \in \Pi(Q_\lambda).$$

Thus ψ is surjective and $\Pi^\lambda(Q)$ is a quotient of $\Pi(Q)$.

Now, we define a left inverse for ψ :

$$\phi: k(\bar{Q}) \longrightarrow k(\bar{Q}_\lambda),$$

given by

$$\begin{aligned} a &\mapsto \begin{cases} a, & \text{if } a \in Q_\lambda \\ 0, & \text{otherwise} \end{cases}, \text{ for } a \in Q, \\ a^* &\mapsto \begin{cases} a^*, & \text{if } a \in Q_\lambda \\ 0, & \text{otherwise} \end{cases}, \text{ for } a \in Q^{op}, \\ e_i &\mapsto \begin{cases} e_i, & \text{if } i \in I_\lambda \\ 0, & \text{otherwise} \end{cases}, \text{ for } i \in I. \end{aligned}$$

We can note that $\phi((c_i | i \in I)) \subseteq (c'_i | i \in I_\lambda)$. In effect, if $i \notin I_\lambda$, then $\phi(c_i) = 0$. By the other hand, if $i \in I_\lambda$, then $\phi(c_i) = c'_i$. This implies that ϕ descends to a map

$$\phi: \Pi^\lambda(Q) \longrightarrow \Pi(Q_\lambda).$$

Finally, $(\phi \circ \psi)(\bar{a}) = \phi(\bar{a}) = \bar{a}$ and $(\phi \circ \psi)(\bar{e}_i) = \phi(\bar{e}_i) = \bar{e}_i$, for $a \in Q$ and e_i any idempotent with $i \in I$. Thus $\phi \circ \psi = id_{\Pi(Q_\lambda)}$, then ψ is an injective map and $\Pi^\lambda \cong \Pi(Q_\lambda)$.

(2) First, note that:

FACT 7.4. If Q_λ is a Dynkin quivers, then Q_λ is a tree.

Now, we will prove by induction on the number of arrows, that for a tree T each simple $\Pi^\lambda(T)$ -module is isomorphic to S_i for some $i \in I_T$, where I_T is the vertex set of T . The initial case is for $|T| = 0$, i.e. T is a quiver with $|I_T|$ vertices and no arrow, then $k\bar{T} = \Pi^\lambda(T) = K^{\oplus |I_T|}$, so the simple modules are the S_i , with $i \in I_T$. We assume, as induction hypothesis that its true for $|T| = n$, and we will prove it for $|T| = n + 1$.

Let 0 a label for a leaf in T . This means that 0 is a vertex connected to a single vertex in T , with $|I_T| = n + 1$. Let 1 be the label for its adjacent vertex. Let $a \in T$ be the arrow between 0 and 1, with $t(a) = 0$ (without loss of generality). Let S a simple $\Pi^\lambda(T)$ -module and write $S = \bigoplus_{i \in I_T} S_{(i)}$, where S_i is the vector space corresponding to the vertex i (remark that it is not the same as the module S_i).

Case 1: $a^* S_{(1)} = 0 \Rightarrow \bigoplus_{i \neq 0} S_{(i)}$ is a $\Pi^\lambda(T)$ -submodule. Then, since S is a simple module:

Subcase 1: $\bigoplus_{i \neq 0} S_{(i)} = 0$, then $S = S_{(0)}$, so $S \cong S_0$.

Subcase 2: $\bigoplus_{i \neq 0} S_{(i)} = S$, then $S_{(0)} = 0$.

This is the only place we use the induction hypothesis. Consider now $\bigoplus_{i \neq 0} S_{(i)}$ and let $\Pi^\lambda(T \setminus \{a\})$, where $T \setminus \{a\}$ is the quiver obtained by removing the vertex 0 and the arrow a from T . For example: The figure 2 illustrates an example of a quiver T , this figure is an orientation for the extended Coxeter-Dynkin graph E_6^1 . The figure 2 illustrates an example of the quiver $T \setminus \{E_6\}$ and it is the Coxeter-Dynkin graph E_6 . (see 7.2.5 in section 7.1 of the preliminary chapter 2)

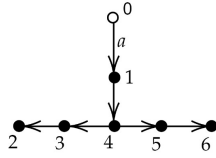


FIGURE 1. The graph E_6^1 with an orientation.

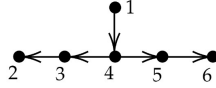


FIGURE 2. The graph E_6 with an orientation.

Since S satisfies the defining relations for $\Pi^\lambda(T)$, then $\bigoplus_{i \neq 0} S_{(i)}$ satisfies the defining relations for $\Pi^\lambda(T \setminus \{a\})$, because there is a relation for each vertex. Then, $\bigoplus_{i \neq 0} S_{(i)}$ is a module of $\Pi^\lambda(T \setminus \{a\})$. So, by hypothesis induction, the simple $\Pi^\lambda(T \setminus \{a\})$ -modules are the S_i , for $i \in I_{T \setminus \{a\}}$. Then $S = \bigoplus_{i \neq 0} S_{(i)} \cong S_j$, for some $j \in I_{T \setminus \{a\}} \subset I_T$.

Case 2: $a^* S_{(1)} \neq 0$. Because of the defining relations for $\Pi^\lambda(T)$ we have $aa^* S_{(1)} = 0$.

Thus $a^* S_{(1)}$ is a submodule and because S is simple, $a^* S_{(1)} = S$, so $S \cong S_0$.

□

We now use reflections to transfer results known for dominant weights to general weights $\lambda \in K^I$, if either Q is a Dynkin quiver, or Q is an extended Dynkin quiver and $\lambda \cdot \delta \neq 0$. In this case, we give some notations:

We write R_λ for the set of roots α with $\lambda \cdot \alpha = 0$, a finite set.

If $\alpha \in R_\lambda$, let s_α denote the automorphism of $\mathbb{Z}^I$ given by $\beta \mapsto \beta - (\alpha, \beta)\alpha$, and let W_λ be the subgroup of W generated by these s_α .

Let Σ_λ be the set of minimal positive elements of R_λ .

LEMMA 7.5. *Let Q be either Dynkin or extended Dynkin and let $\lambda \in K^I$. In the latter case, suppose that $\lambda \cdot \delta \neq 0$.*

- (1) *If Q is Dynkin then λ is conjugate under the Weyl group to a unique dominant weight.*
- (2) *If Q is extended Dynkin then $\lambda/(\lambda \cdot \delta)$ is conjugate under the Weyl group to a unique dominant weight λ^+ . Furthermore, there exists a unique element $w^+ \in W$ of minimal length with $w^+(\lambda/(\lambda \cdot \delta)) = \lambda^+$.*
- (3) *R_λ is the set of roots for a reduced root system (in the vector space it generates). W_λ acts faithfully on R_λ and identifies with the Weyl group for this root system.*
- (4) *Σ_λ is the unique basis of R_λ and consisting of positive roots. Further, $w^+\Sigma_\lambda = \Sigma_{\lambda^+}$.*
- (5) *W_λ is the stabiliser of λ in W .*
- (6) *If λ is dominant, then $\Sigma_\lambda = \{\varepsilon_i \mid i \in I_\lambda\}$ and Q_λ is a disjoint union of Dynkin quivers. (see [26] [25] for more details)*

THEOREM 7.6. *If Q is a disjoint union of Dynkin quivers and $\lambda \in K^I$ then $\Pi^\lambda(Q)$ is Morita equivalent to $\Pi(Q')$ for some subquiver Q' of Q .*

PROOF. By Corollary 5.7 gives us that $\Pi^\lambda \xrightarrow{M} \Pi^{w\lambda}$, where $w \in W$. Then applying Theorem 7.3 for $\Pi^{w\lambda}(Q)$, we obtain that $\Pi^{w\lambda}(Q) \cong \Pi(Q') \xrightarrow{M} \Pi^\lambda$ for some full subquiver Q' of Q . $\square$

THEOREM 7.7. *If Q is extended Dynkin and $\lambda \cdot \delta \neq 0$ then the isomorphism classes of finite-dimensional simple Π^λ -modules are in 1-1 correspondence with the set Σ_λ , under the assignment sending a simple module to its dimension vector.*

PROOF. In the case λ is dominant, by Lemma 7.5 (part 6), $\Sigma_\lambda = \{\varepsilon_i \mid i \in I_\lambda\}$ and Q_λ is a disjoint union of Dynkin diagrams, then Σ_λ can be identified with I_λ . Now, Lemma 7.3 (part 2) implies.

In general, for $\lambda \in K^I$ with $\lambda \cdot \delta \neq 0$, without loss of generality, we may suppose that $\lambda \cdot \delta = 1$. Then, by Lemma 7.5 (part 2) $w^+(\lambda) = \lambda^+$ for a unique element $w^+ \in W$ of minimal length. So, by Corollary 5.7 there is a Morita equivalence from Π^λ -modules to Π^{λ^+} -modules, which acts as w^+ on dimension vectors. Finally, Lemma 7.5 (part 4) implies $w^+\Sigma_\lambda = \Sigma_{\lambda^+}$, so the assertion is proved. $\square$

THEOREM 7.8. *If Q is extended Dynkin then Π^λ is simple if and only if has no non-zero finite-dimensional modules if and only if $\lambda \cdot \alpha \neq 0$ for all roots α .*

PROOF. First, recall that a root is either real or imaginary, and any imaginary root is a multiple of the fundamental imaginary root δ .

$\Rightarrow$ If $\lambda \cdot \alpha = 0$, where α is a real root, i.e. λ is on a wall, by Theorem 6.8 (part 1), Π^λ has a non-zero finite-dimensional representation. While if $\lambda \cdot \delta = 0$, by Lemma 4.5 and Theorem 4.4, Π^λ has a non-zero finite-dimensional representation. Thus, in both cases, Π^λ is not simple.

$\Leftarrow$ Now, if $\lambda \cdot \alpha \neq 0$ for all real root α , i.e. λ is on a wall then, by Theorem 6.8 (part 2), the only possibility for an indecomposable finite-dimensional representation of Π^λ is one with dimension vector an imaginary root, i.e. a multiple of δ , which is impossible by the trace condition.

The result follows from a general fact:

A finitely generated, prime, Noetherian K -algebra R of GK dimension 2 and with no non-zero, finite-dimensional modules is simple. $\square$

8. The case when $\lambda \cdot \delta = 0$

In this section K is an algebraically closed field of characteristic zero and $\Gamma \leq \text{SL}_2(K)$ is a non-trivial subgroup finite. Let Q be the affine quiver corresponding to Γ .

In this section, we present the most relevant theorem in the case when $\lambda \cdot \delta = 0$, it is $\mathcal{O}^\lambda \cong K[\text{Rep}(\Pi^\lambda, \delta)]^{\text{GL}(\delta)}$. In order to understand this result, the sections 6, 7 are necessary.

FACT 8.1. $\text{Rep}(\overline{Q}, \alpha)$ is the cotangent bundle of $\text{Rep}(Q, \alpha)$.

PROOF. Indeed, let V be a K -vector space, its *cotangent bundle* is $T^*V = V^* \times V$. In the case of

$$\text{Rep}(Q, \alpha) = \bigoplus_{a \in Q} \text{Hom}(K^{\alpha_{t(a)}}, K^{\alpha_{h(a)}}),$$

note that $\text{Hom}_K(K^{\alpha_{h(a)}}, K^{\alpha_{t(a)}})$ is isomorphic to $\text{Hom}_K(K^{\alpha_{t(a)}}, K^{\alpha_{h(a)}})^*$, via the following map:

$$\begin{aligned} \text{Hom}_K(K^{\alpha_{h(a)}}, K^{\alpha_{t(a)}}) &\xrightarrow{\sim} \text{Hom}_K(K^{\alpha_{t(a)}}, K^{\alpha_{h(a)}})^* \\ f &\mapsto [g \mapsto \text{tr}(fg)]. \end{aligned}$$

$$\text{Thus, } \text{Rep}(\overline{Q}, \alpha) = \bigoplus_{a \in Q} (\text{Hom}(K^{\alpha_{t(a)}}, K^{\alpha_{h(a)}}) \oplus \text{Hom}(K^{\alpha_{h(a)}}, K^{\alpha_{t(a)}})) \cong T^* \text{Rep}(Q, \alpha). \quad \square$$

On the other hand, $\text{Lie}(\text{GL}(\alpha)) = \text{End}(\alpha) := \bigoplus_{i \in I} \text{End}(K^{\alpha_i})$ is the Lie algebra of $\text{GL}(\alpha)$. More relevant for us is its subalgebra

$$\text{End}(\delta)_0 = \left\{ (\theta_i)_{i \in I} \in \text{End}(\delta) \mid \sum_{i \in I} \text{tr}(\theta_i) = 0 \right\}.$$

Now, there is a map μ :

$$\mu : T^* \text{Rep}(Q, \alpha) = \text{Rep}(\overline{Q}, \delta) \longrightarrow \text{End}(\delta)_0 \subseteq \text{Lie}(\text{GL}(\alpha)), \quad x = (x_a, x_{a^*})_{a \in Q} \mapsto \sum_{a \in Q} [x_a, x_{a^*}].$$

This map has image contained in $\text{End}(\alpha)_0$ since any commutator has trace zero.

Let $\mathfrak{h} = \{\lambda \in K^I \mid \lambda \cdot \delta = 0\}$. Since $\text{tr}(\lambda_i 1_{K^{\delta_i}})_{i \in I} = \sum_{i \in I} \lambda_i \cdot \delta_i = \lambda \cdot \delta$, we can identify λ with $(\lambda_i 1_{K^{\delta_i}})_{i \in I} \in Z(\text{End}(\delta)_0)$ and $\text{Rep}(\Pi^\lambda, \delta) = \mu^{-1}(\lambda)$, with the natural scheme structure.

The coordinate ring of $\text{Rep}(\overline{Q}, \delta)$ is the polynomial ring

$$K[s_{apq}] := K[s_{apq} \mid a \in \overline{Q}, 1 \leq p \leq \delta_{h(a)}, 1 \leq q \leq \delta_{t(a)}],$$

where s_{apq} is the p, q entry of the matrix of x_a , corresponding to $x \in \text{Rep}(\overline{Q}, \delta)$.

LEMMA 8.2. *The scheme $\text{Rep}(\Pi^\lambda, \delta)$ has a coordinate ring $K[s_{apq}]/J_\lambda$, where J_λ is generated by the elements*

$$\sum_{\substack{a \in Q \\ h(a)=i}} \sum_{r=1}^{\delta_{t(a)}} s_{apr} s_{a^* r q} - \sum_{\substack{a \in Q \\ t(a)=i}} \sum_{r=1}^{\delta_{\delta(a)}} s_{a^* pr} s_{ar q} - \delta_{pq} \lambda_i$$

for each vertex i and for $1 \leq p, q \leq \delta_i$.

PROOF. Let

$$\mu^* : K[\text{End}(\delta)_0] \longrightarrow K[\text{Rep}(\overline{Q}, \delta)]$$

$$p \mapsto p \circ \mu$$

be the ring homomorphism corresponding to the affine scheme homomorphism μ .

We write

$$K[\text{End}(\delta)_0] = \frac{\bigoplus_{i \in I} K[\text{End}(K^{\delta_i})]}{\left(\sum_{r=1}^{\delta_i} \sum_{i \in I} t_{rr}^i \right)},$$

where $t_{pq}^i \in K[\text{End}(K^{\delta_i})]$ gives the pq -th entry of an endomorphism of K^{δ_i} .

We define the ideal,

$$I_\lambda = \langle t_{pq}^i - \delta_{pq} \cdot \lambda_i \mid i \in I, 1 \leq p, q \leq \delta_i \rangle \triangleleft K[\text{End}(\delta)_0].$$

Now, let $x = (x_a, x_{a^*})_{a \in Q} \in \text{Rep}(\overline{Q}, \delta)$,

$$\begin{aligned} \mu^*(I_\lambda)(x) &= (I_\lambda \circ \mu)(x) = I_\lambda \left(\sum_{\substack{a \in Q \\ h(a)=i}} x_a x_{a^*} - \sum_{\substack{a \in Q \\ t(a)=i}} x_{a^*} x_a \right) \\ &= \sum_{\substack{a \in Q \\ h(a)=i}} t_{pq}^i (x_a x_{a^*}) - \sum_{\substack{a \in Q \\ t(a)=i}} t_{pq}^i (x_{a^*} x_a) - \delta_{pq} \lambda_i = J_\lambda. \end{aligned}$$

□

8.1. The map $\phi_\lambda : \mathcal{O}^\lambda \rightarrow A^\lambda$. The group $\mathrm{GL}(\delta) = \prod_{i \in I} \mathrm{GL}(K^{\delta_i})$ acts on $\mathrm{Rep}(\overline{Q}, \delta)$ and $\mathrm{Rep}(\Pi^\lambda, \delta) \subseteq \mathrm{Rep}(\overline{Q}, \delta)$ is invariant under this action.

Letting $l = \sum_{i \in I} \delta_i$. Identify $\mathrm{Mat}_{l \times l}(K)$ with $\mathrm{End}_K(\bigoplus_{i \in I} K^{\delta_i})$, then its subset $\mathrm{GL}(\delta)$ acts on $\mathrm{End}_K(\bigoplus_{i \in I} K^{\delta_i})$ by conjugation as follows:

Let $g = (g_i)_{i \in I} \in \mathrm{GL}(\delta)$, and let $m = (m_{ij})_{i,j \in I} \in \mathrm{End}_K(\bigoplus_{i \in I} K^{\delta_i})$, where $m_{ij} \in \mathrm{Hom}_K(K^{\delta_j}, K^{\delta_i})$, then $(g \cdot m)_{ij} = g_i m_{ij} g_j^{-1}$.

Now, observe that

$$\begin{aligned} \mathrm{Mat}_{l \times l}(K[\mathrm{Rep}(\overline{Q}, \delta)]) &\cong \{\text{Polynomial functions from } \mathrm{Rep}(\overline{Q}, \delta) \text{ to } \mathrm{End}_K(\bigoplus_{i \in I} K^{\delta_i})\} \\ &\cong K[\mathrm{Rep}(\overline{Q}, \delta)] \otimes_K \mathrm{End}_K(\bigoplus_{i \in I} K^{\delta_i}). \end{aligned}$$

Since $\mathrm{GL}(\delta)$ acts on $\mathrm{Rep}(\overline{Q}, \delta)$ and also on $\mathrm{End}_K(\bigoplus_{i \in I} K^{\delta_i})$, there is an action of $\mathrm{GL}(\delta)$ on $\{\text{Polynomial functions from } \mathrm{Rep}(\overline{Q}, \delta) \text{ to } \mathrm{End}_K(\bigoplus_{i \in I} K^{\delta_i})\}$ given by:

$$(g \cdot \phi)(x) = g(\phi(g^{-1} \cdot x)), \text{ for } g \in \mathrm{GL}(\delta), \phi \in \mathrm{Mat}_{l \times l}(K[\mathrm{Rep}(\overline{Q}, \delta)]) \text{ and } x \in \mathrm{Rep}(\overline{Q}, \delta).$$

Thus, we define the map

$$\begin{aligned} f : K\overline{Q} &\longrightarrow \mathrm{Mat}_{l \times l}(K[\mathrm{Rep}(\overline{Q}, \delta)]) \\ e_i &\mapsto [(x_a)_{a \in \overline{Q}} \mapsto 1_{\delta_i} \in \mathrm{Hom}_K(K^{\delta_i}, K^{\delta_i})] \\ a &\mapsto [(x_b)_{b \in \overline{Q}} \mapsto x_a \in \mathrm{Hom}_K(K^{\delta_{t(a)}}, K^{\delta_{h(a)}})] \end{aligned}$$

Then $g \cdot f(e_i) = f(e_i)$ and

$$\begin{aligned} (g \cdot f(a))(x_b)_{b \in \overline{Q}} &= g(f(a)(g_{h(b)}^{-1} x_b g_{t(b)}))_{b \in \overline{Q}} \\ &= g(g_{h(a)}^{-1} x_a g_{t(a)}) = x_a = f(a)(x_b)_{b \in \overline{Q}} \\ &\Rightarrow g \cdot f(a) = f(a). \end{aligned}$$

Then, $f(K\overline{Q})$ is invariant under $\mathrm{GL}(\delta)$. Moreover, f induces a map

$$\Pi^\lambda = K\overline{Q}/(c - \lambda) \xrightarrow{f} \mathrm{Mat}_{l \times l}(K[\mathrm{Rep}(\Pi^\lambda, \delta)])^{\mathrm{GL}(\delta)}.$$

Let $\mathcal{O}^\lambda := e_0 \Pi^\lambda e_0 \subseteq \Pi^\lambda$, then

$$f|_{\mathcal{O}^\lambda} =: \phi_\lambda : \mathcal{O}^\lambda \longrightarrow f(e_0) \mathrm{Mat}_{l \times l}(K[\mathrm{Rep}(\Pi^\lambda, \delta)])^{\mathrm{GL}(\delta)} f(e_0).$$

Note that

$$f(e_0) \mathrm{Mat}_{l \times l}(K[\mathrm{Rep}(\Pi^\lambda, \delta)])^{\mathrm{GL}(\delta)} f(e_0) = K[\mathrm{Rep}(\Pi^\lambda, \delta)]^{\mathrm{GL}(\delta)} = (K[s_{apq}]/J_\lambda)^{\mathrm{GL}(\delta)} =: A^\lambda,$$

because $\delta_0 = 1$.

Thus, it gives a map $\phi_\lambda : \mathcal{O}^\lambda \longrightarrow A^\lambda$.

In this section, we shall prove that ϕ_λ is an isomorphism. Our strategy is to first prove it for the generic element of $\mathfrak{h}$, explicitly for elements of the set

$$\mathfrak{h}' = \{\lambda \in \mathfrak{h} \mid \lambda \cdot \alpha \neq 0, \text{ for all real roots } \alpha\},$$

and then to extend to all elements of $\mathfrak{h}$ by the flatness of μ .

We say that $\lambda \in \mathfrak{h}'$ is Weil-generic, i.e. $\mathfrak{h}'$ is an open and dense set in the classic topology.

In order to understand this strategy, we need to be familiar with some results about *flatness, fiber product of affine schemes* and *linearly reductive groups*. (see section 6 in the preliminary chapter 2)

LEMMA 8.3. *The map $\phi_\lambda : \mathcal{O}^\lambda \longrightarrow A^\lambda$ preserves the filtration, and there is a commutative square*

$$\begin{array}{ccc} \mathcal{O}^0 & \xrightarrow{\phi_0} & A^0 \\ \downarrow & & \downarrow \\ \text{gr } \mathcal{O}^\lambda & \xrightarrow{\text{gr } \phi_\lambda} & \text{gr } A^\lambda \end{array}$$

where the left hand vertical map is an isomorphism and the right hand vertical map is surjective.

PROOF. By the Corollary 3.10 there is a natural isomorphism between Π and $\text{gr } \Pi^\lambda$, this isomorphism induces an isomorphism $\mathcal{O}^0 = e_0 \Pi e_0 \longrightarrow \text{gr } \mathcal{O}^\lambda = e_0 \text{gr } \Pi^\lambda e_0$.

Observe that $A^\lambda := \left(K[s_{apq}] / J_\lambda \right)^{\text{GL}(\delta)} \subseteq K[s_{apq}] / J_\lambda := B^\lambda$. $\mathcal{O}^\lambda$ has also a filtration inherited from $K\overline{Q}$, because $\mathcal{O}^\lambda$ is an ideal of a quotient of $K\overline{Q}$.

Because of B^0 and B^λ are quotients of polynomial rings, they have a filtration associated to the degree.

Thus, the obvious short exact sequence

$$0 \rightarrow J_\lambda \hookrightarrow K[s_{apq}] \twoheadrightarrow B^\lambda \rightarrow 0$$

induces the following sequence, also exact,

$$0 \rightarrow \text{gr } J_\lambda \hookrightarrow K[s_{apq}] \twoheadrightarrow \text{gr } B^\lambda \rightarrow 0$$

So, there is a surjection

$$K[s_{apq}] \twoheadrightarrow \text{gr } K[s_{apq}] / J_\lambda$$

.

Moreover, by the Lemma 8.2 there is a surjection

$$K[\text{Rep}(\Pi, \delta)] \twoheadrightarrow \text{gr} K[\text{Rep}(\Pi^\lambda, \delta)]$$

Then, by the linear reductivity of $\text{GL}(\delta)$, since the previous map is a surjective of $\text{GL}(\delta)$ -modules, this induces a surjection

$$K[\text{Rep}(\Pi, \delta)]^{\text{GL}(\delta)} = A^0 \twoheadrightarrow \text{gr} K[\text{Rep}(\Pi^\lambda, \delta)] = \text{gr} A^\lambda$$

.

Since the following diagram commutes,

$$\begin{array}{ccc} K\overline{Q} & \xrightarrow{f} & \text{Mat}_{I \times I}(K[\text{Rep}(\overline{Q}, \delta)]) \\ \downarrow & & \downarrow \\ \text{gr} K\overline{Q} = K\overline{Q} & \xrightarrow{\text{gr } f} & \text{gr Mat}_{I \times I}(K[\text{Rep}(\overline{Q}, \delta)]) \end{array}$$

by descending to the respective quotients, it follows the commutativity of the statement. $\square$

LEMMA 8.4. *The map $\mu : \text{Rep}(\overline{Q}, \delta) \rightarrow \text{End}(\delta)_0$ is surjective and every irreducible component of every fiber has dimension $1 + \sum_{i \in I} \delta_i^2$. Thus, μ is flat.*

PROOF. Let λ such that $\lambda \cdot \delta = 0$, we apply the Lemma 4.5, and we obtain that there is a representation X of Q with dimension vector δ whose endomorphism algebra $\text{End}_{K(Q)}(X) = K$. Then, by the Theorem 4.4, there is a $M \in \text{Rep}(\Pi^\lambda, \delta) = \mu^{-1}(\lambda)$. So, μ is surjective.

This implies that $d = \dim(\text{Rep}(\overline{Q}, \delta)) - \dim(\text{End}(\delta)_0) \leq \dim(C)$, for all C irreducible component of $\mu^{-1}(\lambda)$. Recall

$$\begin{aligned} \dim(\text{Rep}(\overline{Q}, \delta)) &= \sum_{i \rightarrow j} 2\delta_i \delta_j, \\ \dim(\text{End}(\delta)) &= \sum_{i \in I} \delta_i^2 \Rightarrow \dim(\text{End}(\delta)_0) = \sum_{i \in I} \delta_i^2 - 1, \end{aligned}$$

and

$$0 = (\delta, \delta) = 2 \sum_{i \in I} \delta_i^2 - \sum_{i \rightarrow j} \delta_i \delta_j + \sum_j \delta_j \delta_i.$$

Thus

$$d = \sum_{i \rightarrow j} 2\delta_i \delta_j - \sum_{i \in I} \delta_i^2 + 1 = \sum_{i \in I} \delta_i^2 + 1.$$

We consider the projection $\text{Rep}(\overline{Q}, \delta) \rightarrow \text{Rep}(Q, \delta)$ and for $\theta \in \text{End}(\delta)_0$ we first analyze the fibers of its restriction π to $\mu^{-1}(\theta)$

$$\pi : \mu^{-1}(\theta) \rightarrow \text{Rep}(Q, \delta).$$

Consider the fiber of π over any point $x \in \text{Rep}(Q, \delta)$. Note that the map $c = c_x$ of Lemma 4.3 depends on the representation $x \in \text{Rep}(Q, \delta)$. We also note that the map $y \mapsto (x, y)$ is a bijection from $c_x^{-1}(\theta)$ to $\pi^{-1}(x)$. So, the fiber $\pi^{-1}(x)$ is empty or of dimension $\dim(C_x^{-1}(\theta)) = \dim(\ker c_x) = \dim(\text{Ext}^1(X, X))$.

On the other hand, let X the kQ -module corresponding to x . Using the Ringel form $\langle \cdot, \cdot \rangle$ for Q we have:

$$0 = \langle \delta, \delta \rangle = \sum_{i \in I} \delta_i^2 - \sum_{i \rightarrow j} \delta_i \delta_j = \dim(\text{End}(X)) - \dim(\text{Ext}^1(X, X)) \implies \dim(\pi^{-1}(x)) = \dim(\text{End}(X))$$

(see 7.63 in section 7.2 in preliminary chapter 2, for more details see 3.3 in Chapter 7 of [20]). Thus

$$\dim(\pi^{-1}(x)) = \dim(\text{End}(X)) \text{ if } \pi^{-1} \neq \emptyset. \quad (8.1)$$

Now, by Lemma 4.5

$$\text{Rep}(Q, \delta) = Y \cup \bigcup_{i=1}^k \mathcal{O}_i,$$

where $\mathcal{O}_i$ are $\text{GL}(\delta)$ -orbits and Y is the following dense open subset

$$Y = \{x \in \text{Rep}(Q, \delta) \mid \text{End}_{kQ}(X) = K, \text{ where } X \text{ is the corresponding } KQ\text{-module}\}.$$

So, the fibers of the map $\pi^{-1}(Y) \longrightarrow Y$ all have dimension 1. We have that

$$\dim(\mathcal{O}_i) = \dim(\text{GL}(\delta)) - \dim(\text{Stab}(x_i)) = \sum_{i \in I} \delta_i^2 - \dim(\text{End}(X_i)),$$

where $x_i \in \mathcal{O}_i$ and X_i is the corresponding KQ -module (see chapter 8 of [24] or propositions 6.4 and 6.7 in Chapter II of [3]).

This implies that, if $x \in \pi^{-1}(\mathcal{O}_i)$,

$$\dim(\pi^{-1}(x_i)) = \dim(\mathcal{O}_i) + \dim(\pi^{-1}(x_i)) \stackrel{(8.1)}{=} \dim(\mathcal{O}_i) + \dim(\text{End}(X_i)) = \dim(\text{GL}(\delta)) = \sum_{i \in I} \delta_i^2.$$

Since

$$\begin{aligned} \mu^{-1}(\theta) &= \pi^{-1}(Y) \cup \bigcup_{i=1}^k \pi^{-1}(\mathcal{O}_i) \\ &\implies \dim(\mu^{-1}(\theta)) \leq \sum_{i \in I} \delta_i^2 + 1, \end{aligned}$$

which shows that $\dim(\mu^{-1}(\theta)) = d = \sum_{i \in I} \delta_i^2 + 1$.

It follows that μ is flat, because any morphism of smooth varieties such that all irreducible components of all fibers have the same dimension is flat (see Theorem 18.16 of [14], with the choices: $f = \mu$, $X = \text{Rep}(\overline{Q}, \delta)$, $Y = \text{End}(\delta)_0$, $A = \mathcal{O}_{X,x}$, $Q = m_x \triangleleft X$, $R = \mathcal{O}_{Y, f(x)}$ and $P = m_{f(x)} \triangleleft Y$). $\square$

LEMMA 8.5. $\text{Rep}(\Pi^\lambda, \delta) // \text{GL}(\delta)$ has dimension at least 2 for $\lambda \in \mathfrak{h}'$.

PROOF. First: note that since $\lambda \in \mathfrak{h}'$ any representation of Π^λ of dimension vector δ is simple.

Indeed, let M be a representation of Π^λ of dimension vector δ , write a decomposition series for M :

$$M = M_0 \supseteq M_1 \supseteq \dots \supseteq M_p = 0,$$

with M_j/M_{j+1} simple for all $j \in \{0, \dots, p-1\}$. Then

$$\delta = \dim M = \sum_{j=0}^{p-1} M_j/M_{j+1}.$$

Since $\delta_0 = 1$, $\exists! j$ such that $(\dim M_j/M_{j+1})_0 = 1$ and $(\dim M_{j'}/M_{j'+1})_0 = 0$ for all $j' \neq j$.

We write $\lambda = \lambda' + m\epsilon_0$, where λ' is obtained by removing the vertex 0 and $\lambda \cdot \epsilon_0 = \lambda_0 = m \in \mathbb{Z}$ is the 0-th component of λ . Specifically, $\lambda' \in K^I$, with $\lambda'_0 = 0$. Also, we write $\alpha = \beta + k\delta$ for all real root α of Q , where β is a finite root, it's means that it is a root for Q' and $k \in K$. (see Lemma 6.1 in section 6 of Chapter 8 of [20])

Now, since $\lambda \in \mathfrak{h}' \subset \mathfrak{h}$, $0 \neq \lambda \cdot \alpha$, where α is an arbitrary real root, and $0 = \lambda \cdot \delta$, where δ is the fundamental root for the extended Dynkin quiver Q . Then

$$0 = \lambda \cdot \delta = (\lambda' + m\epsilon_0) \cdot \delta = \lambda' \cdot \delta + m\epsilon_0 \cdot \delta = \lambda' \cdot \delta + m$$

$$0 \neq \lambda \cdot \alpha = \lambda \cdot (\beta + k\delta) \text{ where } \beta \text{ is an arbitrary finite root and } k \in \mathbb{Z}$$

$$= (\lambda' + m\epsilon_0) \cdot (\beta + k\delta) = \lambda' \cdot \beta + \lambda' \cdot k\delta + m\epsilon_0 \cdot \beta + m\epsilon_0 \cdot k\delta$$

$$= \lambda' \cdot \beta + m\epsilon_0 \cdot \beta - km + km = \lambda' \cdot \beta + m\epsilon_0 \cdot \beta = \lambda' \cdot \beta$$

Since λ' is identified with an element in $K^{I \setminus 0}$, β is an arbitrary finite root for Q' and Dynkin quivers have no imaginary roots (see Proposition 6.9), $\lambda' \cdot \alpha' \neq 0$, for all real root α' of Q' .

It means that $\lambda' \in K^{I \setminus 0}$ is not on a wall. Then, by Corollary 6.10, $\Pi^{\lambda'}(Q') = 0$. Then, by Corollary 6.11 (see the final of its proof), $\Pi^\lambda(Q)/\Pi^\lambda(Q)e_0\Pi^\lambda(Q) \cong 0$, which implies $\Pi^\lambda(Q) \cong \Pi^\lambda(Q)e_0\Pi^\lambda(Q)$. It follows that $M_{j'}/M_{j'+1} = 0$ for all $j' \neq j$, thus M is simple.

Second: By the representation theory of extended Dynkin quivers (see Chapter 7 in [11]), there is a one-dimensional subvariety C of $\text{Rep}(Q, \delta)$ consisting of non-isomorphic representations with endomorphism algebra K . Now, consider the map π used in the proof of the previous lemma, since $\dim C = 1$, $\dim \pi^{-1}(c) = 1$ for all $c \in C$, we have that $\dim \pi^{-1}(C) = 2$. So $\pi^{-1}(C)$ is a two-dimensional subscheme of $\text{Rep}(\Pi^\lambda, \delta)$, consisting of pairwise non-isomorphic representations.

Third: Recall that $\text{Rep}(\Pi^\lambda, \delta) / \text{GL}(\delta)$ is the set of orbits under the action of $\text{GL}(\delta)$ and $\text{Rep}(\Pi^\lambda, \delta) // \text{GL}(\delta) = \text{Spec} K[\text{Rep}(\Pi^\lambda, \delta)]^{\text{GL}(\delta)}$, where the closed $\text{GL}(\delta)$ -orbits correspond to the closed points. And by [29], the $\text{GL}(\delta)$ -orbits also correspond to the isomorphism classes of semisimple Π^λ -modules of dimension δ .

□

LEMMA 8.6. ϕ_λ is an isomorphism for $\lambda \in \mathfrak{h}'$.

PROOF. Let $x = (x_a)_{a \in \overline{Q}} \in \text{Rep}(\overline{Q}, \delta)$ and let $p = a_1 \dots a_m$ be a path in $K\overline{Q}$ with $t(p) = h(p)$. We define the trace function tr_p , given by

$$\text{tr}_p : \text{Rep}(\overline{Q}, \delta) \longrightarrow K$$

$$\text{tr}_p(x) = \text{tr}(x_{a_1} \cdots x_{a_m}),$$

which is invariant under the action of $\text{GL}(\delta)$.

Indeed, let $p = a_1 \dots a_m$ be a path in $K\overline{Q}$ with $t(p) = h(p)$, $g \in \text{GL}(\delta)$ and $x = (x_a)_{a \in \overline{Q}}$,

$$\begin{aligned} g \cdot \text{tr}_p(x) &= \text{tr}_p(g^{-1} \cdot x) = \text{tr}_p((g_{h(a)} x_a g_{t(a)}^{-1})_{a \in \overline{Q}}) \\ &= \text{tr}(g_{h(a_1)} x_{a_1} g_{t(a_1)}^{-1} g_{h(a_2)} x_{a_2} g_{t(a_2)}^{-1} \cdots g_{h(a_m)} x_{a_m} g_{t(a_m)}^{-1}) \end{aligned}$$

Since p is a path, $t(a_i) = h(a_{i+1})$, for $1 \leq i \leq m-1$ and because $t(p) = h(p)$, $h(a_1) = t(a_m)$.

Thus,

$$g \cdot \text{tr}_p(x) = \text{tr}(x_{a_1} \cdots x_{a_m}) = \text{tr}_p(x).$$

By Theorem 1 in [29], the algebra of polynomial invariants $K[\text{Rep}(\overline{Q}, \delta)]^{\text{GL}(\delta)}$ is generated by the trace functions tr_p (of bounded length).

Now, since $\text{GL}(\delta)$ is a linearly reductive group and $\text{char}(K) = 0$, the surjection

$$K[\text{Rep}(\overline{Q}, \delta)] \twoheadrightarrow K[\text{Rep}(\Pi^\lambda, \delta)]$$

induces a surjection

$$K[\text{Rep}(\overline{Q}, \delta)]^{\text{GL}(\delta)} \twoheadrightarrow K[\text{Rep}(\Pi^\lambda, \delta)]^{\text{GL}(\delta)}$$

(see proposition 6.17 in section 6.3 in the preliminary chapter, and for more details see Chapter IV of [3]). It follows that $K[\text{Rep}(\Pi^\lambda, \delta)]^{\text{GL}(\delta)}$ is generated by images of the trace functions tr_p .

Recall that in this section Q is an affine quiver, because it is the corresponding quiver to $\Gamma \leq \text{SL}_2(K)$. Consider 0 as the extending vertex in Q and the full subquiver Q' given by all the non-zero vertices of Q . Note that $\lambda \in \mathfrak{h}'$ means that λ is not on a wall (see Definition 6.6). Then, we can apply the Corollary 6.11 for Q . Thus, we have $\Pi^\lambda e_0 \Pi^\lambda = \Pi^\lambda$.

So, any element $\overline{a}$ of $e_i \Pi^\lambda e_i = e_i \Pi^\lambda e_0 \Pi^\lambda e_i$ can be written as a sum of paths p_k which pass through vertex 0, i.e. $\overline{a} = \sum_k \alpha_k p_k$, where $\alpha_k \in K$.

Let $c = c_1 \dots c_m$ be a cycle which not pass through vertex 0, with $c_i \in Q$, for all $i = 1, \dots, m$, and let $x = (x_a)_{a \in \overline{Q}} \in \text{Rep}(\Pi^\lambda, \delta)$, we claim that

$$\text{tr}_c(x) = \sum_k \alpha_k \text{tr}_{p_k},$$

if $\overline{c} = \overline{\sum_k \alpha_k p_k}$ and p_k are paths which pass through vertex 0.

Indeed, $\text{tr}_c(x) = \text{tr}(x_{c_1} \cdots x_{c_m})$, which is the trace of the action of c on X , X being the corresponding $K\overline{Q}$ -module.

On the other hand,

$$\sum_k \alpha_k \text{tr}_{p_k} = \sum_k \alpha_k \text{tr}(x_{a_1^k} \cdots x_{a_n^k}) = \text{tr}(\sum_k \alpha_k x_{a_1^k} \cdots x_{a_n^k}),$$

where $p_k = a_1^k \dots a_n^k$ with $a_i^k \in \overline{Q}$ for all $i = 1, \dots, n$ and n depends on the path p_k . The last equality follows from the linearity of the trace function. Finally, note that $\sum_k \alpha_k x_{a_1^k} \cdots x_{a_n^k}$ is also the action of $c = \sum_k \alpha_k p_k$ on X . Thus,

$$\text{tr}_c(x) = \text{tr}(x_{c_1} \cdots x_{c_m}) = \text{tr}(\sum_k \alpha_k x_{a_1^k} \cdots x_{a_n^k}) = \sum_k \alpha_k \text{tr}_{p_k}.$$

Furthermore, if p' is a cyclic permutation of p and p is a path which pass through vertex 0, since $\text{tr}_p = \text{tr}_{p'}$, then $K[\text{Rep}(\Pi^\lambda, \delta)]^{\text{GL}(\delta)}$ is generated by the trace functions tr_p , for paths p starting and ending at vertex 0, i.e.

$$K[\text{Rep}(\Pi^\lambda, \delta)]^{\text{GL}(\delta)} = K\langle \text{tr}_p \mid p \text{ is a cycle in } K\overline{Q}, t(p) = h(p) = 0 \rangle.$$

Next we shall that $\phi_\lambda(\overline{p}) = \text{tr}_p$, for such p that starts and ends at 0. In effect, let $p = a_1 \dots a_m$ be a path such that $h(a_1) = t(a_m) = 0$,

$$f(p) = f(e_0 a_1 \dots a_m e_0) = f(e_0) f(a_1) \cdots f(a_m) f(e_0) \text{ and}$$

$$f(p)(x) = \pi_0 x_{a_1} \cdots x_{a_m} \pi_0 = \text{tr}(x_{a_1} \cdots x_{a_m}) \pi_0 = \phi_\lambda(\overline{p}).$$

The penultimate equality follows from the fact that $x_{a_1} \cdots x_{a_m}$ is a matrix with a single non-zero element at position 00. Thus, this proves that ϕ_λ is surjective.

Now, ϕ_λ is an isomorphism, for since $\mathcal{O}^\lambda$ has GK dimension 2, and any proper quotient of $\mathcal{O}^\lambda$ has dimension at most 1 (see Theorem, of Bergman, as 3.13 in section 3 of preliminary chapter in this thesis, as VI.4.1 in [1] or as 2.5 in [27]). But by the previous lemma, the ring of invariants has dimension at least 2. $\square$

8.2. Situation over the K -algebra R . Until now, we have worked with the fiber of μ given by the following fiber product of affine schemes over K :

$$\begin{array}{ccc} Z & \xrightarrow{\hat{g}} & \text{Spec } K \\ \downarrow & \boxtimes & \downarrow \hat{\lambda} \\ \text{Rep}(\overline{Q}, \delta) & \xrightarrow{\mu} & \text{End}(\delta)_0 \end{array}$$

where μ is a morphism of affine schemes over K , $\lambda \in \mathfrak{h} \subseteq \text{End}(\delta)_0$ induces the application $\hat{\lambda}$ from the morphism of rings:

$$K[\text{End}(\delta)_0] \xrightarrow{\hat{\lambda}} K, \quad p \mapsto p(\lambda),$$

and $\mu^{-1}(\lambda) \cong Z = \text{Rep}(\overline{Q}, \delta) \times_{\text{End}(\delta)_0} \text{Spec } K$ is the fiber of μ .

From the section 4 we have that this pullback of affine schemes is $\mu^{-1}(\lambda) = \text{Rep}(\Pi^\lambda, \delta)$.

We consider a global situation, let R be a commutative ring, we consider the K -algebra $R = K[\mathfrak{h}]$, we shall give R -versions for the situation around the map ϕ_λ .

We define $\text{Rep}(\overline{Q}, \delta)_R$ as the fiber product of affine schemes:

$$\text{Rep}(\overline{Q}, \delta)_R = \text{Rep}(\overline{Q}, \delta) \times_{\text{Spec } K} \text{Spec } R := \text{Spec} \left(K[\text{Rep}(\overline{Q}, \delta)] \otimes_K R \right),$$

it means that $\text{Rep}(\overline{Q}, \delta)_R$ is the base change of $\text{Rep}(\overline{Q}, \delta)$ to the base R . (see example 6.1 in section 6 of the preliminary chapter 2)

Similarly, we define the base change of $\text{End}(\delta)_0$ to the base R :

$$\text{End}(\delta)_{0R} := \text{End}(\delta)_0 \times_{\text{Spec } K} \text{Spec } R := \text{Spec} \left(K[\text{End}(\delta)_0] \otimes_K R \right).$$

The version in R of $\mathfrak{h}$ is

$$\mathfrak{h}_R := \{\Lambda \in R^I \mid \Lambda \cdot \delta = 0\} \subseteq \text{End}(\delta)_{0R},$$

taking Λ as the canonical element in R^I , with $\Lambda \cdot \delta = 0$, Λ is defined by

$$\Lambda = (\Lambda_i)_{i \in I} \in R^I = K[\mathfrak{h}]^I \text{ with } \Lambda_i(\lambda) = \lambda_i \in K.$$

Thus, we have a version of the map μ

$$\mu_R : \text{Rep}(\overline{Q}, \delta)_R \rightarrow \text{End}(\delta)_{0R},$$

a morphism of affine schemes over R .

Now, consider the fiber product

$$\begin{array}{ccc}
Y_{R,\Lambda} & \xrightarrow{\quad} & \text{Spec } K \\
\downarrow & \boxtimes & \downarrow \hat{\Lambda} \\
\text{Rep}(\overline{Q}, \delta)_R & \xrightarrow{\mu_R} & \text{End}(\delta)_{0_R}
\end{array}$$

Again, the map $\hat{\Lambda}$ comes from the evaluation in Λ , now in the global context over R :

$$R[\text{End}(\delta)_{0_R}] \xrightarrow{\Lambda} R = K[\mathfrak{h}], \quad p \mapsto p(\Lambda).$$

And $Y_{R,\Lambda} := \mu^{-1}(\Lambda)_R = \text{Spec}(R[s_{apq}]/J_\Lambda)$ is the fiber of μ_R in $\Lambda \in \mathfrak{h}_R$. Here $R[s_{apq}] = K[s_{apq}] \otimes_K R$ is the base change of $K[s_{apq}]$ to the new base R , and J_Λ is like the J_λ in the context over K except for $\Lambda_i \in R$ in the global version.

There is a natural action of $\text{GL}(\delta)$ over $K[Y_{R,\Lambda}] = R[s_{apq}]/J_\Lambda$. Since $\text{GL}(\delta)$ acts on $K[s_{apq}]$ with J_Λ stable under this action, we define the action $\text{GL}(\delta) \curvearrowright R[s_{apq}]$ as following:

$$g \cdot (f \otimes_K r) = g \cdot f \otimes_K r, \quad \text{for } r \in R \text{ and } f \in K[s_{apq}].$$

Our goal is to work R of the map ϕ_λ . First, for $\Lambda \in \mathfrak{h}_R \subset R^I$ consider the version on R of the deformed preprojective algebra (see up to lemma 2.3 in section 2 of chapter 3)

$$\Pi^{R,\Lambda} := R\overline{Q}/(c - \Lambda),$$

the version on R of $\mathcal{O}^\Lambda$ and identify, such as the case over K ,

$$e_0 \Pi^{R,\Lambda} e_0 =: \mathcal{O}^{R,\Lambda}.$$

This time there is a natural ring morphism

$$R\overline{Q} \rightarrow \text{Mat}_{l \times l} \left(K[\text{Rep}(\overline{Q}, \delta)] \right) \otimes_K R \cong \{ \text{Polynomial functions from } \text{Rep}(\overline{Q}, \delta) \text{ to } \text{End}_K \left(\bigoplus_{i \in I} K^{\delta_i} \right) \} \otimes_K R,$$

which first descends to a morphism

$$f_R : \Pi^{R,\Lambda} \rightarrow \text{Mat}_{l \times l} (K[Y_{R,\Lambda}]),$$

and then restricts to a morphism

$$\phi_{R,\Lambda} := f_R|_{\mathcal{O}^{R,\Lambda}} : \mathcal{O}^{R,\Lambda} \rightarrow A^{R,\Lambda}.$$

LEMMA 8.7. $A^{R,\Lambda}$ is flat over R . (Equivalently $Y_{R,\Lambda} // \text{GL}(\delta) \rightarrow \text{Spec } R$ is flat.)

PROOF. By Lemma 8.4, μ is flat as morphism of affine schemes over K . Let see that μ_R is also flat. By the definition 6.8 and the Proposition 6.10 (in section 6 of the preliminary chapter 2), is the same as saying that $K[\text{Rep}(\overline{Q}, \delta)] \otimes_K R$ is flat as $(K[\text{End}(\delta)_0] \otimes_K R)$ -module:

Let $C \rightarrow D$ be an injective morphism of $(K[\text{End}(\delta)_0] \otimes_K R)$ -modules. Note that, since R and $K[\text{End}(\delta)_0]$ have unity,

$$K[\text{Rep}(\overline{Q}, \delta)] \otimes_K R \otimes_{K[\text{End}(\delta)_0] \otimes_K R} C \cong K[\text{Rep}(\overline{Q}, \delta)] \otimes_{K[\text{End}(\delta)_0]} C,$$

$$K[\text{Rep}(\overline{Q}, \delta)] \otimes_K R \otimes_{K[\text{End}(\delta)_0] \otimes_K R} D \cong K[\text{Rep}(\overline{Q}, \delta)] \otimes_{K[\text{End}(\delta)_0]} C.$$

Then, the induced map

$$K[\text{Rep}(\overline{Q}, \delta)] \otimes_K R \otimes_{K[\text{End}(\delta)_0] \otimes_K R} C \longrightarrow K[\text{Rep}(\overline{Q}, \delta)] \otimes_K R \otimes_{K[\text{End}(\delta)_0] \otimes_K R} D$$

is also injective, because of the flatness of μ . Thus μ_R is a flat morphism between affine schemes.

Because of the flatness is also stable under pullback (see Proposition 6.11 in section 6 of preliminary chapter 2), $Y_{R,\Lambda} \rightarrow \text{Spec} R$ is flat since it is the pullback of μ_R . This implies, by the definition 6.8 and the Proposition 6.10 (in section 6 of the preliminary chapter 2), that $R \rightarrow K[Y_{R,\Lambda}]$ is also flat, it means, that $K[Y_{R,\Lambda}]$ is flat as R -module.

Now, $A^{R,\Lambda} := K[Y_{R,\Lambda}]^{\text{GL}(\delta)}$, then $A^{R,\Lambda}$ is a direct summand of $K[Y_{R,\Lambda}]$, then it is also flat as R -module (see Proposition 6.5 in section 6 of preliminary chapter 2). $\square$

LEMMA 8.8. *If R is a domain with quotient field F , and $\Lambda \cdot \alpha \neq 0$ for all real roots α , then*

$$\phi_{R,\Lambda} \otimes 1 : \mathcal{O}^{R,\Lambda} \otimes F \rightarrow A^{R,\Lambda} \otimes F$$

is an isomorphism. In particular, $A^{R,\Lambda}$ is a domain.

PROOF. It suffices to show that the map is an isomorphism on tensoring with an algebraic closure $\overline{F}$ of F , and then restrict this to $\phi_{R,\Lambda} \otimes 1$. Now, Lemma 8.6 applies with the base field K replaced by $\overline{F}$. Then we obtained the isomorphism

$$\phi_{\overline{F},\Lambda} =: \phi_\Lambda : \mathcal{O}^{\overline{F},\Lambda} \xrightarrow{\sim} A^{\overline{F},\Lambda},$$

where

$$\lambda \in \mathfrak{h}'_{\overline{F}} := \{\lambda \in \mathfrak{h}_{\overline{F}} \mid \lambda \cdot \alpha \neq 0 \text{ for all roots } \alpha\}, \text{ with } \mathfrak{h}_{\overline{F}} =: \{\lambda \in \overline{F}^I \mid \lambda \cdot \delta = 0\}.$$

So, we have the following commutative diagram

$$\begin{array}{ccc} \mathcal{O}^{R,\Lambda} \otimes_R \overline{F} & \xrightarrow{\phi_{R,\Lambda} \otimes 1_{\overline{F}}} & A^{R,\Lambda} \otimes_R \overline{F} \\ \downarrow \wr & & \downarrow \wr \\ \mathcal{O}^{\overline{F},\Lambda} & \xrightarrow{\phi_{\overline{F},\Lambda}} & A^{\overline{F},\Lambda} \end{array}$$

therefore, $\phi_{R,\Lambda} \otimes 1_{\bar{F}}$ is an isomorphism.

For the second assertion, since F is the quotient field of R , there is an inclusion $R \rightarrow F$, then by flatness of $A^{R,\Lambda}$ as an R -module, the induced map $A^{R,\Lambda} = A^{R,\Lambda} \otimes_R R \rightarrow A^{R,\Lambda} \otimes_R F$ is also an injective map (see definition of flatness 6.2 in section 6 of chapter 2).

Then, by the commutative diagram above, $A^{R,\Lambda} \otimes_R F$ is isomorphic to $\mathcal{O}^{F,\Lambda}$, which is a domain, because $\text{gr}\mathcal{O}^{F,\Lambda} \cong F[x, y]^\Gamma$ (see Lemma 1.3 in section 1 of Chapter 3), and this polynomial's ring $F[x, y]^\Gamma$ is a domain since R is one. Thus $A^{R,\Lambda}$ is also a domain. $\square$

LEMMA 8.9. $\text{Rep}(\Pi^\lambda, \delta) // \text{GL}(\delta)$ has dimension 2 for any $\lambda \in \mathfrak{h}$.

PROOF. First: Since $R = K[\mathfrak{h}]$ and R is a domain, then $\mathfrak{h} = \text{Spec}R$ is an irreducible affine scheme. Similarly, since $A^{R,\Lambda}$ is a domain, $\text{Spec}A^{R,\Lambda} = Y_{R,\Lambda} // \text{GL}(\delta)$ is also an irreducible affine scheme. (see Proposition 3.1 and the previous definitions in section 3 of Chapter II of [23]). Then the map

$$Y_{R,\Lambda} // \text{GL}(\delta) \xrightarrow{\psi} \mathfrak{h}$$

is a morphism of irreducible affine schemes. This map is flat, because, $A^{R,\Lambda}$ is flat as R -module, so by the definition 6.8 and the Proposition 6.10 (in section 6 of the preliminary chapter 2) the map considered is flat as a morphism of affine schemes. Also, we have that this map is surjective, because for each $\lambda' \in \mathfrak{h} \subset K^I \subset R^I$, we can define $Y_{R,\Lambda'} // \text{GL}(\delta)$, with $\Lambda'_i(\lambda') = \lambda'_i$. Thus, by Proposition 18.16 in [14], all fibres have the same dimension.

Second: The fibre over λ of ψ is $\text{Rep}(\Pi^\lambda, \delta) // \text{GL}(\delta)$.

Before this, we will prove that $K \otimes_R A^{R,\Lambda} \cong A^\lambda$. Note that K is an R -module via the map $K[\mathfrak{h}] = R \xrightarrow{\lambda} K, p \mapsto p(\lambda)$. We have

$$K \otimes_R A^{R,\Lambda} = K \otimes_R \left(R[s_{apq}] / J_{R,\Lambda} \right)^{\text{GL}(\delta)} \stackrel{(1)}{\cong} \left(K \otimes_R R[s_{apq}] / J_{R,\Lambda} \right)^{\text{GL}(\delta)} \stackrel{(2)}{\cong} A^\lambda \quad (8.2)$$

For the isomorphism (1), we shall prove a more general result:

Let A be an R -module and R be a polynomial ring over the field K , with $K = \bar{K}$ and $\text{char}K = 0$, and let G a linearly reductive group which acts on R trivially, the following holds:

$$K \otimes_R A^G \cong (K \otimes_R A)^G.$$

In effect, note that G acts on A R -linearly and then G also acts on $K \otimes_R A$. Assume $A^G = 0$: We have $R \xrightarrow{\lambda} K$, which is surjective, then by $(- \otimes_R A)$ is a right exact functor, $R \otimes_R A = A \xrightarrow{\lambda \otimes 1} K \otimes_R A$ is also surjective. And because G is linearly reductive $A^G \rightarrow (K \otimes_R A)^G$ is surjective. (see proposition 6.17 in section 6.3 in the preliminary

chapter, and for more details see Chapter IV of [3]) So, $(K \otimes_R A)^G = 0$ because $A^G = 0$. Then

$K \otimes_R A^G = K \otimes_R 0 = 0 = (K \otimes_R A)^G$. Assume now that $A^G \neq 0$: Since G is reductive, we write $A = A^G \oplus A'$, with $A'^G = 0$ (see proposition 6.17 in section 6.3 in the preliminary chapter, and for more details see Chapter IV of [3]). By the previous reasoning $K \otimes_R A'^G = 0$. Now, tensoring with K on the left:

$$K \otimes_R A = (K \otimes_R A^G) \oplus (K \otimes_R A')$$

and there is a split exact sequence

$$0 \rightarrow A^G \rightarrow A^G \oplus A' = A \rightarrow A' \rightarrow 0.$$

By $(K \otimes_R -)$ is a right exact functor, the following sequence is also exact

$$K \otimes_R A^G \rightarrow (K \otimes_R A^G) \oplus (K \otimes_R A') = K \otimes_R A \rightarrow K \otimes_R A' \rightarrow 0.$$

Then, because of the reductivity of G , the sequence of the invariants is right exact

$$(K \otimes_R A^G)^G = K \otimes_R A^G \rightarrow (K \otimes_R A^G \oplus K \otimes_R A')^G = (K \otimes_R A)^G \rightarrow (K \otimes_R A')^G \rightarrow 0.$$

So,

$$0 \cong (K \otimes_R A')^G \cong (K \otimes_R A)^G / K \otimes_R A^G \Rightarrow (K \otimes_R A)^G \cong K \otimes_R A^G.$$

For the isomorphism (2), consider the following exact sequence

$$0 \rightarrow J_{R,\Lambda} \xrightarrow{\iota} R[s_{apq}] \twoheadrightarrow R[s_{apq}] / J_{R,\Lambda} \rightarrow 0,$$

by $(K \otimes_R -)$ is a right exact functor, the following sequence is also exact

$$\begin{aligned} K \otimes_R J_{R,\Lambda} &\xrightarrow{1 \otimes \iota} K \otimes_R R[s_{apq}] \twoheadrightarrow K \otimes_R R[s_{apq}] / J_{R,\Lambda} \rightarrow 0. \\ &\Rightarrow K \otimes_R R[s_{apq}] / J_{R,\Lambda} \cong (K \otimes_R R[s_{apq}]) / (1 \otimes \iota)(K \otimes_R J_{R,\Lambda}). \end{aligned}$$

Let $f_{apq\Lambda_i}$ be the generators of $J_{R,\Lambda}$ and $f_{apq\lambda_i}$ the generator of J_λ , let's calculate $(1 \otimes \iota)(K \otimes_R J_{R,\Lambda})$:

$$(1 \otimes \iota)(k \otimes_R f_{apq\Lambda_i}) = k \otimes_R \iota(f_{apq\Lambda_i}) = k \otimes_R f_{apq\lambda_i}, \quad (8.3)$$

which belongs to $K \otimes_R J_\lambda = J_\lambda$, because the $\Lambda_i \in R$ in the expression for $f_{apq\Lambda_i}$ is moved to the left side of the tensor, via the map $\Lambda_i(\lambda) = \lambda_i$. Thus

$$\Rightarrow K \otimes_R R[s_{apq}] / J_{R,\Lambda} \cong K \otimes_R R[s_{apq}] / K \otimes_R J_\lambda.$$

Third: Now, by the arguments in the final of the proof of Lemma 8.6, the general fibre has dimension 2.

□

DEFINITION 8.10. Let C be an R -module, we define

$$\text{Torsion}(C) = \{c \in C \mid \exists r \in R \setminus \{0\}; rc = 0\}.$$

We say that C is a *torsion* R -module, if $\text{Torsion}(C) = C$.

LEMMA 8.11. Suppose R is a Dedekind domain, $\mathfrak{m}$ is a maximal ideal and C is a torsion R -module, not necessarily finitely generated. If $\text{Tor}_1^R(C, R/\mathfrak{m}) = 0$ then $C \otimes R/\mathfrak{m} = 0$.

PROOF. Assume $C \otimes R/\mathfrak{m} \neq 0$. Let $c \in C \neq 0$, since C is a torsion R -module, there is $r \in R$ such that $r \cdot c = 0$. Then $\text{ann}_R(C) \neq 0$, which implies that $\text{ann}_R(C) = \mathfrak{m}^n$ for some $n \in \mathbb{Z}$.

Write $\mathfrak{m} = \langle \rho \rangle$, for some $\rho \in R$, then $\rho^n \cdot c = 0$ and $\rho^{n-1} \cdot c \neq 0$. Thus $\mathfrak{m} \subseteq \text{ann}_R(\rho^{n-1}c) \triangleleft R$ and $\text{ann}_R(\rho^{n-1}c) \neq R$, then $\text{ann}_R(\rho^{n-1}c) = \mathfrak{m}$.

Thus, the map $R \xrightarrow{\varphi} C, r \mapsto r \cdot \rho^{n-1} \cdot c$, with kernel $\mathfrak{m}$, induce an injection $R/\mathfrak{m} \hookrightarrow C$.

And by the long exact sequence with Tor , considering that Tor_2^R is 0 because R is a Dedekind domain, we obtain that the map $\text{Tor}_1^R(R/\mathfrak{m}, R/\mathfrak{m}) \rightarrow \text{Tor}_1^R(C, R/\mathfrak{m})$ is injective, but $\text{Tor}_1^R(R/\mathfrak{m}, R/\mathfrak{m}) \neq 0$. $\square$

REMARK 8.12. A *Dedekind domain* is a Noetherian normal domain of dimension 1. One class of examples of Dedekind domains is the affine rings of a singular irreducible algebraic curve, for example, $R = \mathbb{C}[x]$ or $R = \mathbb{C}[C]$, where C is an irreducible regular curve over $\mathbb{C}$, and more general $R = K[C]$. (see the definition above Corollary 11.9 in section 11.4 and definitions in section 4.1 of [14]). In our case $R = K[t]$ is a Dedekind domain.

THEOREM 8.13. The map $\phi_0 : \mathcal{O}^0 \rightarrow A^0$ is an isomorphism.

PROOF. We will prove that ϕ_0 is an isomorphism by assuming that ϕ_0 is injective, and then we will exhibit the ideas of the proof that ϕ_0 is injective.

First: Now, fix some $v \in \mathfrak{h}' = \{v \in \mathfrak{h} \mid v \cdot \alpha \neq 0 \text{ for all real roots } \alpha\}$, with $\mathfrak{h} = \{v \in K^I \mid v \cdot \delta = 0\}$.

Let $R = K[t]$ and take $\lambda = vt \in R^I = K[t]$. Note that

$$\lambda \cdot \alpha = vt \cdot \alpha = tv \cdot \alpha \neq 0,$$

then, $\lambda \in \mathfrak{h}'_R \subseteq \mathfrak{h}'_F$, where $F = \text{Frac} R$.

Since $\mathcal{O}^{R,\lambda} := eS^{R,\lambda}e = e \frac{R\langle x,y \rangle \otimes R\Gamma}{(xy-yx-\lambda)} e$ is a free R -module, is flat as R -module. And, by Lemma 8.8, applied to $\lambda = vt \in \mathfrak{h}'_R$, $\phi_{R,\lambda} \otimes 1 : \mathcal{O}^{R,\lambda} \otimes_R F \rightarrow A^{R,\lambda} \otimes_R F$ is an isomorphism. Let C be the cokernel of $\phi_{R,\lambda}$, one has an exact sequence

$$0 \rightarrow \mathcal{O}^\lambda \xrightarrow{\phi_{R,\lambda}} A^{R,\lambda} \rightarrow C \rightarrow 0, \quad (8.4)$$

where $\phi_{R,\lambda}$ is injective, since $\phi_{R,\lambda}$ is a restriction of the isomorphism $\phi_{R,\lambda} \otimes 1$ to the ring $\mathcal{O}^{R,\lambda} : (\phi_{R,\lambda} \otimes 1)|_{\mathcal{O}^{R,\lambda}}(\mathcal{O}^{R,\lambda}) \cong \phi_{R,\lambda}(\mathcal{O}^{R,\lambda})$ via the map

$$(\phi_{R,\lambda} \otimes 1)(e(f(x, y, t) \otimes_R \gamma)e \otimes 1_F) = \phi_{R,\lambda}(e(f(x, y, t) \otimes_R \gamma)e) \otimes 1_F$$

$$\mapsto \phi_{R,\lambda}(e(f(x, y, t) \otimes_R \gamma)e),$$

and because $\mathcal{O}^{R,\lambda}$ and $A^{R,\lambda}$ (see Lemma 8.7) are flat as R -modules, the vertical maps are injective in the following commutative diagram, which ensures injectivity of $\phi_{R,\lambda}$.

$$\begin{array}{ccc} \mathcal{O}^{R,\lambda} \otimes_R R & \xrightarrow{\phi_{R,\lambda}} & A^{R,\lambda} \otimes_R R \\ \downarrow & & \downarrow \\ \mathcal{O}^{R,\lambda} \otimes_R F & \xrightarrow[\sim]{\phi_{R,\lambda} \otimes 1_F} & A^{R,\lambda} \otimes_R F \end{array}$$

Second: $C \otimes_R F = 0$.

By Proposition 6.1 in section 6.3 of [14], F is flat as R -module $\Leftrightarrow \forall M \subset R$ finitely generated ideal of R , $M \otimes_R F \rightarrow F = R \otimes_R F$, $m \otimes f \mapsto mf$ is injective. Let's verify that the latter is true: In our case, $F = \text{Frac} R$, we have that every element of $M \otimes_R F$ is expressible as a simple tensor, $m \otimes f$ for some $m \in M$ and $f \in F$. If $mf = 0$, since F is a quotient field, $m = 0$ or $f = 0$, which is equivalent to $m \otimes f = 0$. Then $M \otimes_R F \rightarrow F$ is injective. Thus, F is flat as R -module.

Then, by flatness of F , tensoring the sequence 8.4 with F , we obtain the following exact sequence

$$0 \rightarrow \mathcal{O}^{R,\lambda} \otimes_R F \xrightarrow{\phi_{R,\lambda} \otimes 1} A^{R,\lambda} \otimes_R F \rightarrow C \otimes_R F \rightarrow 0.$$

Therefore, since $\phi_{R,\lambda} \otimes 1$ is an isomorphism, then

$$\ker(A^{R,\lambda} \otimes_R F \rightarrow C \otimes_R F) = \mathcal{O}^{R,\lambda} \otimes_R F,$$

then $C \otimes_R F = 0$.

Third: C is a torsion R -module, i.e. $C = \text{Torsion}(C)$.

In fact, Let $c \in C$ be an arbitrary element, define $N = Rc$ submodule of C and $L_c = \{r \in R \mid r \cdot c = 0\}$, one has the following exact sequence:

$$0 \rightarrow L_c \rightarrow R \rightarrow N \rightarrow 0,$$

and

$$0 \rightarrow L_c \otimes_R F \rightarrow R \otimes_R F = F \rightarrow N \otimes_R F \rightarrow 0$$

is also exact because of the flatness of F as R -module. Again, by flatness of F , $N \otimes_R F \rightarrow C \otimes_R F = 0$ is injective. Then, $N \otimes_R F = 0$. Thus, $L_c \otimes_R F \cong R \otimes_R F = F$. In particular, $L_c \neq 0$. So, for all $c \in C$ there is $r \in R$ such that $r \cdot c = 0$ (i.e. $r \in L_c$).

Thus, $C = \text{Torsion}(C)$.

Fourth: $\phi_{R,\lambda} \otimes 1_K : \mathcal{O}_R^{R,\lambda} \otimes_R K \rightarrow A_R^{R,\lambda} \otimes_R K$ is an isomorphism.

By tensoring with $K = R/(t)$ the sequence 8.4, we obtain the exact sequence:

$$\mathrm{Tor}_1^R(K, C) \rightarrow \mathcal{O}_R^{R,\lambda} \otimes_R K \xrightarrow{\phi_{R,\lambda} \otimes 1_K} A_R^{R,\lambda} \otimes_R K \rightarrow C \otimes_R K \rightarrow 0.$$

Strategy: If we prove that $\mathrm{Tor}_1^R(K, C) = 0$ and $C \otimes_R K = 0$, then, by the exactness $\phi_{R,\lambda} \otimes 1$ will be an isomorphism. We will use that fact to prove that ϕ_0 is an isomorphism.

By extending further the long exact torsion sequence, we obtain

$$\mathrm{Tor}_1^R(K, A^{R,\lambda}) \longrightarrow \mathrm{Tor}_1^R(K, C) \longrightarrow \mathcal{O}_R^{R,\lambda} \otimes_R K \xrightarrow{\phi_{R,\lambda} \otimes 1_K} A_R^{R,\lambda} \otimes_R K \longrightarrow C \otimes_R K \longrightarrow 0.$$

First, observe that since $A^{R,\lambda}$ is flat as R -module this is equivalent to say that $\mathrm{Tor}_1^R(R/(t), A^{R,\lambda}) = \mathrm{Tor}_1^R(K, A^{R,\lambda}) = 0$ (see Proposition 6.1 in section 6.3 of [14]). Then, by the exactness, $\mathrm{Im}(\mathrm{Tor}_1^R(K, C) \rightarrow K \otimes_R \mathcal{O}_R^{R,\lambda}) = 0$ and $\ker(\mathrm{Tor}_1^R(K, C) \rightarrow K \otimes_R \mathcal{O}_R^{R,\lambda}) = 0$, which implies that $\mathrm{Tor}_1^R(K, C) = 0$. To prove that this is $C \otimes_R K = 0$, we will use the following:

Now, Lemma 8.11 applies with the Dedekind domain $R = K[t]$, the torsion R -module C , and the ideal (t) of R . Thus: If $\mathrm{Tor}_1^R(C, R/(t)) = 0$ then $C \otimes_R R/(t) = C \otimes_R K = 0$.

We conclude that $\phi_{R,\lambda} \otimes 1_K : \mathcal{O}_R^{R,\lambda} \otimes_R K \rightarrow A_R^{R,\lambda} \otimes_R K$ is an isomorphism.

Fifth: Now, We shall prove that $\phi_{R,\lambda} \otimes 1_K : \mathcal{O}_R^{R,\lambda} \otimes_R K \rightarrow A_R^{R,\lambda} \otimes_R K$ is the same map as $\phi_0 : \mathcal{O}^0 \rightarrow A^0$.

First, we define an isomorphism $\omega : K \otimes_R R\langle x, y \rangle \otimes_R R\Gamma \rightarrow K\langle x, y \rangle \otimes_R K\Gamma$. For which it is sufficient to define it on elements of the form $k \otimes_R f(x, y, t) \otimes_R \gamma$, where $k \in K$, $\gamma \in \Gamma$ and $f(x, y, t) \in R\langle x, y \rangle$. And we have

$$\begin{aligned} k \otimes_R f(x, y, t) \otimes_R \gamma &= k \otimes_R \sum_{\beta} g_{\beta}(x, y) t^{\beta} \otimes_R \gamma = \sum_{\beta} (t^{\beta} k \otimes_R g_{\beta}(x, y) \otimes_R \gamma) \\ &\stackrel{(*)}{=} t^0 k \otimes_R g_0(x, y) \otimes_R \gamma = 1 \otimes_R k g_0(x, y) \otimes_R \gamma \end{aligned} \quad (8.5)$$

where the equality marked with (*) is given because the only form to occur

$$\sum_{\beta} t^{\beta} k \in K \text{ is } t^{\beta} = 0 \forall \beta \neq 0.$$

So, we define

$$\begin{aligned} \omega : K \otimes_R R\langle x, y \rangle \otimes_R R\Gamma &\longrightarrow K\langle x, y \rangle \otimes_R K\Gamma, \\ 1 \otimes_R k g_0(x, y) \otimes_R \gamma &\mapsto k g_0(x, y) \otimes_R \gamma. \end{aligned}$$

Now, let see that ω descends to quotients:

Let $k \otimes_R (xy - yx - \sum_{i \in I} \lambda_i \eta_i) \in K \otimes_R R\langle x, y \rangle \otimes_R R\Gamma$, where η_i is an idempotent in $R\Gamma$ as the defined in section 3,

$$\begin{aligned} \omega \left(k \otimes_R (xy - yx - \sum_{i \in I} \lambda_i \eta_i) \right) &= \omega \left(k \otimes_R (xy - yx) - \sum_{i \in I} \lambda_i \otimes_R \eta_i \right) \\ &= k(xy - yx) - \sum_{i \in I} k\varphi(\lambda_i) \otimes_R \eta_i \\ &= k \left(xy - yx - \sum_{i \in I} \varphi(\lambda_i) \eta_i \right) = k \left(xy - yx - \sum_{i \in I} 0 \eta_i \right) \\ &= k(xy - yx) \in K(xy - yx), \end{aligned}$$

where $\varphi : R \rightarrow K$, $\lambda_i \xrightarrow{\varphi} \varphi(\lambda_i) = 0$ is the specialization through which K is an R -module.

Then ω induces a map, also called ω

$$\omega : K \otimes_R \frac{R\langle x, y \rangle \otimes_R R\Gamma}{(xy - yx - \lambda)} \longrightarrow \frac{K\langle x, y \rangle \otimes_R K\Gamma}{(xy - yx)}.$$

Finally, let $e = \frac{1}{|\Gamma|} \sum_{\gamma \in \Gamma} \gamma \in K\Gamma \subset R\Gamma$ be the average of the group elements, and let $k \otimes_R e(f(x, y, t) \otimes_R \gamma) e^{**} = 1 \otimes_R ekg_0(x, y) \otimes_R \gamma e \in \mathcal{O}^{R, \lambda}$, with $k \in K$, $f(x, y, t) \in R\langle x, y \rangle$, $\gamma \in \Gamma$ and (**) is the results of 8.5.

By the other hand, for the isomorphism $K \otimes_R A^{R, \lambda} \cong A^0$, we use the sequence of isomorphism 8.2 in the proof of Lemma 8.9, this time with $\lambda = \nu t$ instead of Λ , $0 \in K$ instead of λ , and in 8.3 $f_{apq\lambda_i}$ is moved to the left side of the tensor, via the specialization $\varphi(\lambda_i) = 0$.

Thus, $K \otimes_R \mathcal{O}^{R, \lambda} \cong \mathcal{O}^0$ and $K \otimes_R A^{R, \lambda} \cong \mathcal{O}^0$, and $1_K \otimes_R \phi_{R, \lambda} = \phi_0$ by the specialization $\lambda \mapsto 0$.

Sixth In order to prove the injectivity of ϕ_0 , we asked W. Crawley-Boevey about it, he answered to us via mail. Some ideas here are referenced by this private communication.

In order to prove that ϕ_0 is injective, is suffices to prove that the homomorphism

$$F : \Pi \rightarrow \text{Mat}_{I \times I}(K[\text{Rep}(\Pi, \delta)])$$

is injective, because $\mathcal{O}^0 \subset \Pi$ and $A^0 \subseteq \text{Mat}_{I \times I}(K[\text{Rep}(\Pi, \delta)])$.

From Lemma 3.11 in section 3, we have that Π is Noetherian. Also, Π is a PI-ring (see 10 in [27]), since $S^0 = \frac{K\langle x, y \rangle \otimes_K K\Gamma}{(xy - yx)}$ is. And Π is a K -algebra finitely generated, because $K\bar{Q}$ is finitely generated over K .

Fact 1: Π is prime.

Indeed, let A the quotient field of $K[x, y]$, then $K[x, y] \otimes_K K\Gamma$ is an order in $A \otimes_K K\Gamma$ ([9]). Then $K[x, y] \otimes_K K\Gamma$ is a prime PI-ring (see 5.3.10 in [31]). So, $\text{gr} S^0 \cong K[x, y] \otimes_K K\Gamma$ is a prime ring (we have this isomorphism from the Theorem 1.3 in section 1). Now,

by 1.6.6 in [31], S^0 is also a prime ring. Finally, since being a prime ring is a property Morita invariant, and by 3.9 $S^0 \stackrel{M}{\sim} \Pi$, then Π is prime.

Fact 2: Π has GK dimension 2 and any proper quotient of Π has GK dimension at most 1.

Indeed, by Corollary 10.7 in [27], since Π is a finitely generated PI-algebra, then $\text{GKdim}\Pi < \infty$. And by Corollary 10.16 in [27], since Π is a Noetherian PI-ring finitely generated with $\text{GKdim}\Pi < \infty$, then $\text{GKdim}\Pi = \text{cl.Kdim}\Pi \in \mathbb{Z}_{\geq 0}$ and the same for any proper quotient of Π : $\text{GKdim}(\Pi/I) = \text{cl.Kdim}(\Pi/I) \in \mathbb{Z}_{\geq 0}$ for all $0 \neq I \triangleleft \Pi$. So, since Π has Classical Krull dimension 2, $\text{GKdim}\Pi = 2$.

Now, since Π is a prime ring, for any proper quotient of Π , $\text{cl.Kdim}(\Pi/I) < \text{cl.Kdim}(\Pi) = 2$. So, any proper quotient of Π has Classical Krull dimension at most 1, so has GK dimension at most 1.

Assume, by reduction to the absurd that that $\text{Ker}F \neq 0$.

By 8.9 $\dim(\text{Rep}(\Pi, \delta) // \text{GL}(\delta)) = 2$. Then, the closed points in $\text{Rep}(\Pi, \delta)$ depend on 2 parameters. But the closed points in $\text{Rep}(\Pi, \delta)$ correspond to the closed $\text{GL}(\delta)$ -orbits in $\text{Rep}(\Pi, \delta)$, which correspond to the semisimple representations of Π of dimension vector δ (see Corollary 5.4 in [20]). Thus, the semisimple Π -modules of dimension vector δ depend on 2 parameters. Since these modules all arise by specializing F to points of $\text{Rep}(\Pi, \delta)$, it follows that $\Pi/\text{Ker}F$ also has semisimple representations of dimension δ depending on 2 parameters. Now, all but one of the composition factors of such a representation has dimension vector vanishing at the extending vertex 0, so is a simple representation for the preprojective algebra of the corresponding Dynkin quiver ([9]). But this algebra has only finitely many simple representations. We deduce that $\Pi/\text{Ker}F$ has simple modules of some dimension n , say, depending on 2 parameters ([9]).

Now, we will use the above to arrive at a contradiction:

Let $\text{Mod}(\Pi/\text{Ker}F, n)$ be the variety of $\Pi/\text{Ker}F$ -module structures on K^n , and let $\text{Simp}(\Pi/\text{Ker}F, n)$ be the open subset of simple modules. The group $\text{GL}(n)$ acts on $\text{Mod}(\Pi/\text{Ker}F, n)$, and since any simple module X has endomorphism algebra K (see chapter 6 in [20]), the orbits on $\text{Simp}(\Pi/\text{Ker}F, n)$ have dimension $\dim \text{GL}(n) - \dim \text{End}_{KQ}(X) = n^2 - 1$ (see 5.1 in chapter 5 of [20]).

Thus, the claim is that $\text{Simp}(\Pi/\text{Ker}F, n)$ has dimension at least $n^2 + 1$, which is a contradiction.

Indeed, if $S \in \text{Simp}(\Pi/\text{Ker}F, n)$, then its annihilator $\text{ann}S \triangleleft \Pi/\text{Ker}F$ is a prime ideal ([9]), so, $\text{ann}S$ contains a minimal prime ideal P of $\Pi/\text{Ker}F$. Thus S is a simple module for $B = \Pi/\text{Ker}F/P$ ([9]).

Since B is a prime ring of GK dimension at most 1 (by the Fact 2) and finitely generated as a K -algebra, B is finitely generated as a module over its centre Z by [34] ([9]). Then, by 13.9.10 in [31], Z is finitely generated as K -algebra and Z has Krull dimension at most 1.

Now, there is a correspondence between simple modules of B and maximal ideals in Z , sending a simple module S to its annihilator as a Z -module. Then it is deduced that the simple modules for B depend on at most 1 parameter. This is $\text{Simp}(B, n)$ has dimension at most n^2 . But,

$$\text{Simp}(\Pi/\text{Ker}F) = \bigcup_{B=\Pi/\text{Ker}F/P} \text{Simp}(B, n),$$

where P runs through the finite number of minimal prime ideals in $\Pi/\text{Ker}F$ ([9]). Then

$$\dim \text{Simp}(\Pi/\text{Ker}F) \leq \sum_{B=\Pi/\text{Ker}F/P} n^2,$$

so, $\dim \text{Simp}(A, n)$ has dimension at least $n^2 + 1$.

□

COROLLARY 8.14. *For any $\lambda \in \mathfrak{h}$ the map $\phi_\lambda : \mathcal{O}^\lambda \rightarrow A^\lambda$ is an isomorphism.*

PROOF. By the commutative diagram in Lemma 8.3 $\text{gr}\phi_\lambda : \text{gr}\mathcal{O}^\lambda \rightarrow \text{gr}A^\lambda$ is surjective. And, by Lemma 8.9, $\text{GKdim}(A^\lambda) = 2$ for all $\lambda \in \mathfrak{h}$. Then $\text{GKdim}(\text{gr}A^\lambda) = \text{GKdim}(A^\lambda) = 2$ (see Definition 6.9 and Theorem 6.19 in [31]). Also $\text{gr}\mathcal{O}^\lambda$ has dimension 2, because by the Theorem 1.3 $\text{gr}\mathcal{O}^\lambda \cong K[x, y]^\Gamma$ and $\text{GKdim}(K[x, y]^\Gamma) = 2$ (see Example 3.1 in section 3 of preliminary chapter 2 and Proposition 4.3 in Chapter 4 of [27]). Thus, $\text{gr}\phi_\lambda : \text{gr}\mathcal{O}^\lambda \rightarrow \text{gr}A^\lambda$ is an isomorphism, which implies that ϕ_λ is an isomorphism (see proposition 1.12 in section 1 of preliminary chapter 2). □

Final comments.

The article by Crawley-Boevey and Holland provides us with a tool to find simple representations of the deformation of the semidirect product $C[x_0, \dots, x_n] \rtimes W_n(\Gamma)$ (see [5]), based on the classification in Theorem 7.7 and the Morita equivalence of Corollary 3.9. With which a fundamental conjecture appears for progress towards the solution of the problem zero fiber ring dimension.

Bibliography

- [1] M. ARTIN, *Class notes. Noncommutative rings.*, 1999.
- [2] D. BAER, W. GEIGLE, AND H. LENZING, *The preprojective algebra of a tame hereditary artin algebra*, Communications in Algebra, 15 (1987), p. 425–457.
- [3] A. BOREL, *Linear Algebraic Groups*, Springer New York, 1991.
- [4] N. BOURBAKI, *Lie Groups and Lie Algebras. Chapters 4-6*, Springer International Publishing, 2008.
- [5] L. CARTAYA AND S. GRIFFETH, *Zero fiber of quaternionic quotient singularities*, arXiv, jan 2024.
- [6] S. C. COUTINHO, *A Primer of Algebraic D-Modules*, London Mathematical Society Student Texts, Cambridge University Press, 1995.
- [7] W. CRAWLEY-BOEVEY, *Lectures on representations of quivers*.
- [8] W. CRAWLEY-BOEVEY, *Preprojective algebras, differential operators and a conze embedding for deformations of kleinian singularities*, Commentarii Mathematici Helvetici, 74 (1999), p. 548–574.
- [9] W. CRAWLEY-BOEVEY, *Private communication.*, (2024).
- [10] W. CRAWLEY-BOEVEY AND M. P. HOLLAND, *Noncommutative deformations of kleinian singularities*, Duke Mathematical Journal, 92 (1998).
- [11] H. DERKSEN AND J. WEYMAN, *An Introduction to Quiver Representations*, American Mathematical Society, Nov. 2017.
- [12] V. DLAB AND C. M. RINGEL, *Indecomposable representations of graphs and algebras*, 1976.
- [13] D. S. DUMMIT AND R. M. FOOTE, *Abstract Algebra*, Prentice Hall, 1991.
- [14] D. EISENBUD, *Commutative Algebra*, Springer New York, 1995.
- [15] G. GONZALEZ-SPRINGER AND J.-L. VERDIER, *Construction géométrique de la correspondance de McKay*, Annales scientifiques de l’École normale supérieure, 16 (1983), pp. 409–449.
- [16] I. GORDON, *On the quotient ring by diagonal invariants*, Inventiones Mathematicae, 153 (2003), p. 503–518.
- [17] I. GORDON AND S. GRIFFETH, *Catalan numbers for complex reflection groups*, 2012.
- [18] S. GRIFFETH, *Towards a combinatorial representation theory for the rational cherednik algebra of type $g(r, p, n)$* , Proceedings of the Edinburgh Mathematical Society, 53 (2010), p. 419–445.
- [19] S. GRIFFETH, *The diagonal coinvariant ring of a complex reflection group*, Algebra and Number Theory, 17 (2023), p. 2033–2053.
- [20] C. GRUSON AND V. SERGANOVA, *A Journey Through Representation Theory*, Springer International Publishing, 2018.
- [21] M. HAIMAN, *Hilbert schemes, polygraphs and the macdonald positivity conjecture*, Journal of the American Mathematical Society, 14 (2001), p. 941–1006.
- [22] M. HAIMAN, *Vanishing theorems and character formulas for the hilbert scheme of points in the plane*, Inventiones Mathematicae, 149 (2002), p. 371–407.
- [23] R. HARTSHORNE, *Algebraic Geometry*, Springer New York, 1977.
- [24] J. E. HUMPHREYS, *Linear Algebraic Groups*, Springer New York, 1975.

- [25] J. E. HUMPHREYS, *Reflection Groups and Coxeter Groups*, Cambridge University Press, June 1990.
- [26] V. G. KAC, *Infinite-Dimensional Lie Algebras*, Cambridge University Press, 3rd ed., 1995.
- [27] G. KRAUSE AND T. LENAGAN, *Growth of Algebras and Gelfand-Kirillov Dimension*, Graduate studies in mathematics, American Mathematical Society, 2000.
- [28] S. M. LANE, *Categories for the Working Mathematician*, Springer New York, 1978.
- [29] L. LE BRUYN AND C. PROCESI, *Semisimple representations of quivers*, Transactions of the American Mathematical Society, 317 (1990).
- [30] S. MAC LANE, *Homology*, Springer Berlin Heidelberg, 1963.
- [31] J. MCCONNELL AND J. ROBSON, *Noncommutative Noetherian Rings*, American Mathematical Society, feb 2001.
- [32] J. J. ROTMAN, *An Introduction to Homological Algebra*, Springer New York, 2009.
- [33] J.-P. SERRE, *Linear Representations of Finite Groups*, Springer New York, 1977.
- [34] L. SMALL AND R. WARFIELD, *Prime affine algebras of gelfand—kirillov dimension one*, Journal of Algebra, 91 (1984), p. 386–389.